



ESCUELA DE ADMINISTRACIÓN
FACULTAD DE ECONOMÍA Y ADMINISTRACIÓN

INTERNATIONAL FINANCE

GENERAL INFORMATION

Jose Tessada

August 2024

- The final grade in this course will be based on two exams (30% each), case reports (30%), and engagement (10%).

- The formula to calculate the final grade is:

$$Grade = 0.3 \times Exam1 + 0.3 \times Exam2 + 0.3 \times C + 0.1 \times E$$

where *Exam1* and *Exam2* are the grades in the midterm and final exams, respectively. *E* is the engagement grade. *C* corresponds to the grade in the cases.

Classes

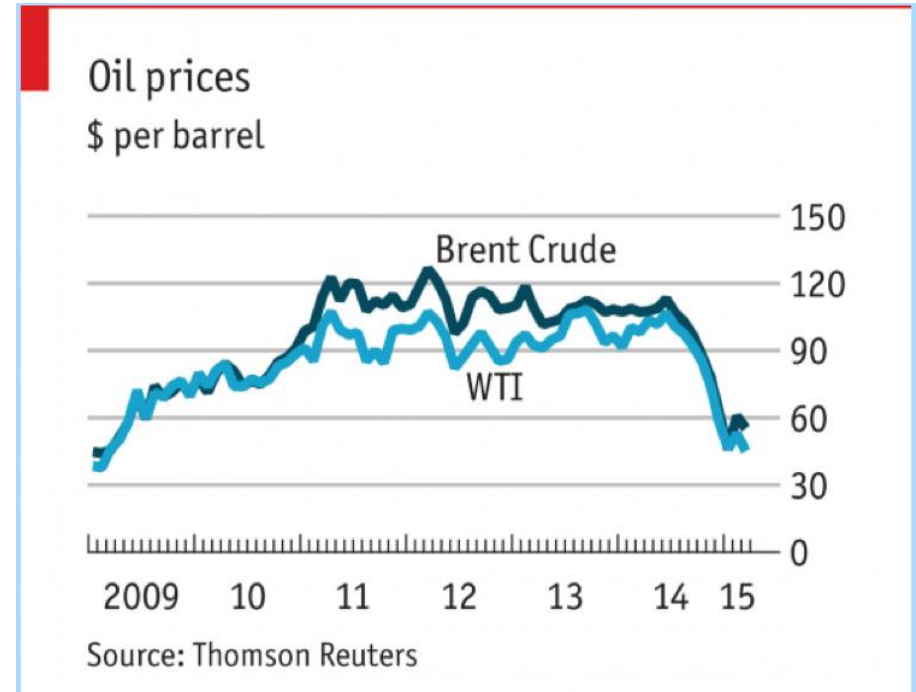
- Tuesday and Thursday, 12:20pm to 1:30pm, room AE207
- Recitations : Th. 2:50pm to 4pm, room AE204
- Calendar in Canvas
- Cases and readings will be available in Canvas
 - Questions for the cases will be uploaded at least one week before they are due
- For questions , email me: jtessada@uc.cl
- Office hours : Wednesdays 5:30pm to 7pm, via Teams , or by appointment .

Rules general

- Please keep cell phones in silence and in your pocket
- Class notes will be uploaded the night before the class (except for the first class)
- Can use a tablet to take notes
 - Also can bring computer laptop but preferably use a tablet

News Example

- The oil price: so much for the rally. World oil stockpiles are rising by 1.6m barrels a day — and drowning hopes of a price rally... Demand is soggy (notably due to China's slowing economy), and supply is strong. Libya has confounded expectations by doubling its production in the past few weeks



ANY QUESTIONS ?



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INTERNATIONAL FINANCE GLOBALIZATION AND BUSINESS DECISIONS

Jose Tessada

August 2024

FIRMS, GLOBALIZATION AND MULTINATIONALS

Multinational Companies What are they?

- Multinational company (MN): company that has a subsidiary, office or franchise operating in other countries
- What other companies operate in international markets without being MN?

Globalization and Companies MN

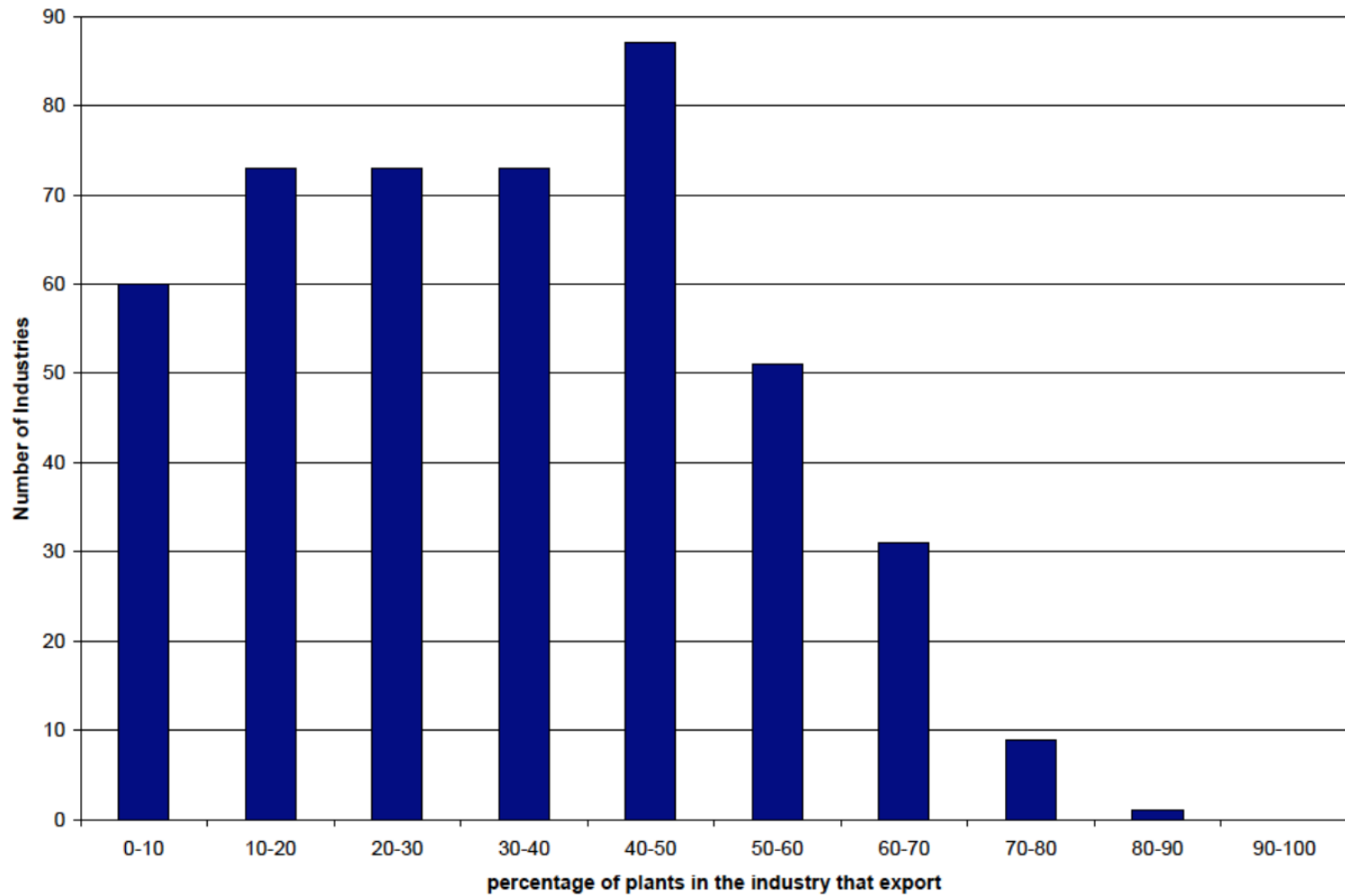
- Globalization
 - Increase in connectivity and integration between countries, people and companies
 - MN operate (produce and/or sell) in more than one country
- How globalized are we?

Exporting By U.S. Manufacturing Firms, 2002

<i>NAICS industry</i>	<i>Percent of firms</i>	<i>Percent of firms that export</i>	<i>Mean exports as a percent of total shipments</i>
311 Food Manufacturing	6.8	12	15
312 Beverage and Tobacco Product	0.7	23	7
313 Textile Mills	1.0	25	13
314 Textile Product Mills	1.9	12	12
315 Apparel Manufacturing	3.2	8	14
316 Leather and Allied Product	0.4	24	13
321 Wood Product Manufacturing	5.5	8	19
322 Paper Manufacturing	1.4	24	9
323 Printing and Related Support	11.9	5	14
324 Petroleum and Coal Products	0.4	18	12
325 Chemical Manufacturing	3.1	36	14
326 Plastics and Rubber Products	4.4	28	10
327 Nonmetallic Mineral Product	4.0	9	12
331 Primary Metal Manufacturing	1.5	30	10
332 Fabricated Metal Product	19.9	14	12
333 Machinery Manufacturing	9.0	33	16
334 Computer and Electronic Product	4.5	38	21
335 Electrical Equipment, Appliance	1.7	38	13
336 Transportation Equipment	3.4	28	13
337 Furniture and Related Product	6.4	7	10
339 Miscellaneous Manufacturing	9.1	2	15
Aggregate manufacturing	100	18	14

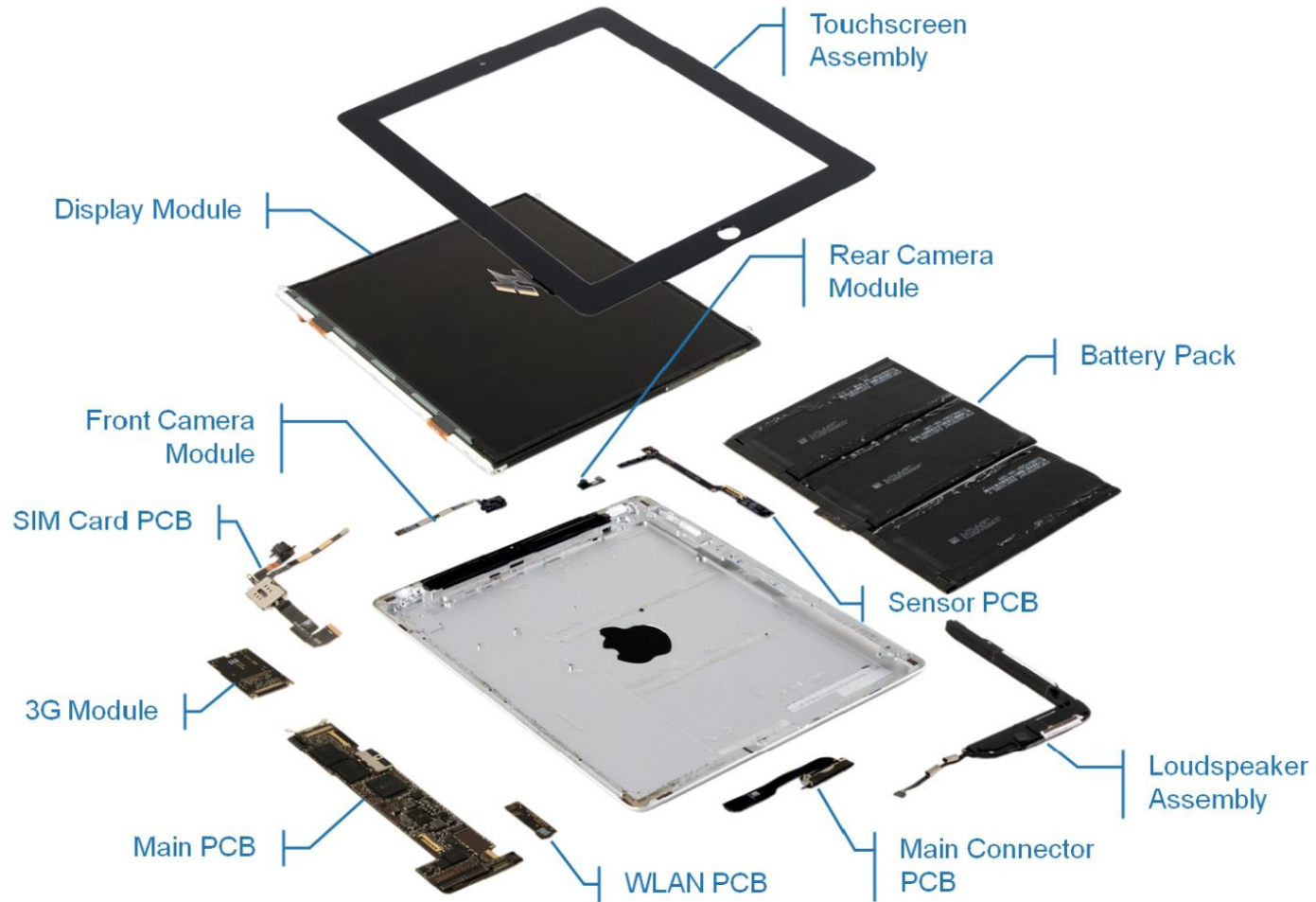
Sources: Data are from the 2002 U.S. Census of Manufactures.

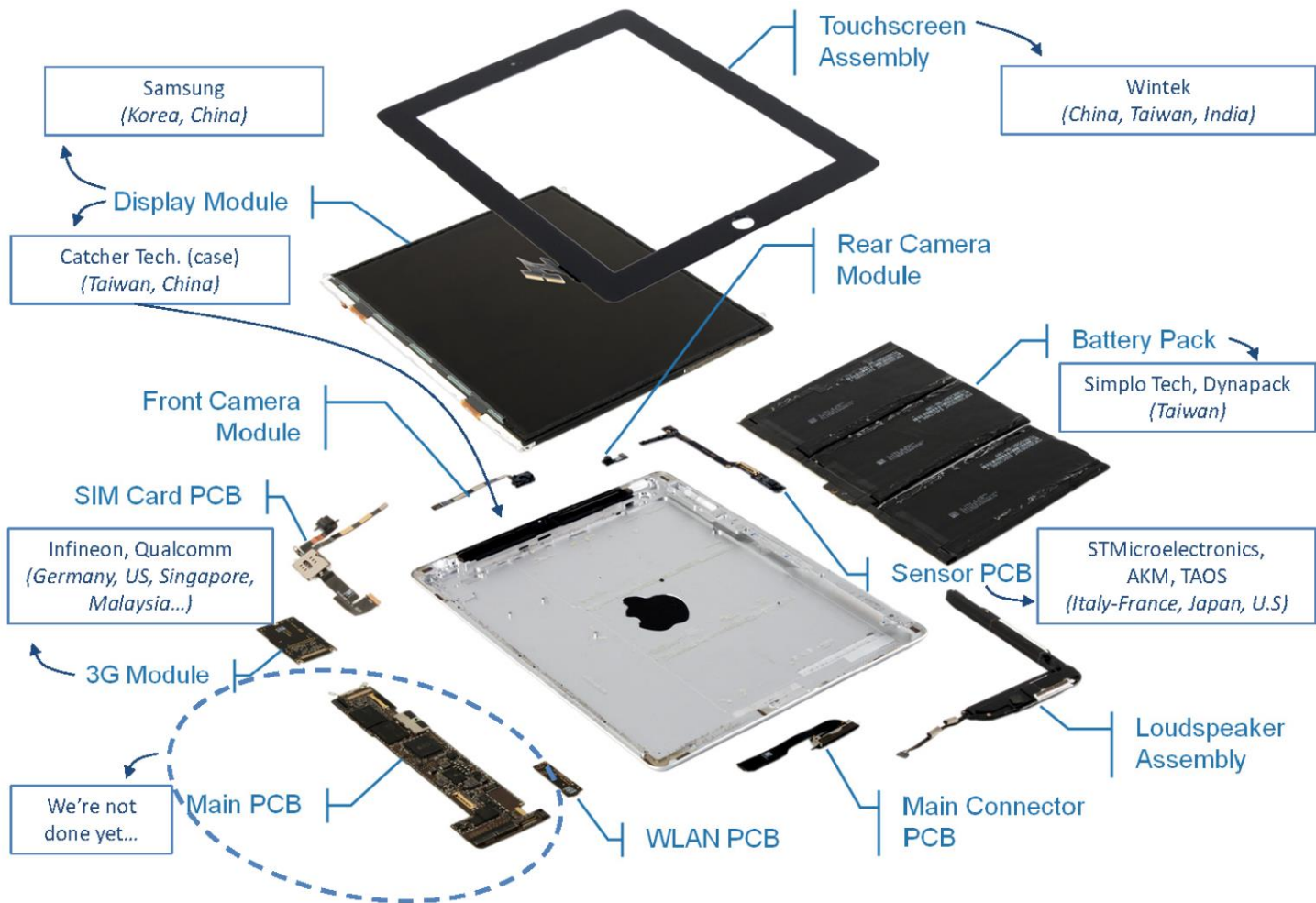
Notes: The first column of numbers summarizes the distribution of manufacturing firms across three-digit NAICS manufacturing industries. The second reports the share of firms in each industry that export. The final column reports mean exports as a percent of total shipments across all firms that export in the noted industry.



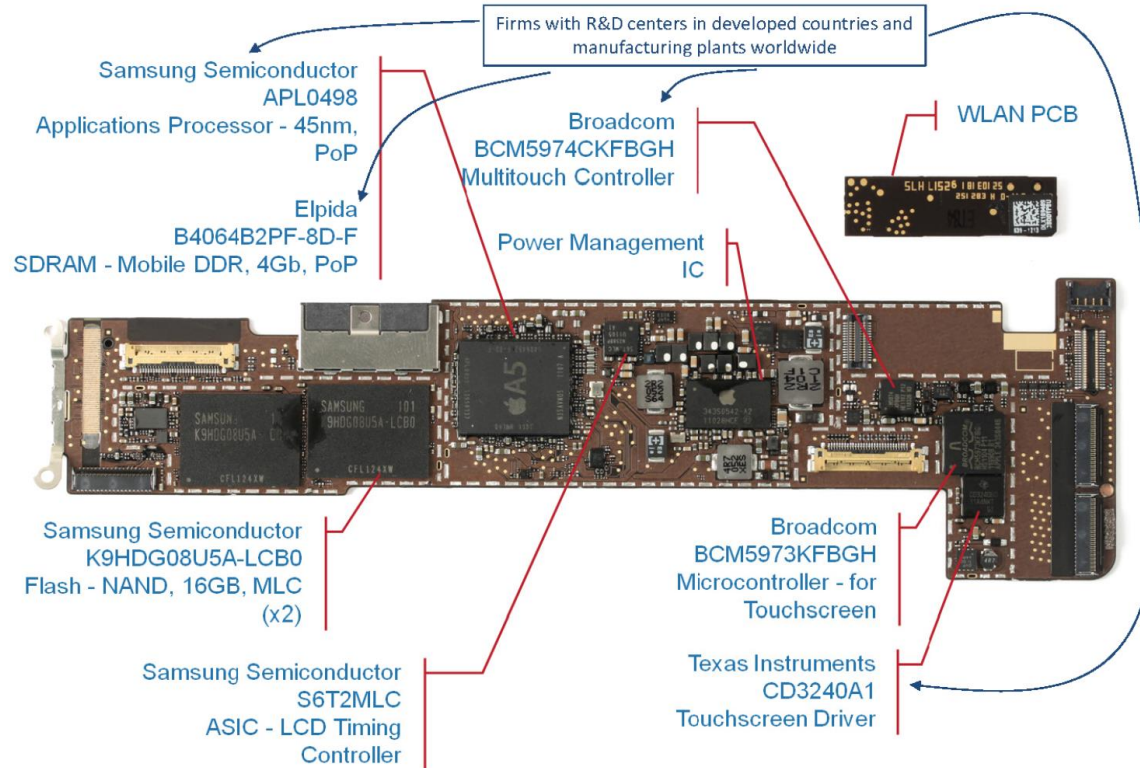


Designed by Apple in California, Assembled in China





Using magnifying glass



UNDERSTANDING INTERNATIONAL EXPANSION

Comparative advantage

- Let us start from the simplest environment, assume
 - Free trade
 - Perfect competition
 - Without uncertainty
 - Perfect information
 - Without government intervention

Comparative advantage

- The point here is who is relatively better at producing
- It is not an absolute issue, but a comparative one.
 - A produces M instead of N if and only if it is relatively better than B at producing M instead of N
 - It doesn't matter if B is better at everything
- Since the productive factors do not move, trade allows us to specialize
- Adam Smith talked about Portuguese wine and English cloth

Comparative advantage

- What variables really matter?
 - Terms of exchange
 - Lithium today, 2 years ago and 10 years ago
 - Supply and demand determine what each country produces (and how much)
 - No country loses and at least one must win but...
 - Both do not necessarily earn the same.

Comparative Advantage... but...

- It's not wine for cloth anymore... trade in tasks
- It is not always comparative advantages that determine this
 - government intervention
 - Movement of factors
 - Many factors...
 - Dynamics of comparative advantage
 - Differentiated products
- Or are some of these elements a source of comparative advantage?

Comparative Advantage... Today...

- We should probably also think about services...
- And that leads us to think about what additional factors can affect comparative advantage
 - “Institutions”
 - Contracts and specialized production
 - Infrastructure:
 - Communications
 - Human capital?

Huachipato dejará de operar tras 70 años: las razones tras el cierre pese a la sobretasa al acero chino

La compañía siderúrgica, filial del grupo CAP, apuntó que no pudo subir los precios y anunció que cerrará de aquí a septiembre. El ministro de Economía apuntó sus críticas a la empresa y a su principal cliente, Molycop, quienes negociaron hasta la mañana de este miércoles. El mercado bursátil celebró con un alza en las acciones de CAP.

How could you use this for the class?

First, can anyone tell us what the issue is?

Impact on stock prices

- One potential spin with a finance side for the company

Acción de CAP salta casi 5% y firma mide efecto financiero



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●●● Aunque llegó a subir más de 8% durante la jornada, la acción de CAP cerró el día con un aumento de 4,86%, la mayor alza de las empresas del Ipsa. Su controladora, Invercap ganó 4,46%.

El gerente de estudios de Rentat4, Guillermo Araya, recordó que en 2023 mientras el área de minería de hierro de CAP, mediante la Compañía Minera del Pacífico, aportó US\$ 400 millones al resultado neto, Huachipato perdió US\$ 385 millones. "Es decir, el área siderúrgica destruye casi todo el valor económico que aporta el área de minería de hierro, por lo tanto, el cierre de CSH es bien recibido por el mercado", dijo.

A popular trading strategy just blew up in investors' faces



Analysis by Allison Morrow, CNN

🕒 4 minute read · Updated 11:09 AM EDT, Wed August 7, 2024

New York (CNN) — Sometimes, markets tumble because of a big, unambiguous event, like a pandemic or a war or an economic slowdown. Sometimes, they fall because of less visible forces, like high-level financial shenanigans designed to generate profits out of thin air.

This week, markets have been grappling with both.

The selloff on Wall Street began last week with some fundamental concerns: Tech earnings were disappointing, and there were red flags in the US labor market that could indicate a serious slowdown in the world's biggest economy.

But the financial shift that poured gasoline on the fire is more obscure, and so technical that even some longtime investors struggle to explain it succinctly.

The carry trade, explained

Put simply: A carry trade is when you borrow money in a place where interest rates are low and use it to invest elsewhere in assets that generate some kind of return.

For years, the place to get money on the cheap has been Japan, where interest rates have held steadily at or near zero.

An investor could borrow Japanese yen (for a small fee) and use that yen to buy, say, US tech stocks or government bonds or the Mexican peso — all of which have offered solid returns in recent years. In theory, as long as the yen remains low against the dollar, you can pay back what you borrowed and still walk away with a tidy profit.

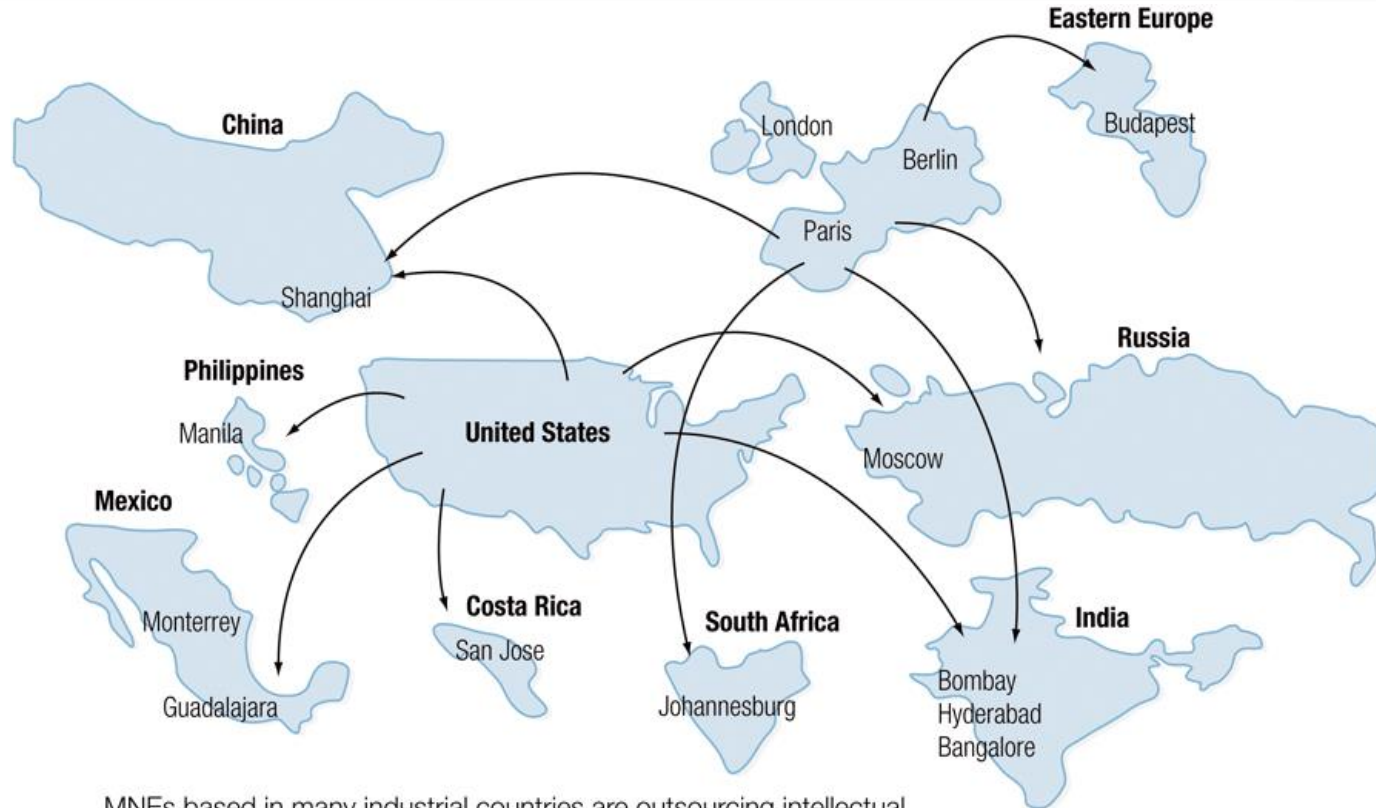
“For the century so far, you would have made more money in the yen-peso carry trade than you would have done in the S&P 500,” Bloomberg columnist John Authers said on the Big Take Daily podcast. “That is bonkers.”

The yen carry trade proved especially popular in the last four years, because Japan was the only major economy in the world offering essentially free money. (While the US, Europe and others were raising interest rates to fight inflation, Japan has had the opposite problem, and it kept borrowing rates low to encourage economic growth.)

Borrowing for next to nothing and getting a 5% return on a US Treasury sounds like a no-brainer.

MULTINATIONAL COMPANIES

Outsourcing and Comparative Advantage



MNEs based in many industrial countries are outsourcing intellectual functions to providers based in traditional emerging market countries.

Between firms or within a firm?

- Data for the United States
- Variation when comparing countries
- Much variation between sectors

Why a MN company?

- You need to obtain a product for your operation
 - Corn
 - Aluminum parts
- Do you sign a contract with a supplier?
- Do you create a subsidiary that purchases in the country of origin?
- Do you create a subsidiary and produce yourself?

According to *Diario Financiero*, data shows that hazelnut plantings in Chile have multiplied ten times in the last decade and now account for the fourth-largest area of any type of fruit tree in Chile.

Many people are unaware that a percentage of these hazelnuts will support the international production of brands like Nutella, Kinder Bueno and Rocher chocolates – all of which are part of the Ferrero portfolio.

The Italian group operates in Chile through its subsidiary AgriChile, whose agricultural branch has over 4,000 hectares of hazelnut trees. Its industrial branch is responsible for processing the nuts to prepare them for the company's final products before shipment.

«The business situation in Chile is promising; hazelnut production is steadily increasing,» said Marco Botta, CEO of Ferrero Hazelnut Company CEO, from the company's Luxembourg headquarters.

Consequently, the company plans to maintain its commitment to the country moving forward: «There is tremendous room to grow. Demand is growing, and I believe the industry will continue to do the same. Chile is fantastic in terms of production. Quality is extremely high (...) Ferrero always operates with the long term in mind.»

<https://blog.investchile.gob.cl/ferrero-chilean-hazelnuts-investment>

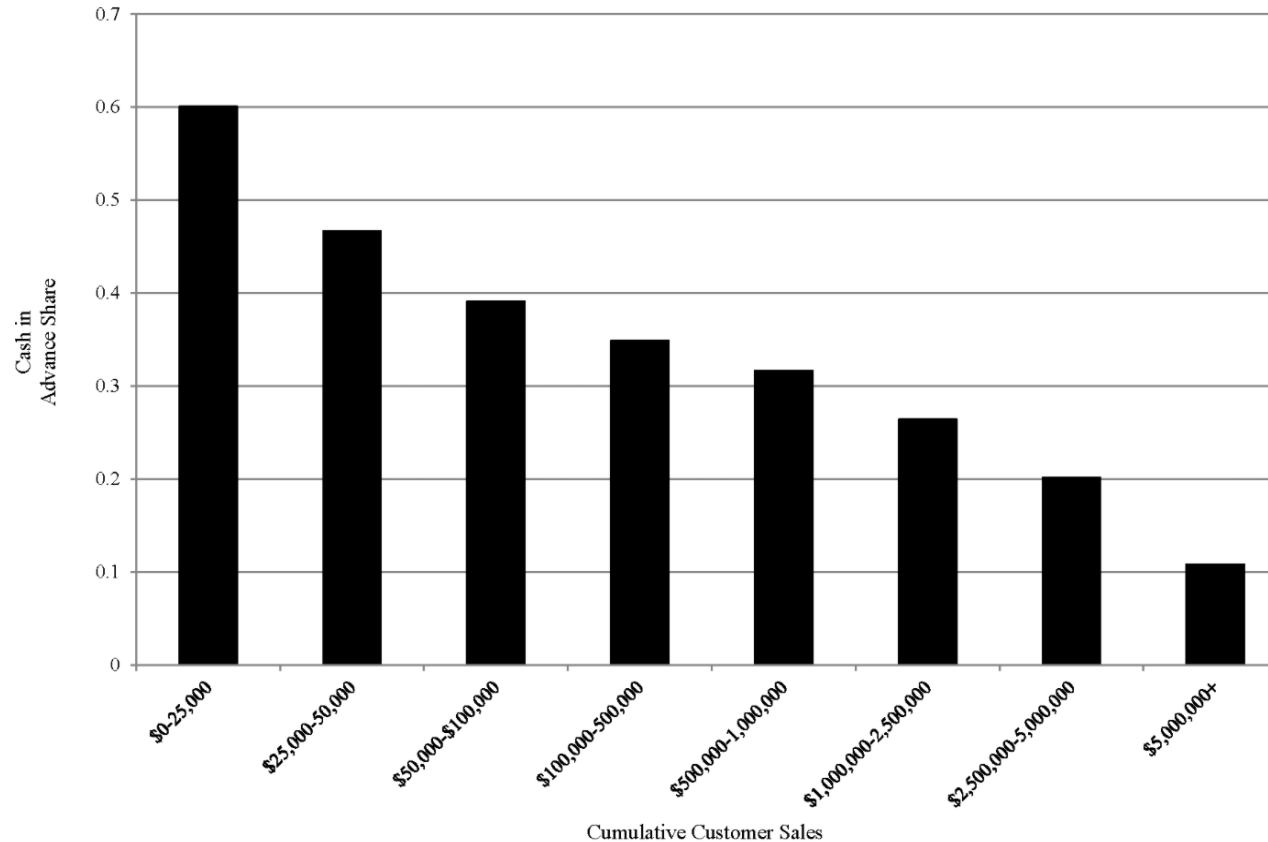
What is an MN company looking for?

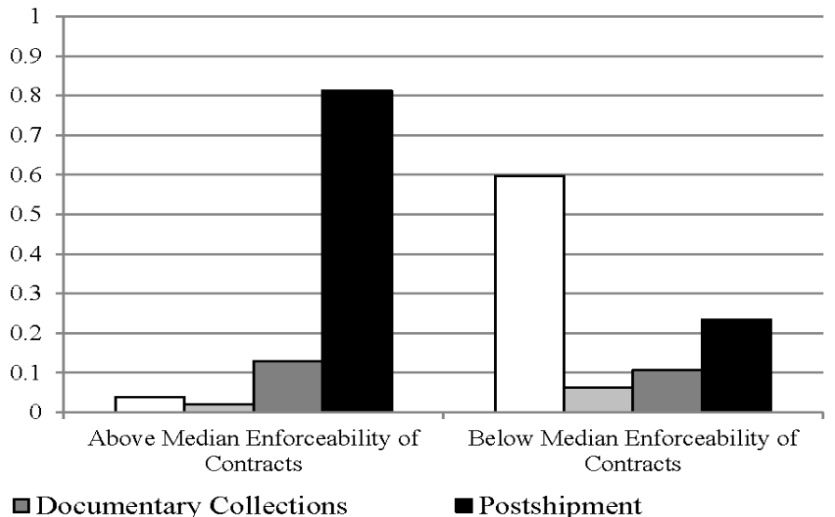
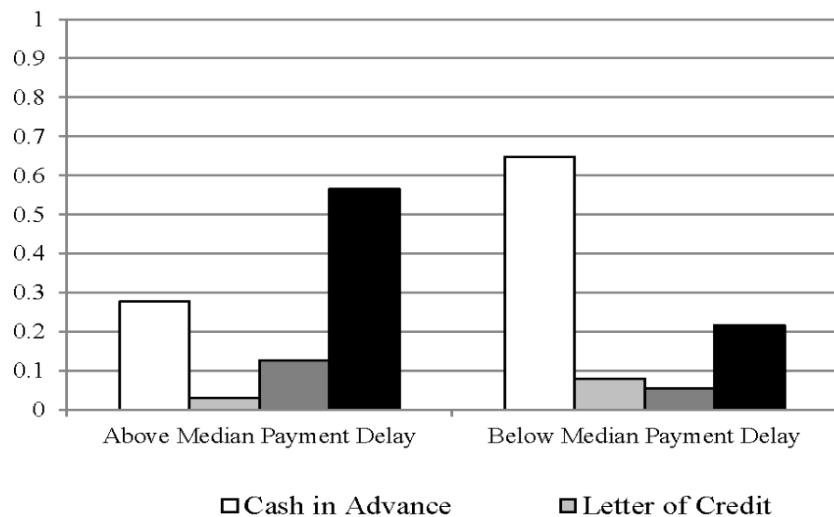
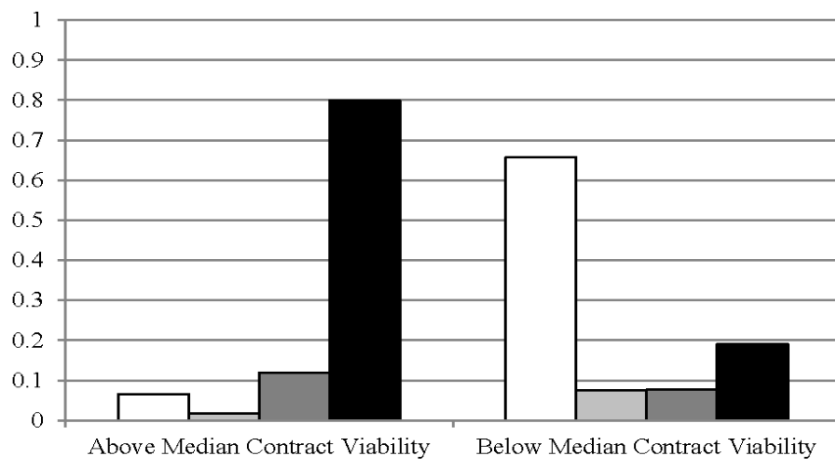
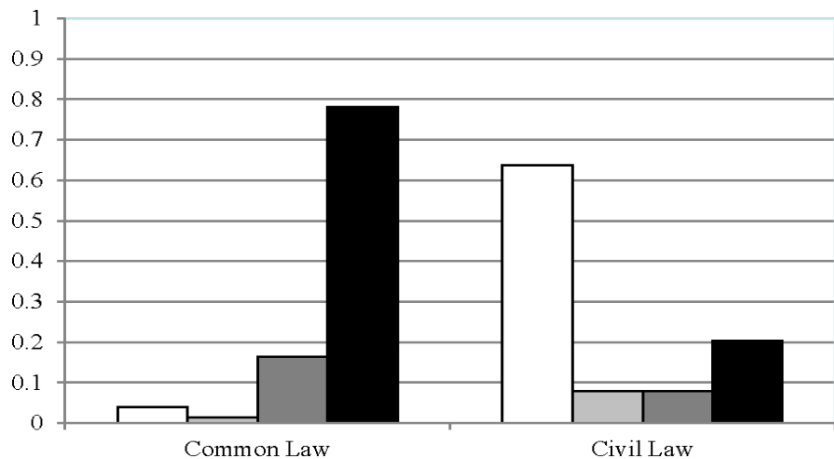
- Opportunities
- Strategic reasons
 - Search new markets
 - Search primary materials
 - Efficiency gains
 - Knowledge and technology
 - Political security
- You can be proactive or defensive (when there is imperfect competition)

Options and decisions

- Domestic production with international parts or inputs
 - Currency for contracts
 - Contract enforcement
 - Buying and selling... affects the way you trade
- Investment versus presence abroad
 - I produce and export, or I invest and produce abroad
 - I control assets or license
 - Joint venture or full ownership
 - If I am an owner, starting from scratch or buying an existing company

How is international trade financed?





IS FINANCE A SOURCE OF ADVANTAGE FOR A COUNTRY OR COMPANY?



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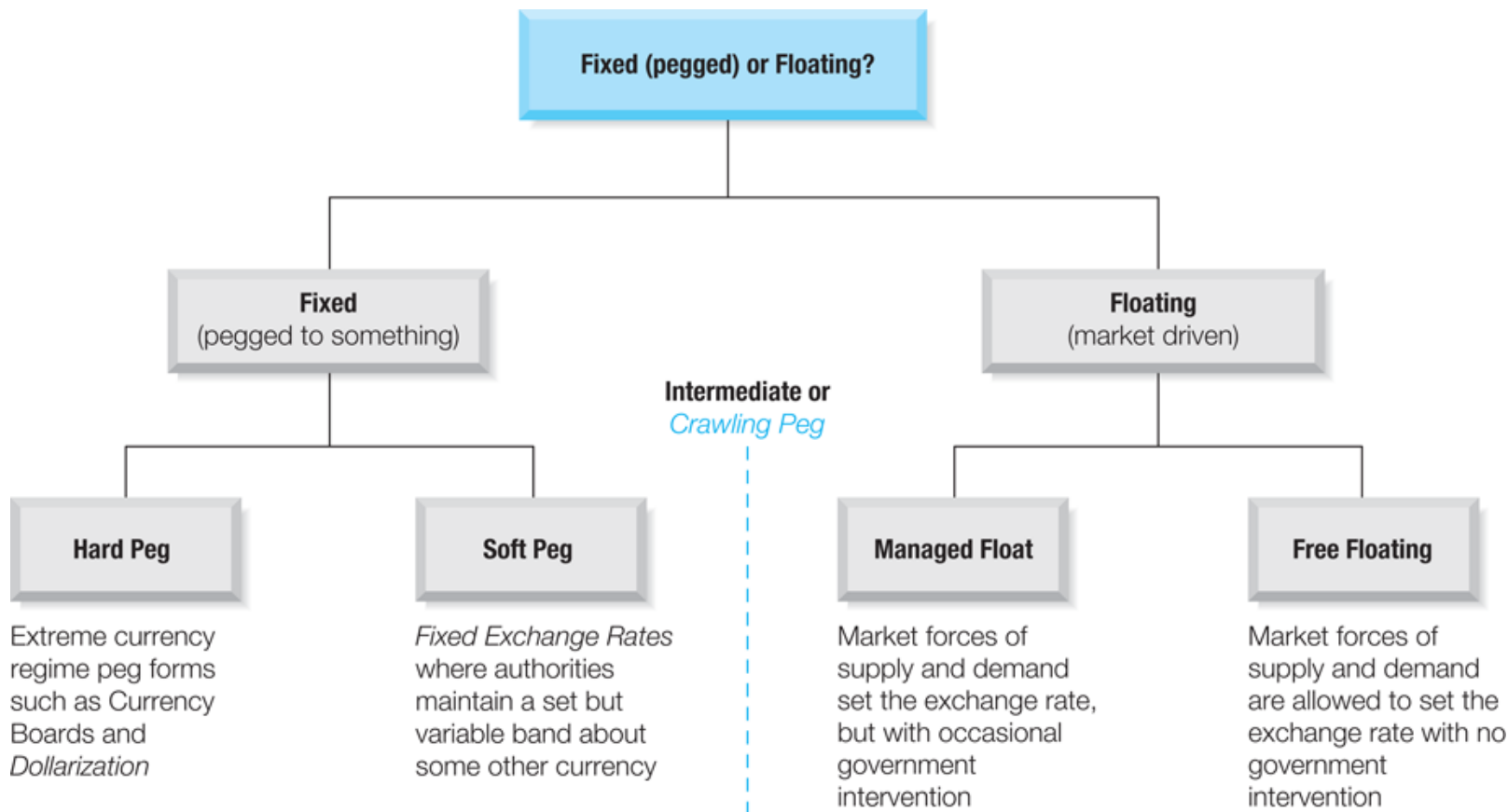
INTERNATIONAL FINANCE BALANCE OF PAYMENTS

José A. Tessada

August 2024

TAKING STOCK...

Exchange Rate Regimes (IMF)



- Country's "optimal" exchange rate regime?
- In practice, a mix of several elements
 - Macroeconomic, trade, financial factors
 - Institutional factors
- Trilemma or Impossible Trinity (remember the triangle)
 - Connection between markets matters even more
- Choice of exchange rate regimes implies a trade-off
 - Countries must balance risks
 - This lesson was learnt the hard way for developing countries (some still learning it)

¿Exchange rate regimes and risks?

- Floating exchange rate
- Exchange rate band
- Fixed exchange rate
- Currency board
- Dollarization
- Dirty float or managed float
- What about countries with inflation targets?

BALANCE OF PAYMENTS

THE BALANCE OF PAYMENTS (BOP)

- Reflects all international economic transactions between the residents of a country and foreign residents is called the **Balance of Payments (BOP)**
- BOP data is important for **policymakers** and **MNEs** as it contains information of a nation's economic fundamentals
- For a MNE, home and host country BOP data is important as:
 - An indication of **pressure** on a country's foreign **exchange rate**
 - A forecast of a country's **market potential**
 - A signal of the imposition or removal of **controls** in various sorts of payments (dividends, interest, and other cash disbursements)

An example...

Exchange rate market yesterday in Chile

Spot market (we will define that in more detail soon) shows large variation...

How big is this swing?

Why the copper price?

International market...

Dólar cae más de \$15 al inicio de la tarde en medio de avance del cobre y descenso internacional de la divisa

Los futuros de cobre superaban los US\$ 4,2 en el mercado Comex.



El dólar bajaba con fuerza al inicio de la tarde del lunes, en medio de un avance en el precio del cobre y un retroceso en la cotización internacional de la divisa, y luego de datos de PIB y déficit en la cuenta corriente en Chile que estuvieron dentro de lo esperado. Poco antes de las 14 horas,...

TYPICAL BOP TRANSACTIONS

- You buy something on Aliexpress or Amazon with a Chilean credit card
- The Chilean subsidiary of a multinational pays profits (dividends) back to a parent in its home country
- The BOP *must* balance → It is incorrect to state that the BOP is in disequilibrium (with one caveat)
- The BOP is a **cash flow statement**

GENERIC BALANCE OF PAYMENTS

A. Current Account

1. Net exports/imports of goods (balance of trade)
2. Net exports/imports of services
3. Net income (income from direct and portfolio investment plus employer compensation)
4. Net transfers (e.g. remittances)

B. Capital Account

Foreign aid, transactions that involve intangible assets (brands, copyrights)

C. Financial Account

1. Net foreign direct investment
2. Net portfolio investment
3. Other financial items: trade credit, currency and deposits

D. Error and Omissions

Missing data such as illegal transfers

E. Reserves

Changes in official monetary reserves including gold, foreign Exchange, and IMF position

Current Account

- Includes all international economic **transactions with income or payment flows** occurring within a period. It consists of the following four subcategories:
 - Export and import of goods
 - Export and import of services
 - Income
 - Current transfers
- The Current Account is typically dominated by exports/imports of goods, which is known as the Balance of Trade (BoT) even though it excludes service trade
 - Technological change shaped the composition of the BoT and services
 - What is the impact on currency flows? Exchange rates?

Think of one example of trade in services that has become very common in the last 10-15 years

Capital and Financial Account

- All international economic transactions of financial assets
- The Capital Account is smaller when compared to the Financial Account
- The latter one focuses on the degree of investor control over assets
 - Direct Investment – in which the investor exerts some explicit degree of control over the assets (10 percent or more voting power)
 - Portfolio Investment – in which the investor has no control over the assets, motivated by a search for returns (equity or debt)
 - Other Investment – currency deposits, bank deposits, trade credit, cross-border loans, and other related to cross-border trade
- Capital flow swings impact the financial account
 - Might look here for evidence of swings in market conditions

- Reserves are held by official monetary authorities within the country
- These reserves are normally composed of the major currencies used in international trade and financial transactions
- Relevance depends on whether the country is operating under a fixed exchange rate regime or a floating exchange rate system
- The Net Errors and Omissions account ensures that the BOP actually balances
- Dynamics of reserves were key element in first models of fixed exchange rate crisis (Krugman, 1979, Journal of Money, Credit and Banking)

Chile's Balance of Payments

Series description	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Current account	-8.981,59	-6.631,18	-6.534,48	-7.615,78	-13.265,29	-14.505,47	-4.952,33	-22.962,27	-26.161,98	-11.899,04
Trade of goods and services	1.074,24	-1.995,12	-527,95	1.879,85	-2.932,91	-5.068,93	11.444,34	-2.188,47	-11.869,38	4.541,74
Goods (X-M)	6.730,62	3.575,84	4.950,68	7.490,39	4.408,55	3.016,11	18.916,84	10.305,31	3.729,44	15.323,43
Exports	75.324,40	62.120,25	60.769,49	68.904,19	74.838,12	68.792,35	74.024,36	94.604,15	98.556,64	94.557,42
Imports (FOB)	68.593,78	58.544,41	55.818,82	61.413,80	70.429,57	65.776,23	55.107,52	84.298,84	94.827,20	79.233,99
Trade on services	-5.656,39	-5.570,96	-5.478,62	-5.610,54	-7.341,46	-8.085,04	-7.472,50	-12.493,78	-15.598,83	-10.781,69
Income	-11.574,94	-5.984,04	-6.935,56	-10.755,64	-12.161,77	-10.411,01	-15.864,68	-17.946,60	-14.224,06	-17.009,14
Investment income	-11.331,12	-5.656,55	-6.654,51	-10.330,15	-11.607,76	-9.757,64	-15.315,30	-17.376,11	-13.617,84	-16.397,58
Current transfers	1.519,12	1.347,97	929,03	1.260,01	1.829,38	974,47	-531,98	-2.827,20	-68,53	568,36
Capital account	10,66	695,36	8,67	89,01	42,77	1.024,88	1,14	2,06	2,19	56,96
Lending / financing	-8.970,93	-5.935,82	-6.525,80	-7.526,77	-13.222,52	-13.480,58	-4.951,18	-22.960,21	-26.159,79	-11.842,08
Financial account	-9.342,59	-3.916,90	-4.231,27	-6.157,82	-11.338,64	-10.789,69	-7.717,11	-25.194,93	-27.931,59	-10.345,32
Direct investment	-15.447,96	-1.915,15	-3.486,98	-2.702,25	-6.096,03	-3.234,01	-5.049,11	-603,86	-5.030,31	-15.459,84
Portfolio investment	-3.483,41	-2.152,16	-2.493,30	4.801,24	-2.695,83	-10.563,16	-13.090,12	-32.645,17	-5.313,11	1.771,00
Financial derivatives	1.610,46	722,22	689,51	64,69	881,73	1.500,62	2.536,40	-17,42	42,13	-1.979,98
Other investment	6.921,46	-782,93	-745,60	-5.571,90	-4.825,23	1.659,34	10.780,68	-4.139,95	-8.429,14	-1.464,31
Reserve assets	1.056,86	211,12	1.805,09	-2.749,60	1.396,71	-152,48	-2.894,95	12.211,47	-9.201,17	6.787,82
Errors and omissions	-371,66	2.018,91	2.294,53	1.368,96	1.883,88	2.690,89	-2.765,92	-2.234,72	-1.771,80	1.496,76
Settlement of payments balance	1.056,86	211,12	1.805,09	-2.749,60	1.396,71	-152,48	-2.894,95	12.211,47	-9.201,17	6.787,82

How does the BoP work?

- Suppose you import 100 million dollars in basketball shoes and pay that with a transfer upon receiving shipment in Chile
 - Where in the BoP do you register that transaction?
- Does the exchange rate regime matter for the BoP accounting?

How does the BoP work?

- You import 100 dollars worth of Pokemon cards and pay with a Chilean credit card
 - Register this transaction in the BoP
- What if instead of Pokemon cards you are paying a subscription to Amazon cloud or Dropbox?

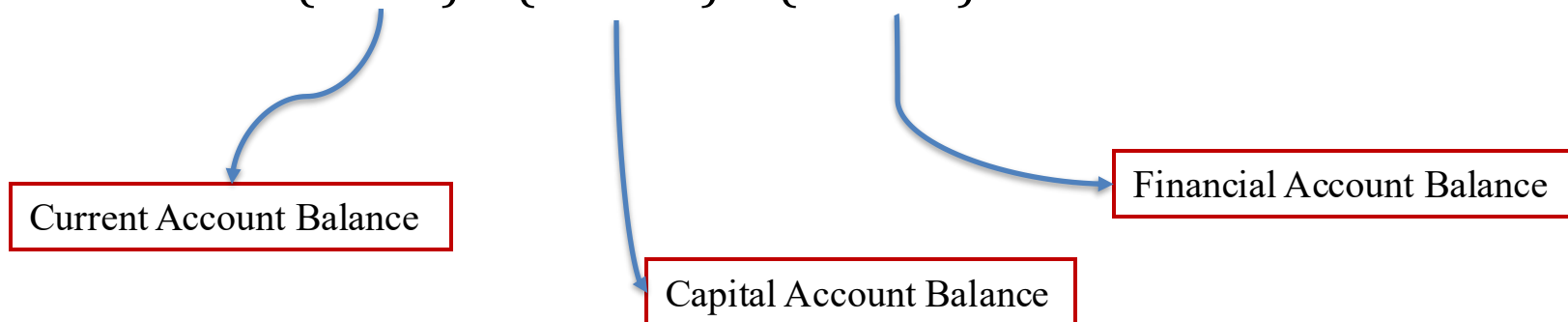
A GOVERNMENT CAN PEG the exchange value of its currency in a variety of ways. In a country with highly developed financial markets it can use open-market operations, intervention in the forward exchange market, and direct operations in foreign assets to defend an exchange parity (see [2] for an analysis of central bank operations and their effects on the exchange rate); the list could be extended to include such other instruments as changes in bank reserve requirements. But all of these policy instruments are subject to limits. A government attempting to keep its currency from depreciating may find its foreign reserves exhausted and its borrowing approaching a limit. A government attempting to prevent its currency from appreciating may find the cost in domestic inflation unacceptable. When the government is no longer able to defend a fixed parity because of the constraints on its actions, there is a “crisis” in the balance of payments.

common process must be at work. A “standard” crisis occurs in something like the following manner. A country will have a pegged exchange rate; for simplicity, assume that pegging is done solely through direct intervention in the foreign exchange market. At that exchange rate the government’s reserves gradually decline. Then at some point, generally well before the gradual depletion of reserves would have exhausted them, there is a sudden speculative attack that rapidly

Balance of Payments and Exchange Rates

- A country's BOP can have a significant impact on the level of its exchange rate and vice versa

$$(X - M) + (CI - CO) + (FI - FO) + FXR = BOP$$



Under a fixed exchange rate, governments ensure that the BOP is near zero

Under a floating exchange rate, governments have no responsibility to peg its foreign exchange rate

Under a managed float, governments often find it necessary to take action to maintain desired ER

Balance of Payments and Interest Rates

- Relatively low real interest rates should normally stimulate an outflow of capital seeking higher rates elsewhere
 - However, in the case of the U.S., the opposite has occurred due to perceived safety – allowing it to finance its large fiscal deficit
- The degree to which capital moves freely across borders is critically important to a country's balance of payments
 - The United States' financial account surplus has at least partially offset the current account deficits over the last +20 years
 - China has run a surplus in each of these accounts in recent years

BALANCE OF PAYMENTS AND MACROECONOMIC EQUILIBRIUM

Current account and the macroeconomy

- It is important to separate accounting analysis from causal analysis
 - Particularly for policy analysis

- So, let's move from accounting to a more economic analysis

- Consider the definition of national income (Y)

$$Y = GDP + NFI$$

- NFI is net foreign income
- Remember that $GDP = C + I + G + NX$, that is private consumption, fixed investment, government consumption and net exports

Current account and the macroeconomy

- Domestic expenditure is $C + I + G$
- Use identities and definitions
$$Y - (C + I + G) = GDP + NFI - (C + I + G) = NX + NFI$$
- Right-hand side is the current account, the left-hand side is national income minus domestic expenditure
- Thus, current account deficit is expenditure is higher than income
- Furthermore,
$$Savings - Investment = Current\ account$$
- You run a current account surplus if you lend money to the rest of the world

A bit of current account history...

	Brazil	China	India	Indonesia	Korea	Malaysia	Philippines	Singapore	Russia	Thailand
1990	-0.7	3.1	-2.4	-2.5	-0.7	-2.1	-6.1	8.0	N/A	-8.3
1996	-2.8	0.8	-1.6	-2.9	-4.0	-4.4	-4.6	14.7	2.8	-7.9
2000	-3.8	1.7	-1.0	4.8	2.3	9.0	-2.9	10.8	18.0	7.6
2004	1.8	3.6	0.1	0.6	3.9	12.1	1.9	17.1	10.1	1.7
2008	-1.7	9.6	-2.0	0.0	-0.6	17.5	2.2	18.5	6.2	0.6
2010	-2.6	4.7	-3.1	0.9	2.6	14.7	4.1	20.5	4.7	3.6

Notes: Data are from the IMF's *World Economic Outlook*, October 2010. Data for 2010 are IMF estimates. N/A, not available.



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INTERNATIONAL FINANCE DERIVATIVES AND PARITIES

José A. Tessada

September 2024

TAKING STOCK...

Where are we at?

- Why study international financial management?
- Why do firms operate abroad?
- Balance of payments (and macro equilibriums)
- Exchange rate systems
- ...

DERIVATIVES

Some data

A.2 Non-financial firms					
	Obs.	Share	Notional Median	Maturity Median	Non- delivery
	(#)	(%)	(\$ 000)	(days)	(%)
Instrument	(6)	(7)	(8)	(9)	(10)
Forwards	639,736	88.3	1308.5	90.5	65.1
Futures	356	0	1728.8	85.6	82.6
Call	21,414	3	716.1	164.4	91.2
Put	13,224	1.8	852	175.1	93.6
FX swaps	15,650	2.2	3901.7	77.7	37
CC Swaps	34,033	5	8,104	2375	62
Total	724,413	100.0	1352.6	122.2	63

Some data

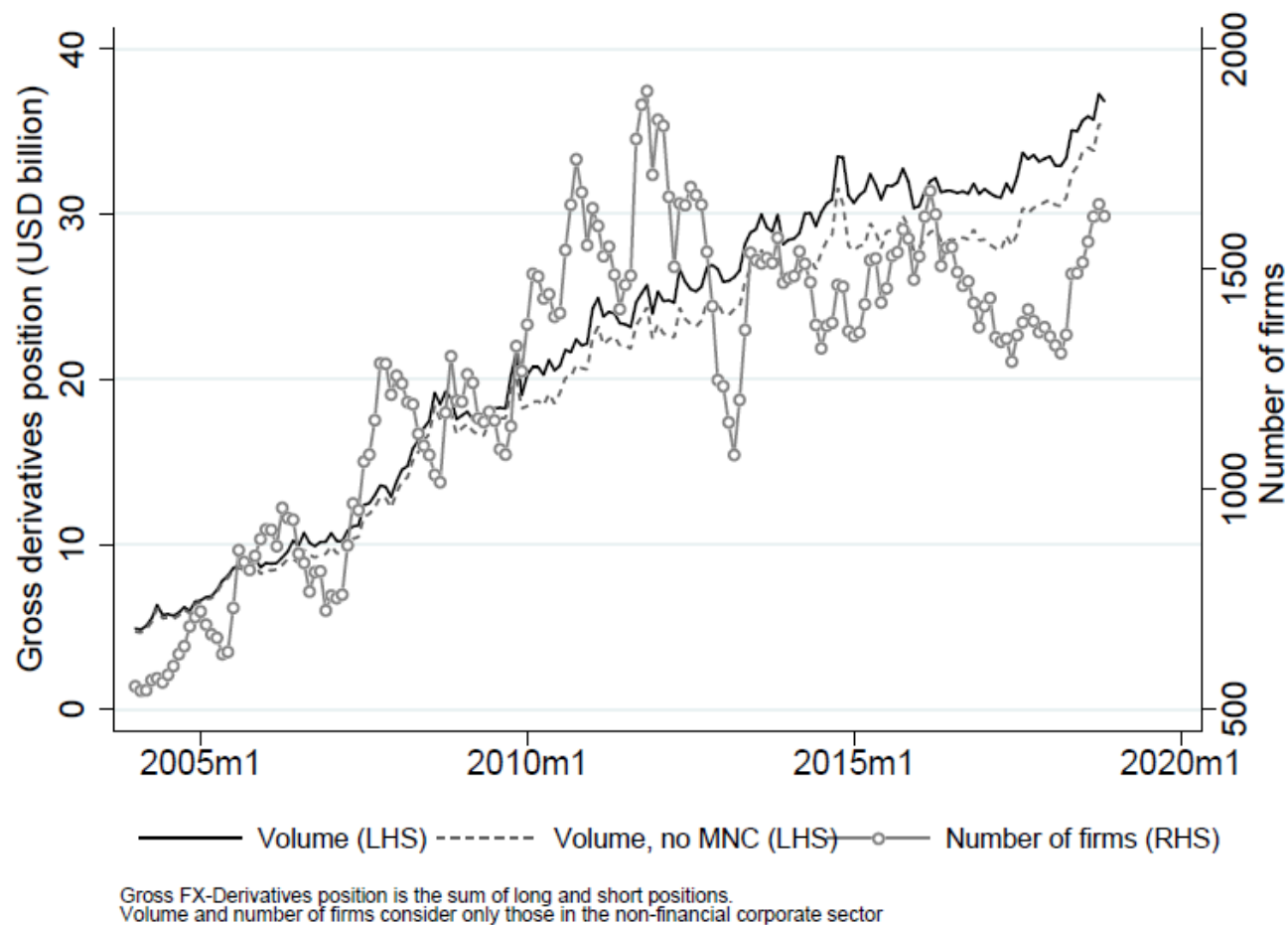


Figure 1: Number of firms and gross FX Derivatives positions

Some data

Table 1: Descriptive statistics FX derivatives contracts

Panel A: By market												
	A.1. All Market						A.2. Non-financial firms					
	Contracts	Share	Value	Share	Maturity	Share	Contracts	Share	Value	Share	Maturity	Share
	(#)	contracts	median	Value	median	NDFX	(#)	contracts	median	Value	median	NDFX
		(%)	(thousand \$)	(%)	(days)	(%)		(%)	(thousand \$)	(%)	(days)	(%)
Forwards	1,518,688	80.4	5,630.0	68.7	71.1	83.5	695,612	88.8	1,326.8	85.8	92.4	66.5
Futures	2,211	0.1	1,684.4	0.0	43.3	96.8	357	0.0	1,724.4	0.1	85.3	82.6
Call options	24,974	1.3	1,436.4	0.3	159.2	91.6	22,038	2.8	823.1	1.7	165.2	90.8
Put options	15,677	0.8	1,936.0	0.2	167.6	93.0	13,609	1.7	977.1	1.2	176.2	93.5
Swaps	502	0.0	7,887.2	0.0	1,382.4	74.3	334	0.0	5,381.6	0.2	1,233.8	43.4
FX swaps	271,427	14.4	12,723.1	27.8	77.2	90.6	17,453	2.2	3,619.8	5.9	75.0	38.3
CC Swaps	55,474	2.9	6,505.3	2.9	1,051.4	31.4	34,018	4.3	1,639.3	5.2	686.3	5.7
Total	1,888,953	100.0	6,584.8	100.0	103.0	83.2	783,421	100.0	1,373.2	100.0	121.7	64.4

Granular Corporate Hedging Under Dominant Currency, by Alfaro et al. (2023)

United States Dollar to Chilean Peso

942.8400 ↑ 32.52% +231.3500 5Y

Sep 5, 4:11:00 PM UTC · Disclaimer



Euro to Chilean Peso

1,047.0800

↑35.04% +271.7100 5Y

Sep 5, 11:20:00 AM UTC · Disclaimer



Brazilian Real to Chilean Peso

168.5298 ↓ 3.68% -6.4402 5Y

Sep 5, 4:11:00 PM UTC · Disclaimer

1D 5D 1M 6M YTD 1Y 5Y MAX



United States Dollar to Brazilian Real

5.5947 ↑ 38.23% +1.5472 5Y

Sep 5, 4:14:00 PM UTC · Disclaimer

1D 5D 1M 6M YTD 1Y 5Y MAX



- Intl financial management involves financial derivatives
- These derivatives, so named because their values are derived from underlying assets, are a powerful tool used in business today
- These instruments can be used for two very distinct management objectives:
 - Speculation –to take a position in the expectation of a profit
 - Hedging –to reduce the risks associated with the everyday management of corporate cash flow
- Foreign exchange derivatives are derivatives with currencies as underlying assets

- A *foreign currency option* is a contract giving the option purchaser (the buyer) the right, but not the obligation, to buy or sell a given amount of foreign exchange at a fixed price per unit for a specified time period (until the maturity date).
- There are two basic types of options: *puts* and *calls*.
 - A call is an option to buy foreign currency
 - A put is an option to sell foreign currency
- The buyer of an option is termed the *holder*, while the seller of the option is referred to as the *writer* or *grantor*

- Every option has three different price elements:
 - The *exercise* or *strike* price – the exchange rate at which the foreign currency can be purchased (call) or sold (put)
 - The *premium* – the cost, price, or value of the option itself
 - The underlying or actual spot exchange rate in the market
- An *American* option gives the buyer the right to exercise the option at any time between the date of writing and the expiration or maturity date
- A *European* option can be exercised only on its expiration date, not before
- Relation between exercise price and spot price: *at-the-money* (ATM), *in-the-money* (ITM), and *out-of-the money* (OTM)
- Flows of an option
 - Example: European put option

Put option

Call option

Forwards and futures

- Both represent a commitment to carry out a transaction of a defined good on a future date at a given price
- Forwards: commitment today to carry out a certain transaction in the future
- Futures: standardized and formalized contracts traded in financial markets
- Over-the-counter versus market

- The forward exchange rate is related to the current spot exchange rate and the ratio of interest rates of the same maturity for the two subject currencies
 - Why? Arbitrage
- This is related to the covered interest rate parity
- Not to be confused with the uncovered interest rate parity
 - (More on this soon...)
- Let us see the flows of a forward contract first...

Forwards

- A foreign currency futures contract is an alternative to a forward contract that calls for future delivery of a standard amount of foreign exchange at a fixed time, place and price
- Contract specifications are established by the exchange on which futures are traded.
- Major features that are standardized are:
 - Contract size
 - Method of stating exchange rates
 - Maturity date
 - Last trading day
 - Collateral and maintenance margins
 - Settlement
 - Commissions
 - Use of a clearinghouse as a counterparty

- Foreign currency futures contracts differ from forward contracts in a number of important ways:
 - Futures are standardized in terms of size while forwards can be customized
 - Futures have fixed maturities while forwards can have any maturity (both typically have maturities of one year or less)
 - Trading on futures occurs on organized exchanges while forwards are traded between individuals and banks
 - Futures have an initial margin that is market to market on a daily basis while only a bank relationship is needed for a forward
 - Futures are rarely delivered upon (settled) while forwards are normally delivered upon (settled)

INTEREST RATE PARITY CONDITIONS...

COVERED AND UNCOVERED

Covered parity

- Central element: arbitrage
- You have a cash flow of \$1 million today and you would only need to use it for at least 90 days more
- What can you do with it?
 1. Deposit in \$, 90/day maturity
 2. Deposit in US\$, 90/day maturity
- Let us compute the flows in 90 days...
- Some notation:
 - \$ interest rate: i
 - US\$ interest rate: i^*
 - Spot exchange rate: E
 - Forward exchange rate: F

The mechanics... And the intuition

- US\$ deposit gives you $(1+i^*)$ return... In dollars
- And you need dollars today
 - Exchange \$ into US\$ today --> you get $(1/E)$ million US\$ today
- In 90 days, you will have $(1+i^*)/E$ million dollars
- If you had invested in pesos you would have $(1+i)$ million pesos
- How can you compare those two flows?
 - Alt 1: wait and convert US\$ to \$ at the then spot rate
 - Alt 2: sign a forward contract today!

Covered parity

- You sign then a contract with forward exchange rate F
- And now you compare apples and apples! Pesos in this case...
- Invest pesos:

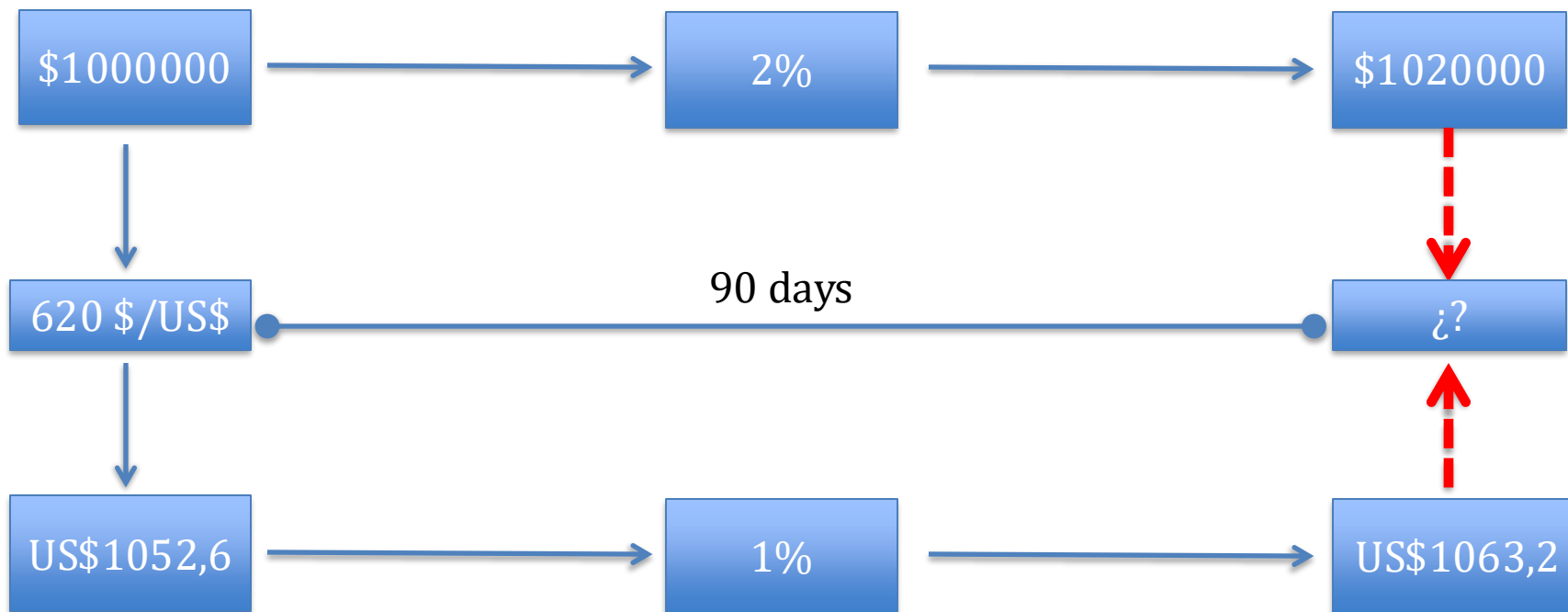
$(1+i)$ millions

- Invest dollars:

$(1+i^*)(F/E)$ millions

- The key: all prices are known today!
- Thus, the covered interest rate parity

$$(1+i) = (1+i^*)(F/E)$$



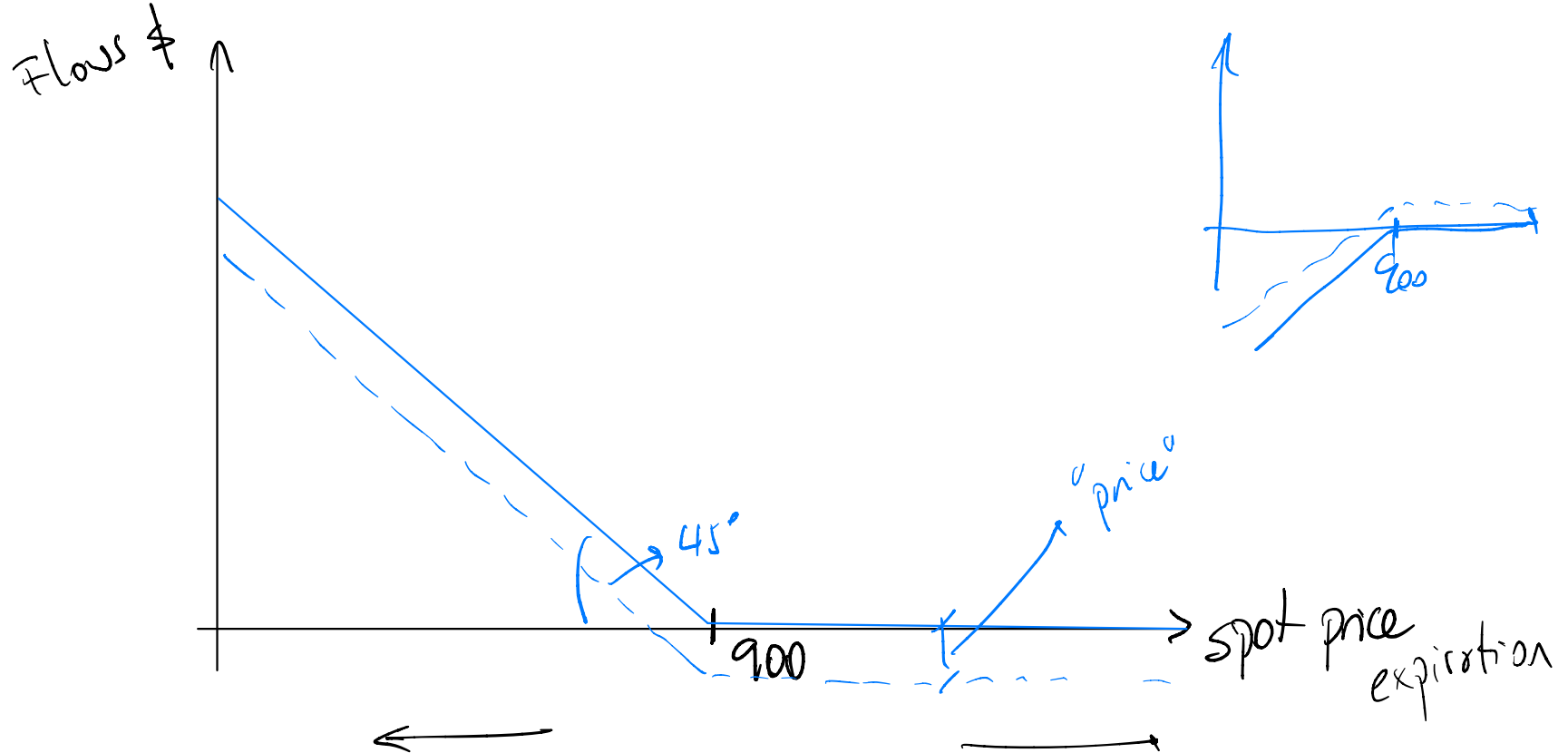
Uncovered interest rate parity

- What is the difference?
- The way you (do not) tie the flow in the future...
- In the uncovered case, you do not tie it today!
 - Wait and use the spot exchange rate
 - So you can compute today the *expected* return...
 - That's the condition you likely have seen in macro courses
- Is this an arbitrage condition? Could you arbitrage it?

-
- What if we put them together?
 - Does this mean the forward exchange rate is an unbiased predictor of the spot exchange rate?
 - Do these parities hold?

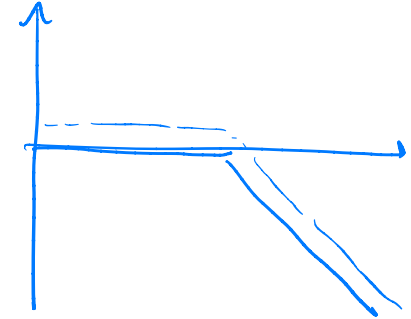
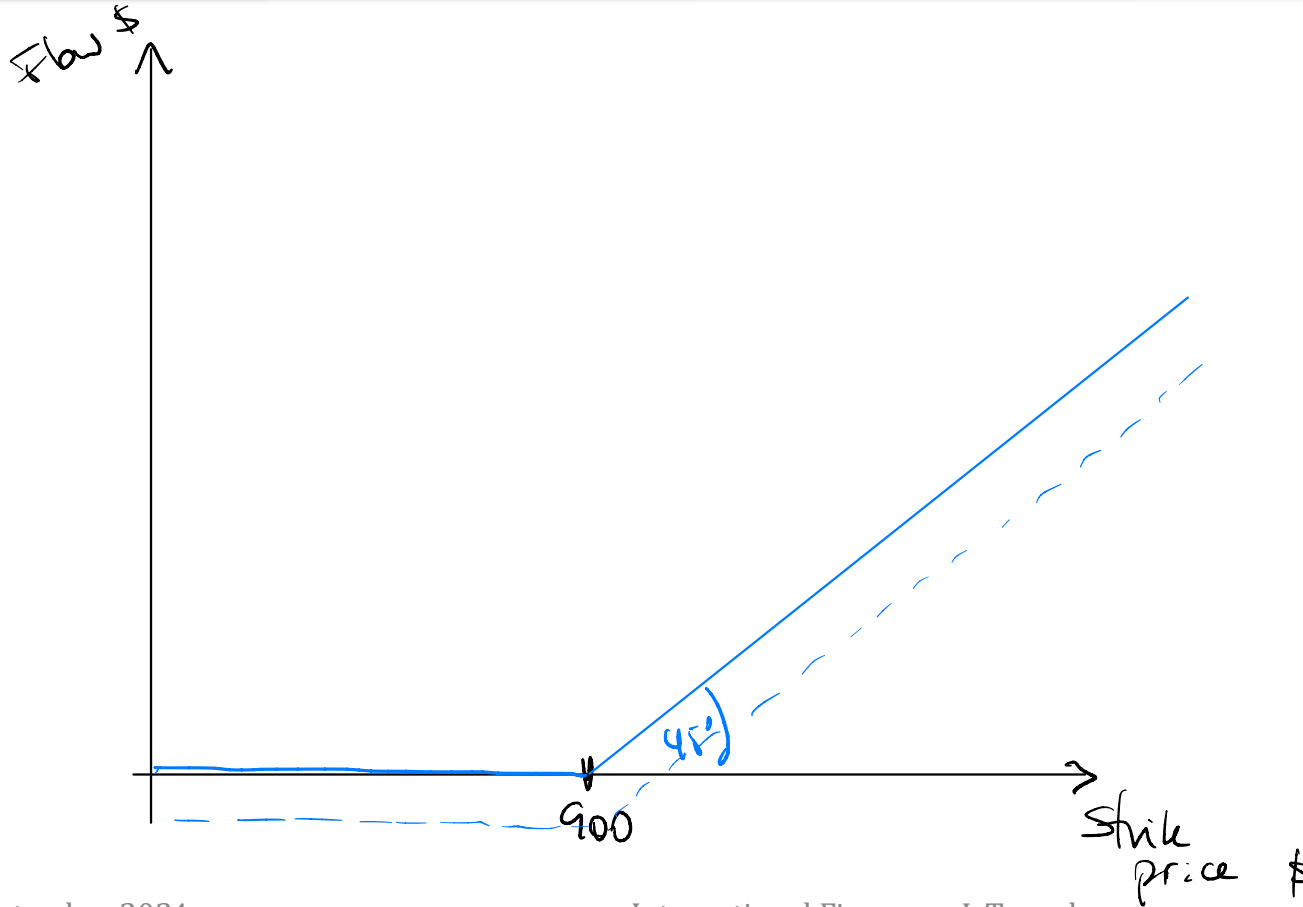
Put option

Sell



Call option

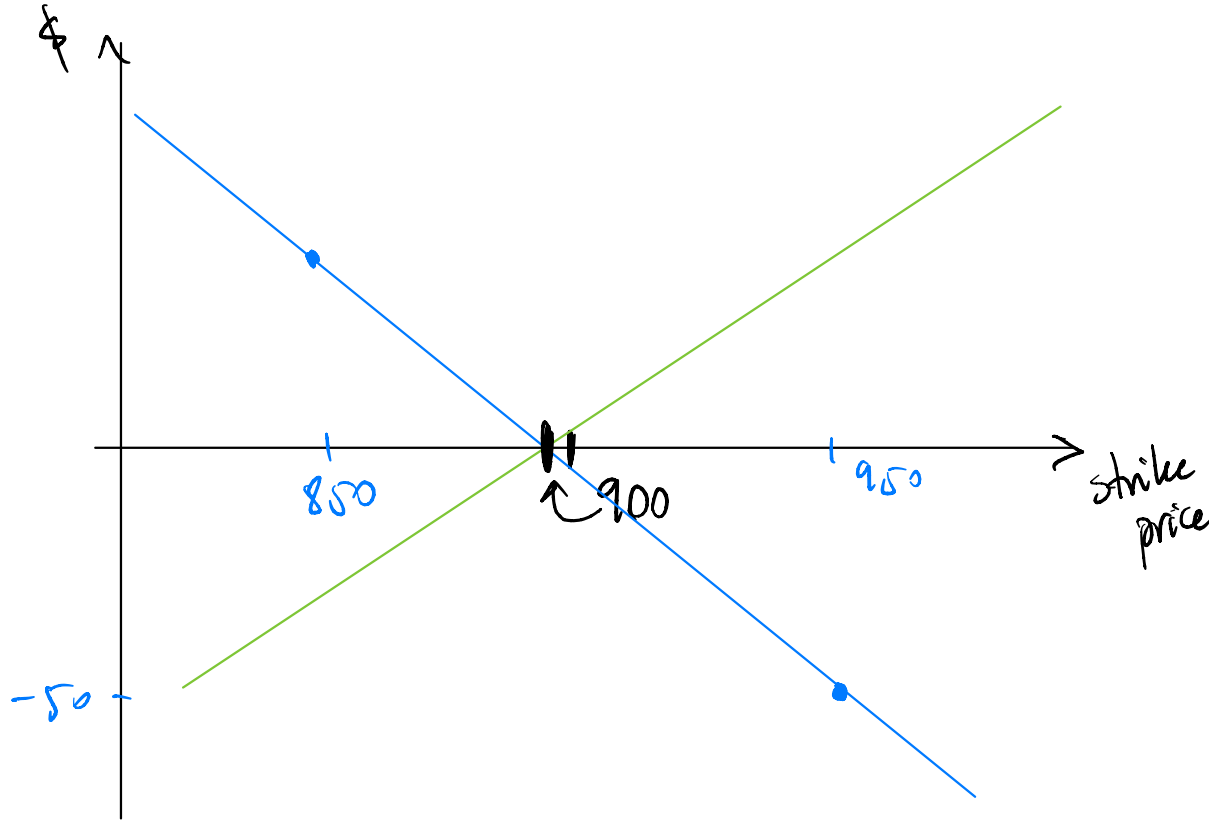
BUY



Forwards

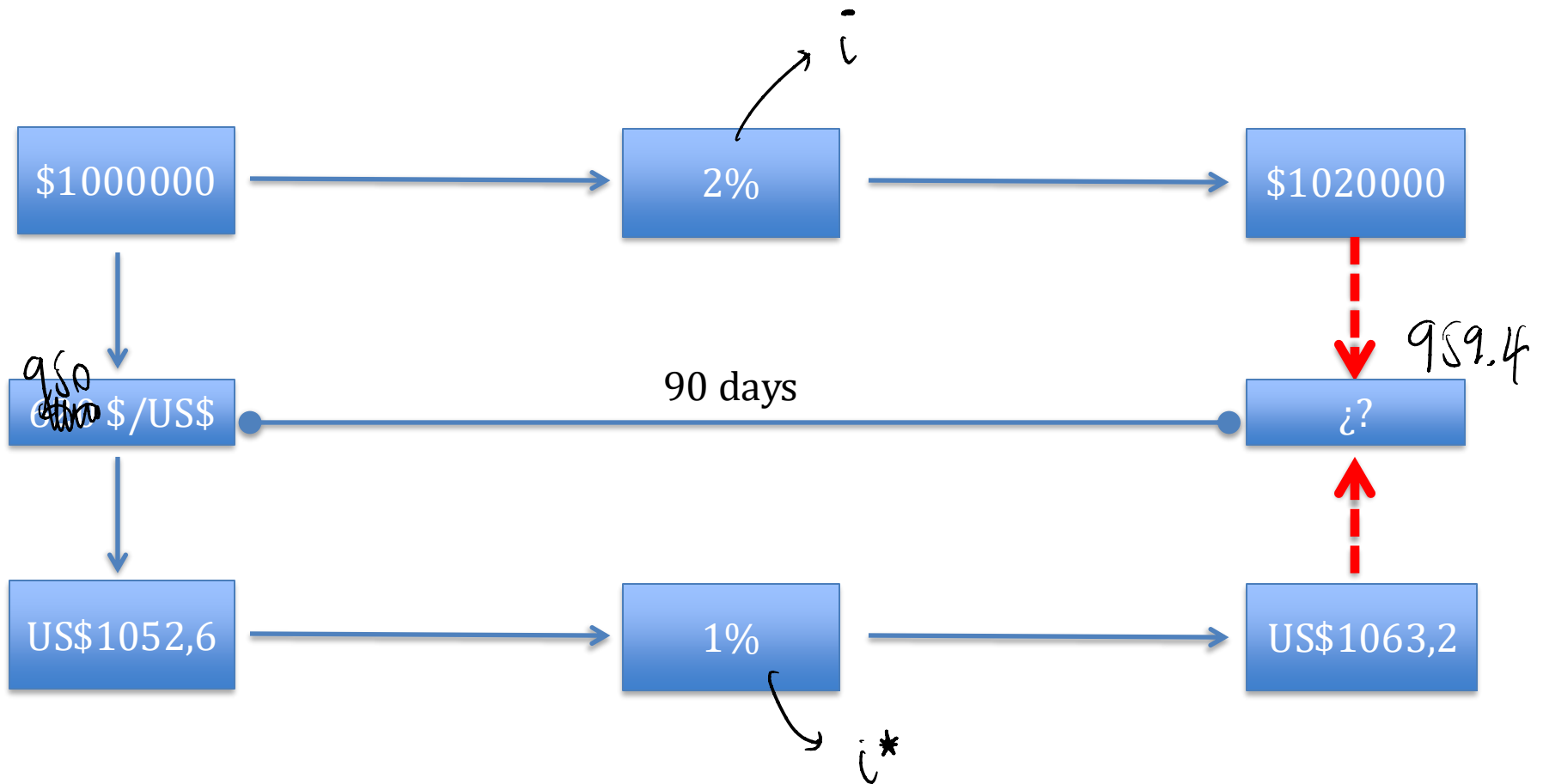
venta

Buy



Covered parity

- Central element: arbitrage
- You have a cash flow of \$1 million today and you would ^{not}~~only~~ need to use it for at least 90 days more
- What can you do with it?
 1. Deposit in \$, 90/day maturity
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- Let us compute the flows in 90 days...
- Some notation:
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$$1+i = (1+i^*) \frac{F_{t+1}^e}{E}$$



ESCUELA DE ADMINISTRACIÓN
FACULTAD DE ECONOMÍA Y ADMINISTRACIÓN

FOREIGN EXCHANGE EXPOSURE TRANSACTION EXPOSURE

José A. Tessada

September 2024

TAKING STOCK...

Interest rate parities

- What is the difference?
- The way you (do not) tie the flow in the future...
- In the uncovered case, you do not tie it today!
 - Wait and use the spot exchange rate
 - So you can compute today the *expected* return...
 - That's the condition you likely have seen in macro courses
- Is this an arbitrage condition? Could you arbitrage it?

-
- What if we put them together?
 - Does this mean the forward exchange rate is an unbiased predictor of the spot exchange rate?
 - Do these parities hold?

FOREIGN EXCHANGE EXPOSURE

- *Foreign exchange exposure* is a measure of the potential for a firm's *profitability, net cash flow, and market value* to change because of a change in exchange rates
- You need to
 - Measure foreign exchange exposure and to Manage it to maximize the profitability, net cash flow, and market value of the firm

Resulting from Accounting

Transaction Exposure

Impact of settling outstanding obligations entered into before change in exchange rates but to be settled after change in exchange rates

Translation Exposure

Changes in income and owners' equity in consolidated financial statements caused by a change in exchange rates

Resulting from Economics

Operating Exposure

Change in expected future cash flows arising from an unexpected change in exchange rates

Changes in future cash flows arising from firm and competitor firm responses

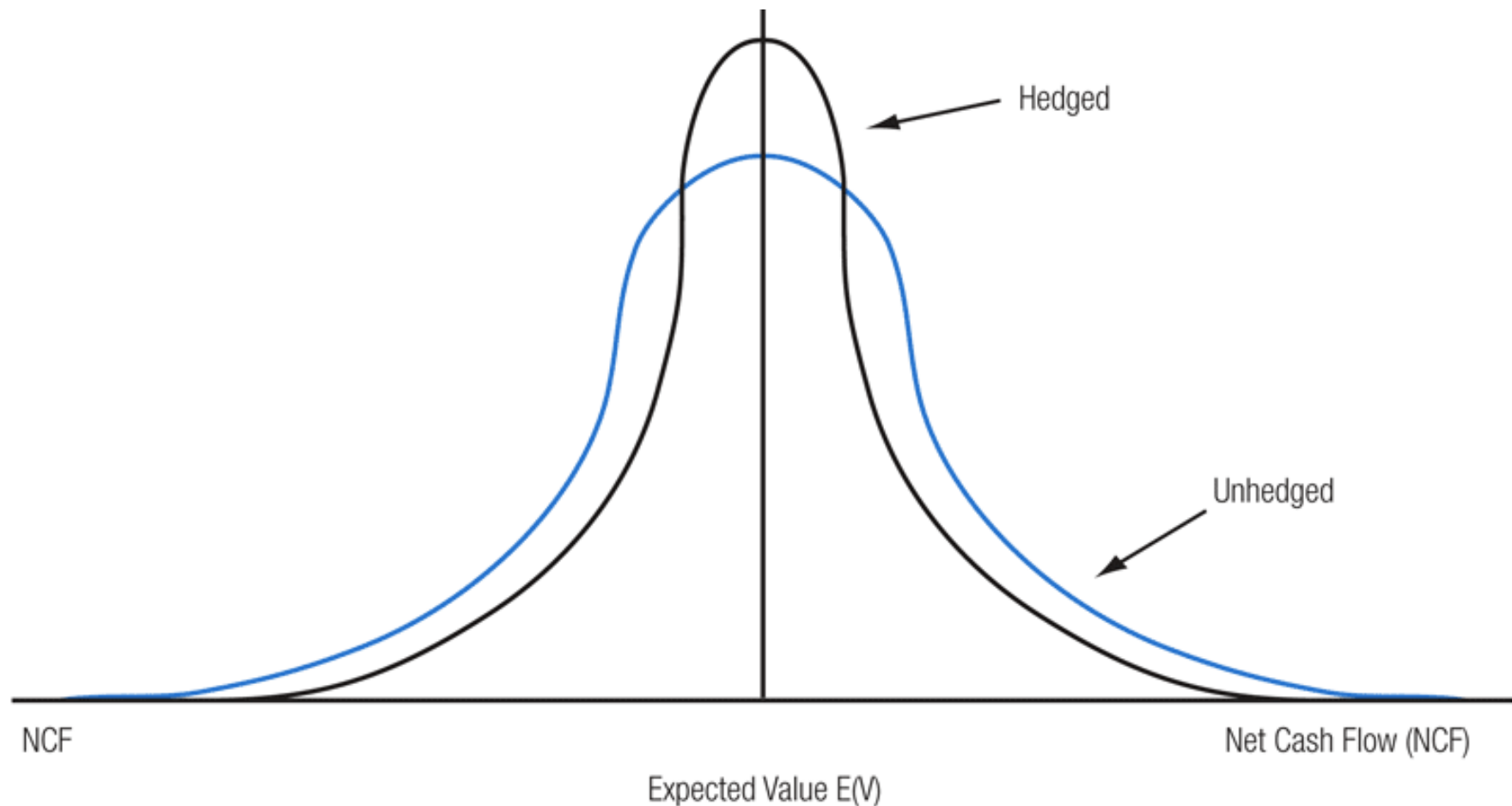
Time and Exchange Rate Changes

Types of exposure

- *Transaction exposure* measures changes in the value of outstanding financial obligations incurred prior to a change in exchange rates but not due to be settled until after the exchange rates change
 - Thus, this type of exposure deals with changes in cash flows the result from existing contractual obligations
- *Operating exposure*, also called *economic exposure*, *competitive exposure*, or *strategic exposure*, measures the change in the present value of the firm resulting from any change in future operating cash flows of the firm caused by an *unexpected* change in exchange rates
- *Translation exposure*, also called *accounting exposure*, is the potential for accounting-derived changes in owner's equity to occur because of the need to “translate” foreign currency financial statements of foreign subsidiaries into a single reporting currency to prepare worldwide consolidated financial statements

Exposure and hedging

- These three financial price risks are the subject of the growing field of *financial risk management*
- Many firms attempt to manage their currency exposures through hedging
- Hedging is the taking of a position, acquiring either a cash flow, an asset, or a contract (including a forward contract) that will rise (fall) in value and offset a fall (rise) in the value of an existing position
- While hedging can protect the owner of an asset from a loss, it also eliminates any gain from an increase in the value of the asset hedged against



Hedging reduces the variability of expected cash flows about the mean of the distribution. This reduction of distribution variance is a reduction of risk.

To hedge or not to hedge

- However, is a reduction in the variability of cash flows sufficient reason for currency risk management?
- (Some) arguments against hedging:
 - Shareholders are much more capable of diversifying currency risk than the management of the firm
 - Currency risk management does not increase the expected cash flows of the firm
 - Management often conducts hedging activities that benefit management at the expense of the shareholders (*agency* conflict)
 - Managers cannot outguess the market

To hedge or not to hedge

- In favor of hedging:
 - Reduction in risk in future cash flows improves the planning capability of the firm
 - Reduction of risk in future cash flows reduces the likelihood that the firm's cash flows will fall below a necessary minimum (the point of *financial distress*)
 - Management has a comparative advantage over the individual shareholder in knowing the actual currency risk of the firm
 - Management is in better position to take advantage of disequilibrium conditions in the market

TRANSACTION EXPOSURE

Time and Events



Seller quotes
a price to buyer
(verbal or written form)

Buyer places
firm order with
seller at price
offered at time t_1

Seller ships
product and
bills buyer
(becomes A/R)

Buyer settles A/R
with cash in amount
of currency
quoted at time t_1

**Quotation
Exposure**

**Backlog
Exposure**

**Billing
Exposure**

Time between quoting
a price and reaching a
contractual sale

Time it takes to
fill the order after
contract is signed

Time it takes to
get paid in cash after
A/R is issued

- Foreign exchange transaction exposure can be managed by *contractual*, *operating*, and *financial hedges*
- The main contractual hedges employ the forward, money, futures, and options markets
- Operating and financial hedges employ the use of risk-sharing agreements, leads and lags in payment terms, swaps, and other strategies
 - The term *natural hedge* refers to an off-setting operating cash flow, a payable arising from the conduct of business
 - A financial hedge refers to either an off-setting debt obligation (such as a loan) or some type of financial derivative such as an interest rate swap
 - Care should be taken to distinguish *operating* hedges from *financing* hedges

How can derivatives help you hedge?

- Forwards?
- Futures?
- Options?
- *Money* hedge?

- Is it just currency mismatch?

- Decisions according to the timing of the exposure

Some comments

- Transaction exposure management programs are generally divided along an “option-line,” those that use options and those that do not
- Firms that do not use currency options rely almost exclusively on forward contracts and money market hedges
- Policies generally set standard fractions to be hedged plus discretionary decisions per each transaction
- Most firms do hedge billing exposure or when the transaction is in the books
- Tax treatment: opportunity cost vs losses

Oct 10

Jan 10

Money hedge

$$\frac{\text{US\$ } 500,000}{(1+i^*)}$$

→ $\$ E \cdot \left(\frac{\text{US\$ } 500,000}{1+i^*} \right)$

$$\left. \begin{array}{l} \text{US\$ } 500,000 \\ (\text{US\$ } 500,000) \end{array} \right\} \text{Net} = 0$$



Forward

$$\$ 500,000 \times F$$

← $1+i$



ESCUELA DE ADMINISTRACIÓN
FACULTAD DE ECONOMÍA Y ADMINISTRACIÓN

FOREIGN EXCHANGE EXPOSURE ECONOMIC EXPOSURE

José A. Tessada

September 2024

TAKING STOCK

Before the break...

- Transaction exposure
- Currency derivatives
- Hedging

- AIFS
 - Hedge or not to hedge...
 - What to hedge
 - How to hedge
 - Hedging cannot leave you exposed to another shock
 - More subtle, exposure arose from strategic decisions and positioning

A SMALL DETOUR TRANSLATION EXPOSURE

Definition

- It arises because financial statements of foreign subsidiaries stated in foreign currency must be restated in the parent's reporting currency to prepare consolidated financial statements
- Thus, *translating* involves converting these foreign subsidiaries financial statements into local currency denominated statements
- **Translation exposure is the potential for an increase or decrease in the parent's net worth and reported net income caused by a change in exchange rates since the last translation**
- In some cases, different exchange rates are used in remeasuring different line items
 - Translation principles in many countries are often a complex compromise between historical and current market valuation

Managing it?

- The main technique to minimize translation exposure is called a balance sheet hedge
- A balance sheet hedge requires an equal amount of exposed foreign currency assets and liabilities on a firm's consolidated balance sheet
- If this can be achieved for each foreign currency, net translation exposure will be zero
- Is this needed?

ECONOMIC (OR OPERATING) EXPOSURE

Understanding its origins with some examples

Examples

1. A Chilean avocado producer (and exporter?) and the \$/US\$ exchange rate
2. NHL and the Canadian vs . US dollar exchange rate fluctuation
3. A firm that owns and operates cabins for tourism in different regions of Chile and the effect of the \$/US\$ exchange rate

ECONOMIC EXPOSURE

Definition

Definition

- The effect of (unexpected) variation in the exchange rates on the value of the firm
 - Notice the difference with transaction exposure
- Impact might not only come from sales and competition in the market but from other operational flows and even from financing flows
 - Cost structure (case #2!)
 - Interest payments in other currencies
- Including the word *unexpected* is not minor, why?

Definition

- Where does it come from?
- Cash flows
 - Operational
 - Intra o inter firm
 - Operation: payables and receivables, rental, etc
 - Financial
 - Intra or inter firm
 - Linked to loans, equity issuance, dividends
- Asset valuation

Measuring exposure

- One simple (summary statistic-like) approach is to quantify the relation between value (or cash flows) and exchange rate

$$P = a + b \times E + e$$

where P is the share price and E is the exchange rate

- It is a very concrete and practical way... But,
 - Which exchange rate to use?
 - Expected versus unexpected
 - Again, it is simple!
- This is called currency exposure
- But we still need to understand where the covariance comes from

Some comments

- Economic exposure is far more important for the long-run health of a business than changes caused by transaction or translation exposure
- However, operating exposure is inevitably subjective because it depends on estimates of future cash flow changes over an arbitrary time horizon
- Planning for operating exposure is a total management responsibility because it depends on the interaction of strategies in finance, marketing, purchasing and production
- What about different horizons?
 - Does this matter for policies to deal with economic exposure?

Horizon	Time	Prices?	Volumes?	Structural?

Examples

- Meat importer to Chile (from US)
 - Currency for the costs and revenues
 - Competitors?
 - Pass-through to prices

- Car importer to Chile
 - Currency for the costs and revenues
 - Competitors?
 - Pass-through to prices

HOW TO DEAL WITH IT... IF AT ALL

Effects and responses

- Following the separation between operational and financial cash flows, we can think of looking at measures dealing with operations and then with financing
- Responses or measures include decisions beyond finance only
 - Strategy, operations, etc give rise to some of the exposure and thus can be part of the solution
- Example
 - Where does the firm operate?
 - Strategies and operations intra and inter firm

Effects and responses

- What do we want to achieve?
 - Reduce the volatility of operational cash flows in response to exchange rate(s) shocks
- This has a strategic dimension
 - Effects arise due to your positioning and your operation
 - Thus, strategy is part of the solution
 - This escapes the control of the CFO alone
- You must consider decisions beyond just hedging... Even though that might be part of the response
- Let us look first at strategic and operations, then at some financial decisions

Responses

1. Market diversification
2. Product differentiation
3. Cost structure and production decision
 - a. Supply chain
 - b. Location

Responses

- Financial decisions might also help
- You work with *financial aspects* or *financial consequences* of international exposure
- You complement operations and strategy with financial adjustments
- Some of them point towards building hedging or diversification into the normal functioning of the firm

1. Risk sharing – currency clauses
 - An alternate method for managing a long-term cash flow exposure between firms is risk-sharing
 - This is a contractual arrangement in which the buyer and seller agree to “share” or split currency movement impacts on payments between them
 - This agreement is intended to smooth the impact on both parties of volatile and unpredictable exchange rate movements

2. Matching currency cash flows

- One way to offset an anticipated continuous long exposure to a particular currency is to acquire debt denominated in that currency (matching)
- An alternative would be for the local firm to seek out potential suppliers of raw materials or components abroad as a substitute for local
- In addition, the company could engage in currency switching, in which the company would pay suppliers with the same currency it receives

3. Back-to-back loans

- A back-to-back loan, also referred to as a parallel loan or credit swap, occurs when two business firms in separate countries arrange to borrow each other's currency for a specific period of time
 - At an agreed terminal date they return the borrowed currencies
 - Such a swap creates a covered hedge against exchange loss, since each company, on its own books, borrows the same currency it repays
-
- Easy to achieve?

A detour: Swaps

- Swaps are contractual agreements to exchange or swap a series of cash flows
 - Most commonly the interest payments associated with debt service
- The swap itself is not a source of capital, but rather an alteration of the cash flows associated with payment
- The fixed- to floating-rate interest rate swap existing in each currency allow firms to swap across currencies

4. Currency swap

- A currency swap resembles a back-to-back loan except that it does not appear on a firm's balance sheet
- In a currency swap, a firm and a swap dealer or swap bank agree to exchange an equivalent amount of two different currencies for a specified amount of time

Horizon	Time	Prices?	Volumes?	Structural?
SHORT	6 months	Fixed (contracts)	constant	NO
MEDIUM	In between	increase pass-through maybe partial	↓ by consumers	increase from competitors
LONG RUN	> 3-5 years	increase	↓	more impact new entrants

FLEXIBILITY ↓

Examples

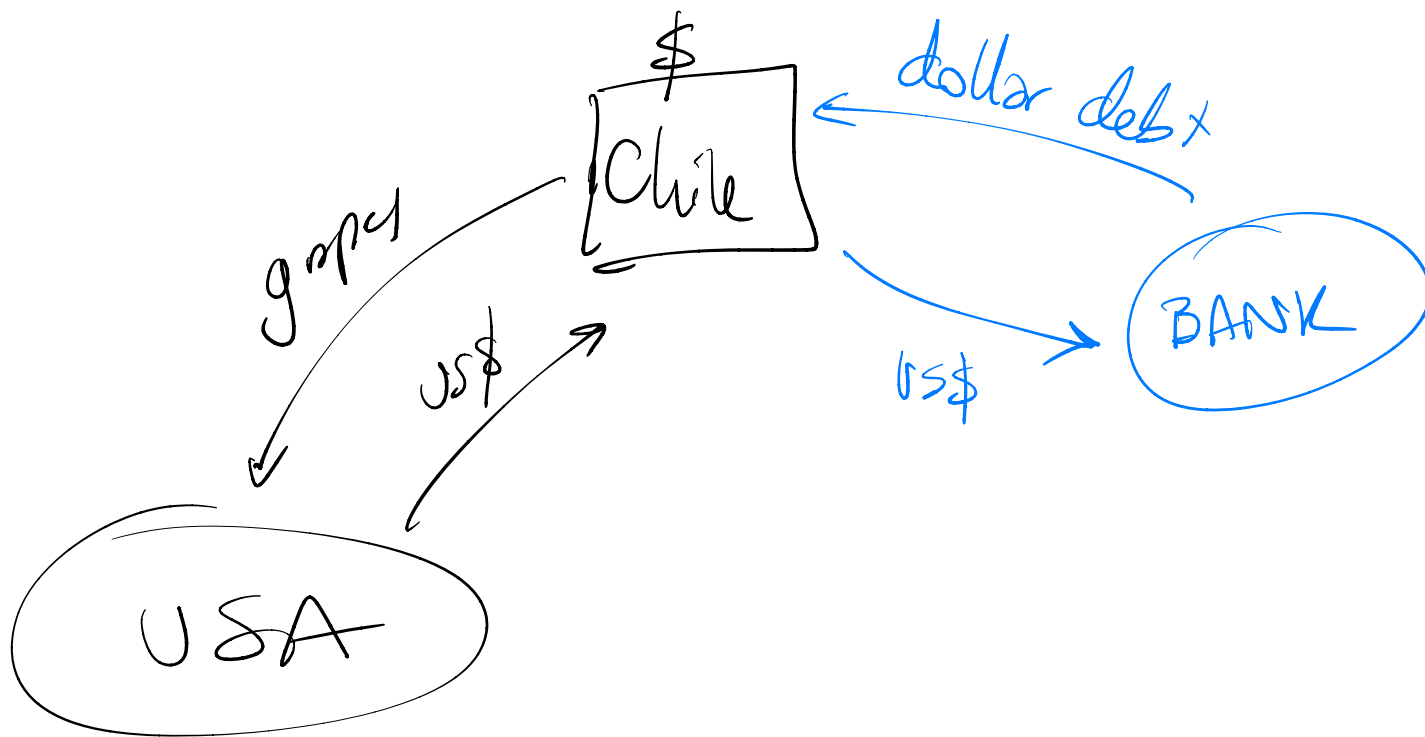
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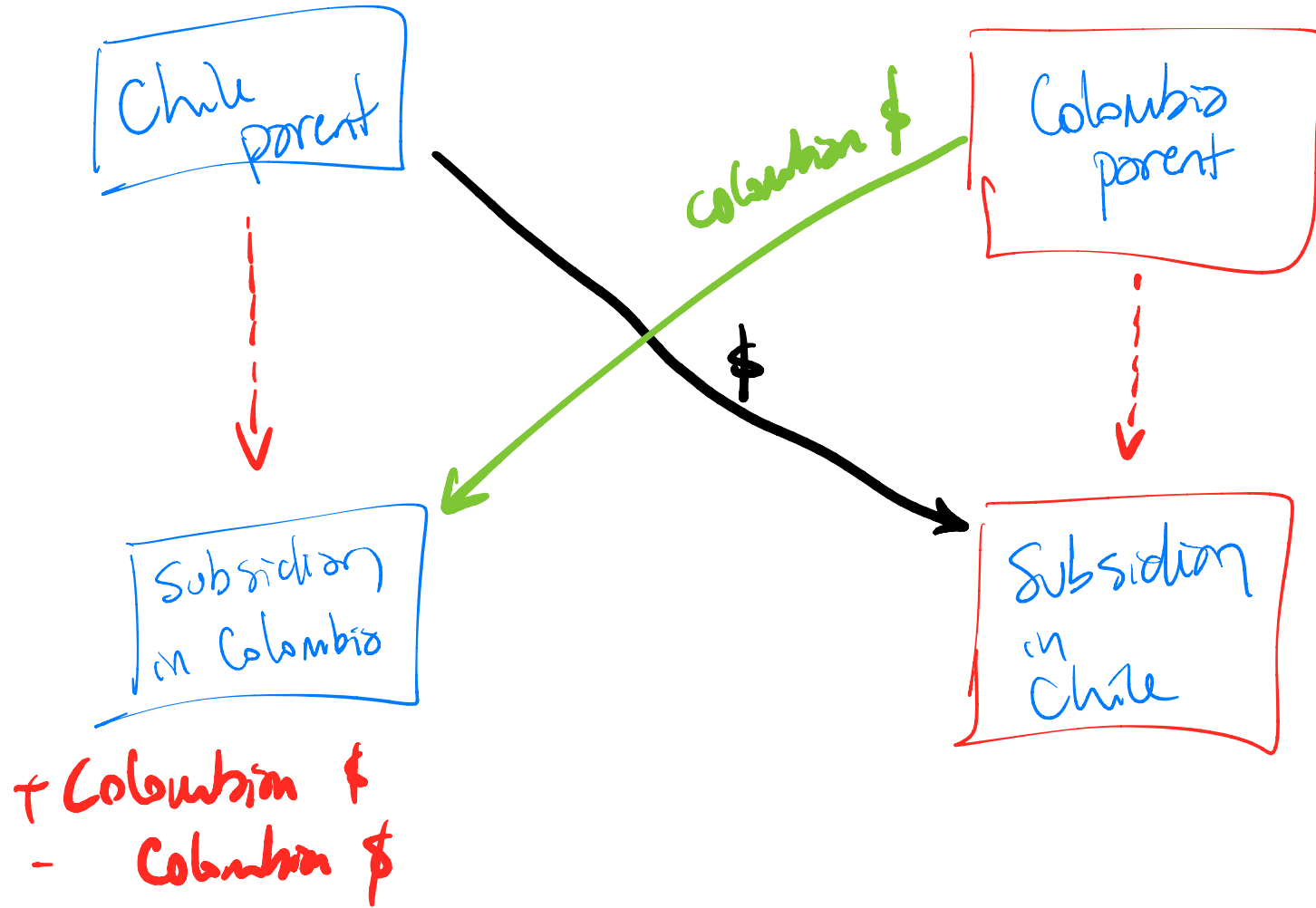
$\$/\text{US}\$$ ↑

exchange rate
pass-through

Effects and responses

- What do we want to achieve?
 - Reduce the volatility of operational cash flows in response to exchange rate(s) shocks
- This has a strategic dimension
 - Effects arise due to your positioning and your operation
 - Thus, strategy is part of the solution
 - This escapes the control of the CFO alone
- You must consider decisions beyond just hedging... Even though that might be part of the response
- Let us look first at strategic and operations, then at some financial decisions





Chile

Activo Asset	Pasivos + Pat. Liability
Sells to Colombia	In \$

gets \$

Colombian \$

Swap
dealer

Colombia

Activo Asset	Pasivos + Patrimonio Liability
Sells to Chile	in Colombian \$

Colombian \$



ESCUELA DE ADMINISTRACIÓN
FACULTAD DE ECONOMÍA Y ADMINISTRACIÓN

INVESTMENTS AND COST OF CAPITAL IN INTERNATIONAL MARKETS

José A. Tessada

October 2024

TAKING STOCK

So far...

-
- Not so good...?
 - Transaction exposure
 - Translation exposure
 - Economic exposure

INTERNATIONAL DIVERSIFICATION

Risk and diversification

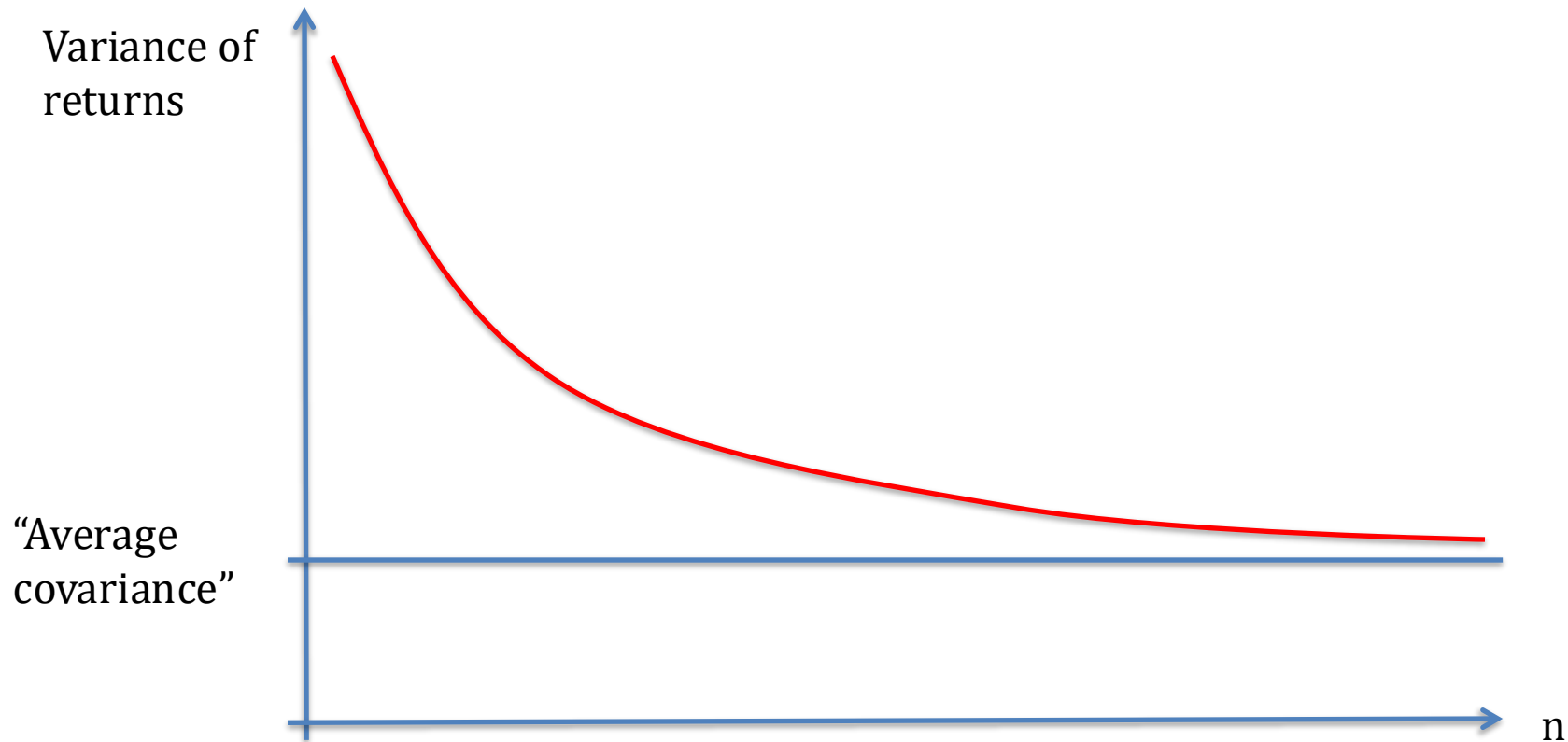
- Diversification: reducing risk of a portfolio by dividing the amount invested into a larger number of assets

$$V(Ret_p) = \sigma_p^2 = \sum_{i=1}^n \sum_{j=1}^n \omega_i \omega_j \sigma_{ij}$$

- Where the ω are the weights of the assets
- Does it matter if I can invest in global markets?

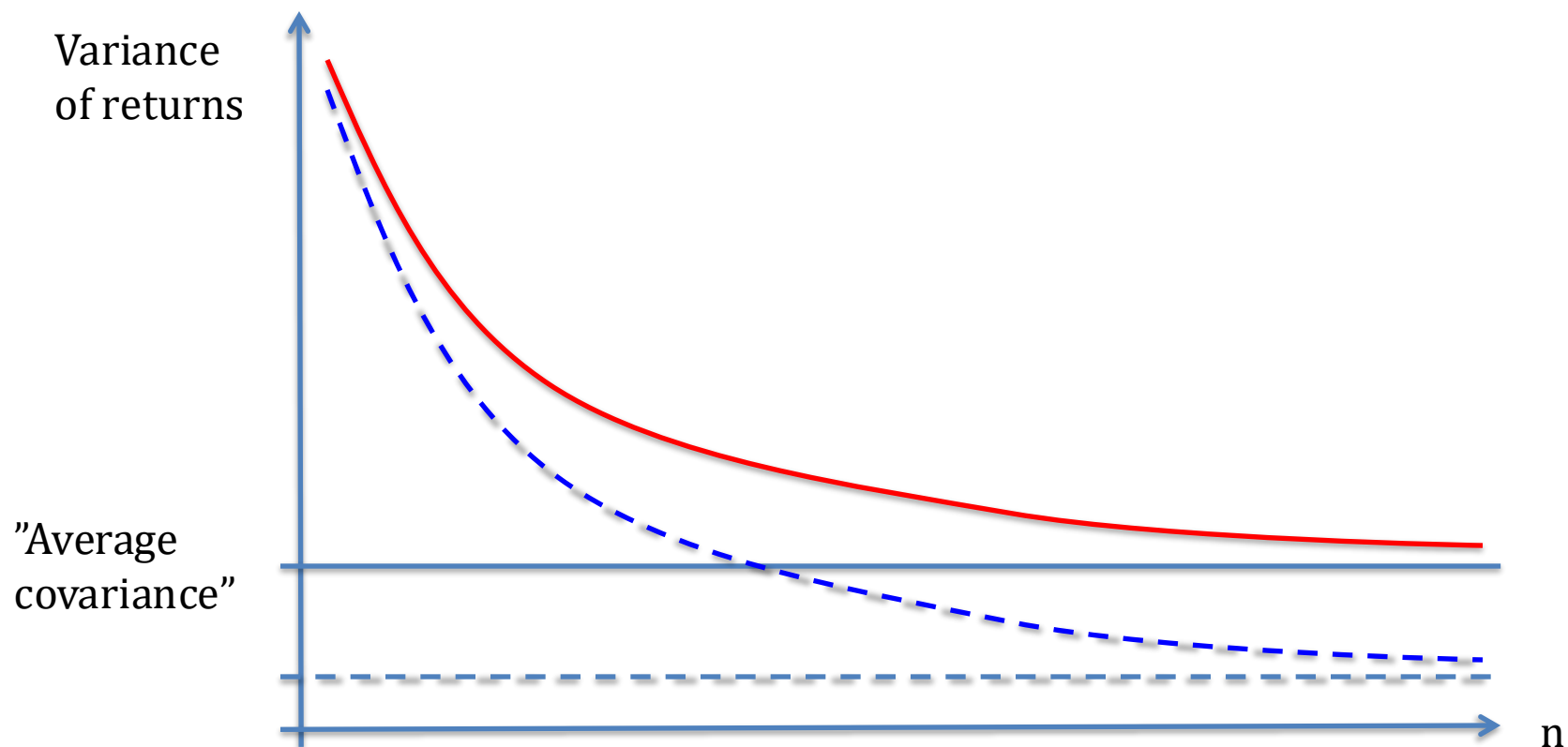
Risk and diversification

- Let us consider something very mechanical
- Portfolio with n assets, weights are $1/n$...
- You reduce the variance of returns as n increases...
 - But there is a limit...
 - Sort of an “average covariance”



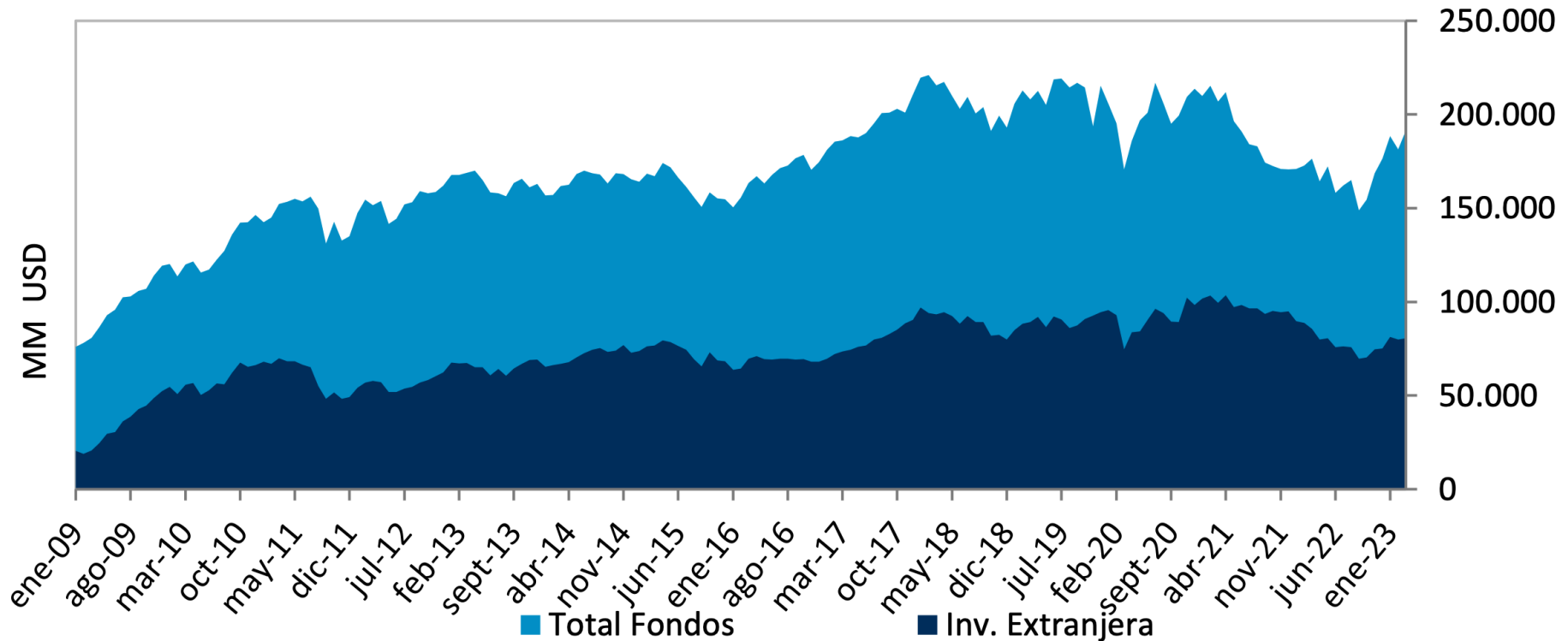
Risk and diversification

- The concept is the usual one, total risk has two components
 - Diversifiable
 - Non diversifiable
- As increases you get rid of the first one, but you are stuck with the latter one
- How does access to international markets play out here?



One example... Chilean pension funds

Gráfico N° 1
Evolución de los Fondos de Pensiones



Fuente: Superintendencia de Pensiones

Cartera de Inversiones por Tipo de Fondo

Composición de las carteras de los Fondos, marzo 2023

Tipo de instrumento	A	B	C	D	E	TOTAL
Nacional	18,0%	36,0%	57,1%	78,9%	91,5%	57,9%
Renta Variable	13,0%	10,3%	6,8%	3,2%	2,2%	6,8%
Acciones	11,6%	9,4%	5,9%	2,8%	2,2%	6,1%
Fondos de inversión y Otros	1,4%	0,9%	0,9%	0,4%	0,0%	0,7%
Renta Fija	3,8%	24,2%	48,9%	74,0%	87,9%	49,6%
Instrumentos Estatales	1,0%	5,4%	21,6%	34,2%	40,6%	21,5%
Bonos	1,5%	17,2%	25,4%	38,2%	45,5%	26,4%
Depósitos	0,2%	0,2%	0,3%	0,4%	0,7%	0,4%
Otros	1,1%	1,3%	1,5%	1,3%	1,1%	1,3%
Otros Nacionales	1,2%	1,5%	1,5%	1,7%	1,4%	1,5%
Extranjera	82,1%	64,1%	42,9%	21,1%	8,5%	42,1%
Renta Variable	64,4%	43,8%	24,6%	9,7%	2,0%	26,8%
Fondos Mutuos	39,0%	26,1%	16,0%	4,7%	0,1%	16,1%
Otros	25,4%	17,7%	8,6%	4,9%	2,0%	10,7%
Renta Fija	16,8%	19,3%	17,3%	10,2%	5,5%	14,3%
Otros Extranjeros	0,8%	0,9%	1,0%	1,2%	1,0%	1,0%
Total	100%	100%	100%	100%	100%	100%

Fondos de inversión y Otros (RV) incluye activos alternativos. Otros (RF) incluye letras hipotecarias, fondos mutuos y de inversión, activos alternativos y disponible. Otros Nacionales incluye derivados y otros. Otros (RV) incluyen activos alternativos y otros. Otros Extranjeros incluyen derivados y otros.

Tabla N° 9

Principales emisores de instrumentos de renta variable

	Emisor	Administrador	Instrumento	Valor MM USD	% Activos Totales	% Inversión Extranjera
1	SCHRODER INTERNATIONAL SELECTION FUND CHINA A	SCHRODER	CMEV	2.593	1,35%	3,22%
2	VANGUARD INDEX FUNDS VANGUARD VALUE ETF	VANGUARD	ETFA	2.154	1,13%	2,67%
3	ISHARES S&P SMALLCAP 600 INDEX FUND	BLACKROCK	ETFA	1.968	1,03%	2,44%
4	SCHRODER INTERNATIONAL SELECTION FUND EMERGING ASIA	SCHRODER	CMEV	1.942	1,01%	2,41%
5	SCHRODER INTERNATIONAL SELECTION FUND ASIAN OPPORTUNITIES	SCHRODER	CMEV	1.736	0,91%	2,15%
6	ABERDEEN STANDARD SICAV I CHINA A SHARE SUSTAINABLE EQUITY FUND	ABRDN_PLC	CMEV	1.563	0,82%	1,94%
7	BAILLIE GIFFORD OVERSEAS GROWTH FUNDS ICVC BAILLIE GIFFORD PACIFIC FUND	BAILLIE_GIFFORD	CMEV	1.412	0,74%	1,75%
8	DFA INVESTMENT DIMENSIONS GROUP INC US LARGE CAP VALUE PORTFOLIO	DIMENSIONAL	CMEV	1.390	0,73%	1,72%
9	ISHARES (DE) I INVESTMENTAKTIENGESELLSCHAFT MIT TEILGESELLSCHAFTSVERMOGEN ISHARES MSCI BRAZIL UCITS ETF (DE)	BLACKROCK	ETFA	1.223	0,64%	1,52%
10	INVESCO EUROPEAN INVESTMENT SERIES INVESCO EUROPEAN EQUITY FUND(UK)	INVESCO	CMEV	1.105	0,58%	1,37%
				17.085	8,93%	21,19%

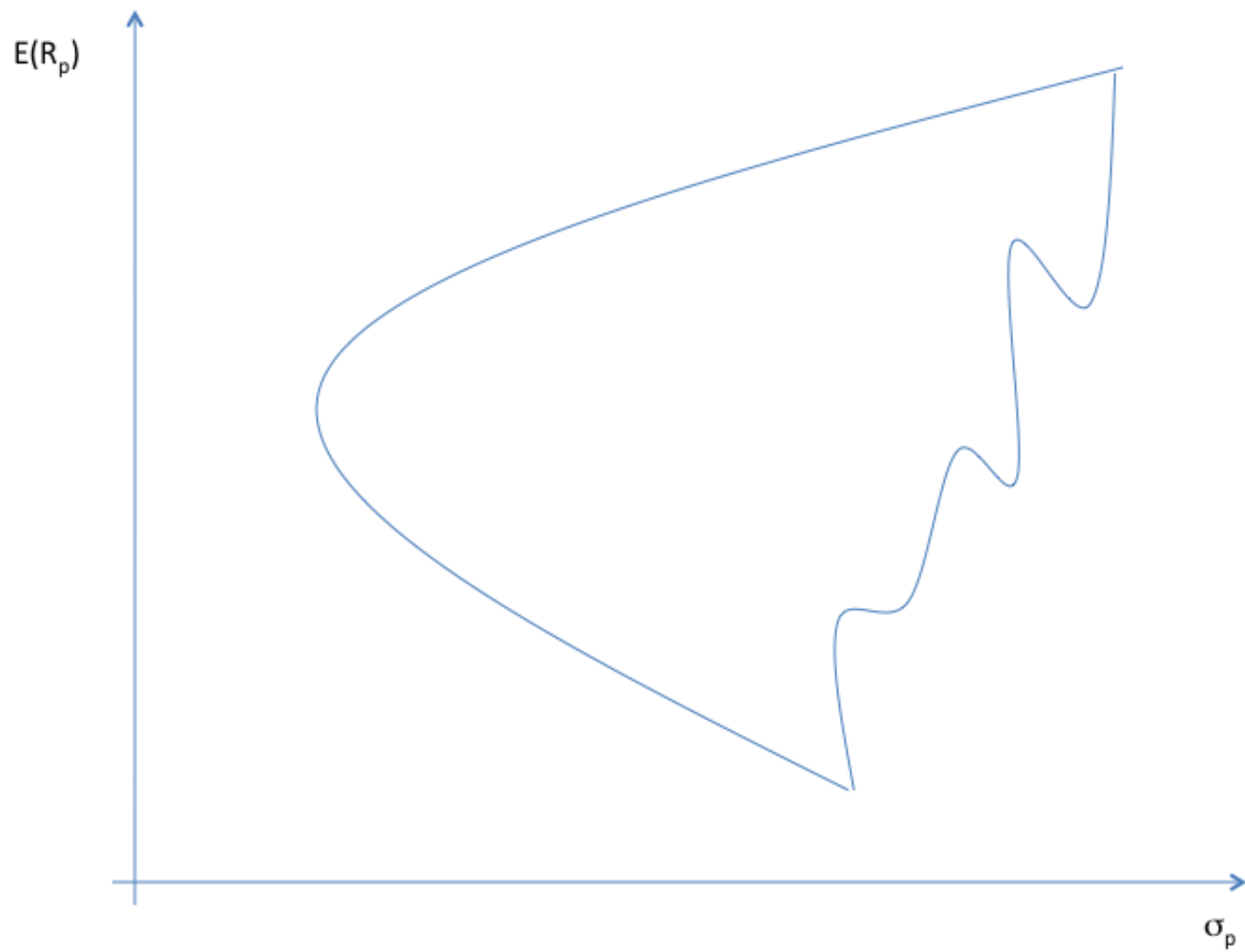
CMEV: fondos mutuos de renta variable; ETFA: títulos representativos de índices financieros

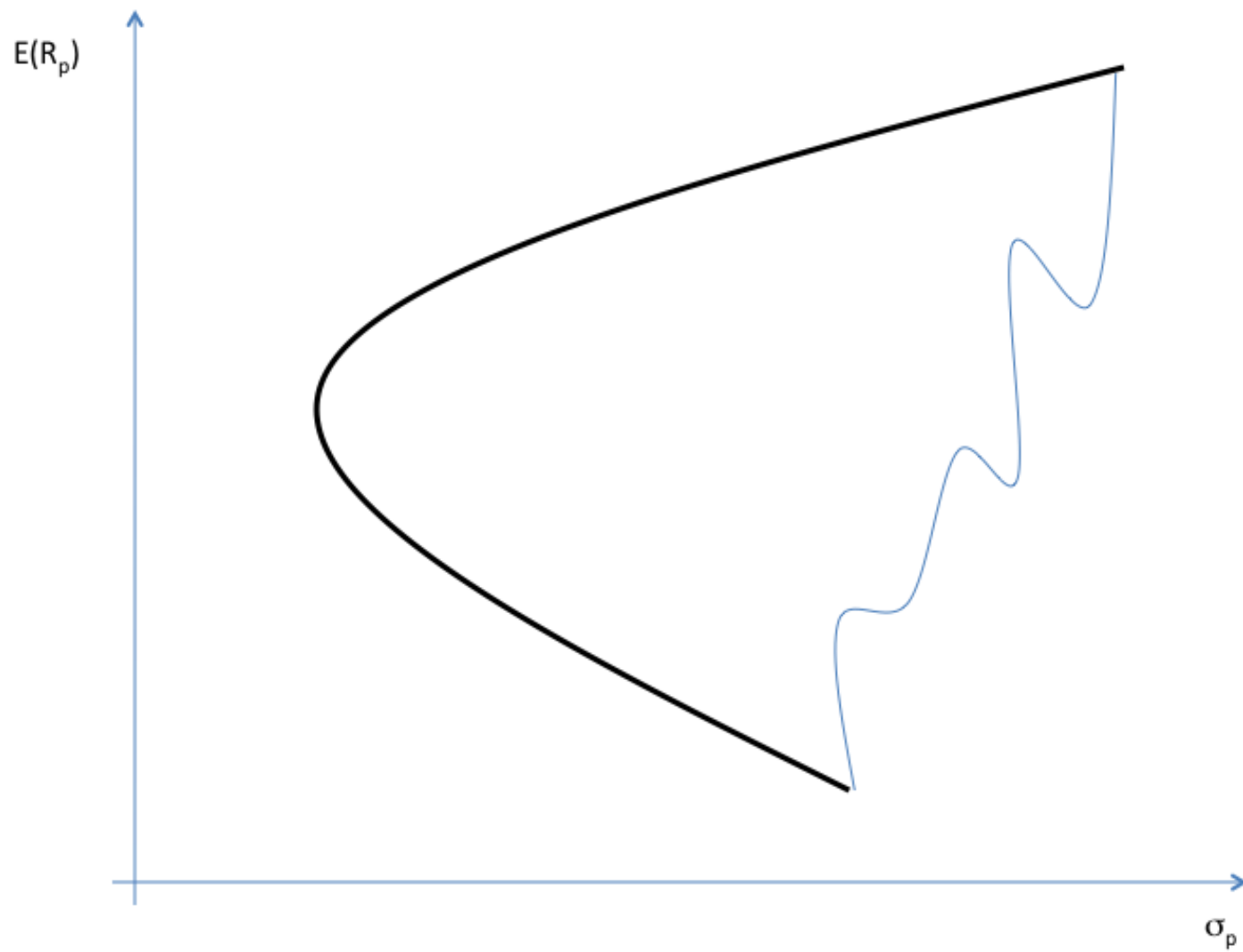
Fuente: Superintendencia de Pensiones

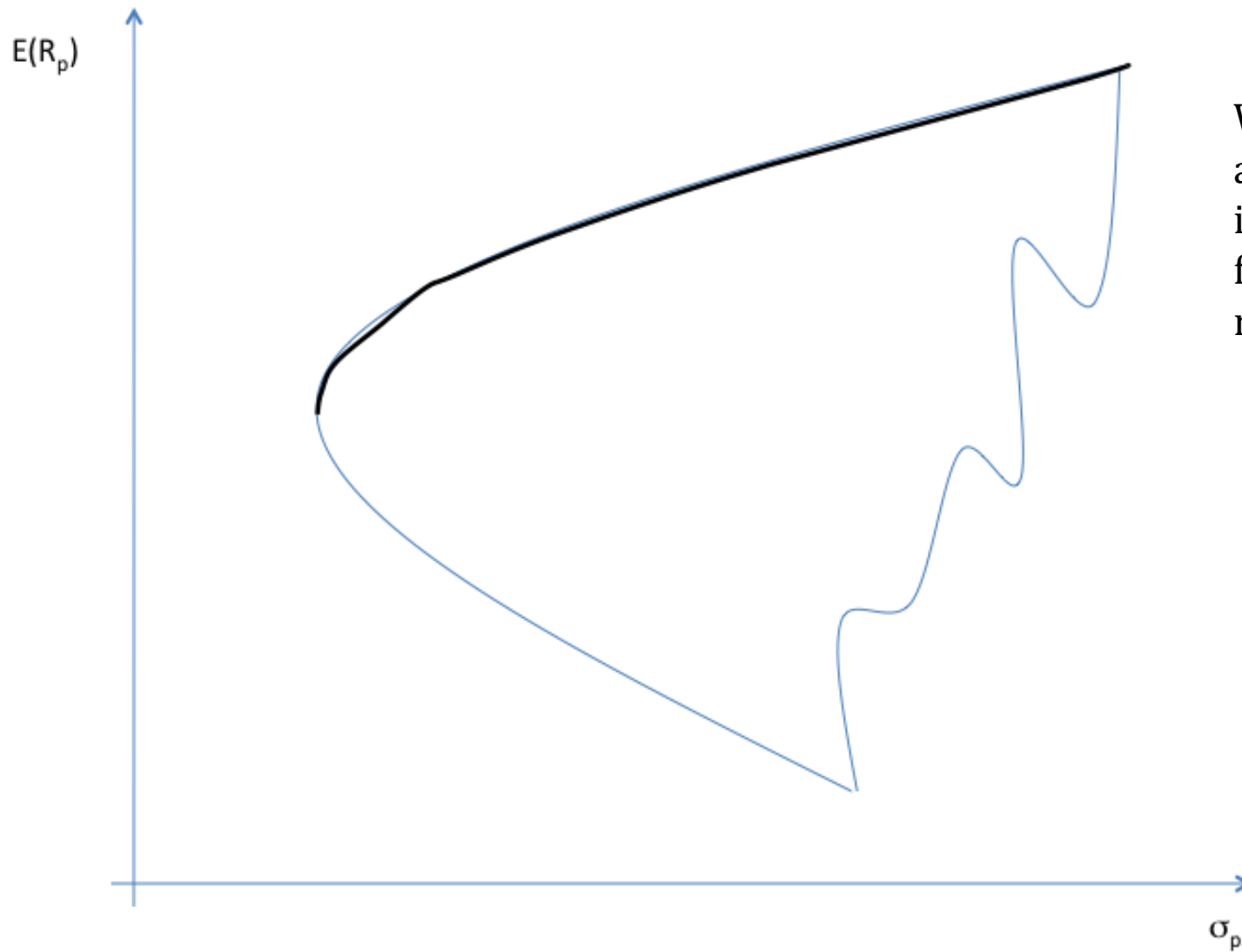
INTERNATIONAL DIVERSIFICATION: PORTFOLIO THEORY

Portfolio Theory

- As usual, risk-averse investors
- Let us stick with mean-variance preferences
 - What does this mean?
- Now, start from the usual local analysis
 - Local --> just Chile







What does
access to
international
financial
markets do?

International diversification

- What do we obtain as investors by accessing international financial markets?
 1. More assets (larger n)
 2. ...

Furthermore...



Value of countries' listed firms*

Matched to companies with equivalent market capitalisation
\$bn, January 2014



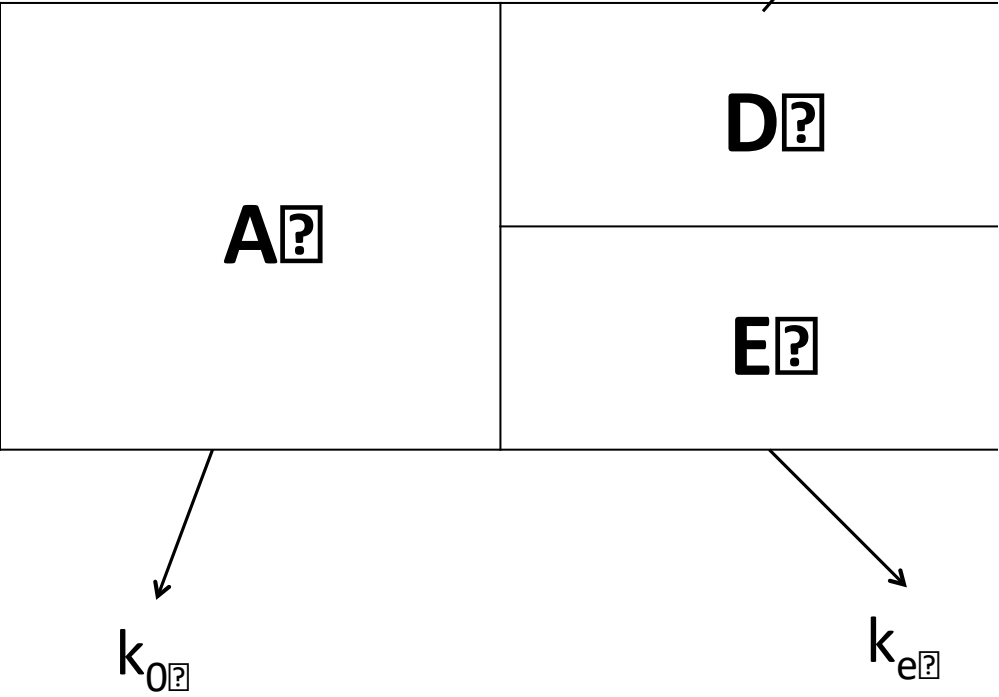
Sources:
Bank of America Merrill
Lynch; Bloomberg;
The Economist

*Freely-traded shares only, for firms accounting for 85% of market value

GLOBAL COST AND AVAILABILITY OF CAPITAL

A quick refresher

- What do we want to measure with the cost of capital?
 - Returns we could obtain by investing the resources in the market
 - Adjusted by risk
 - Discount future cash flows for valuation
- For a company then we need to consider the cost and benefit of financing abroad



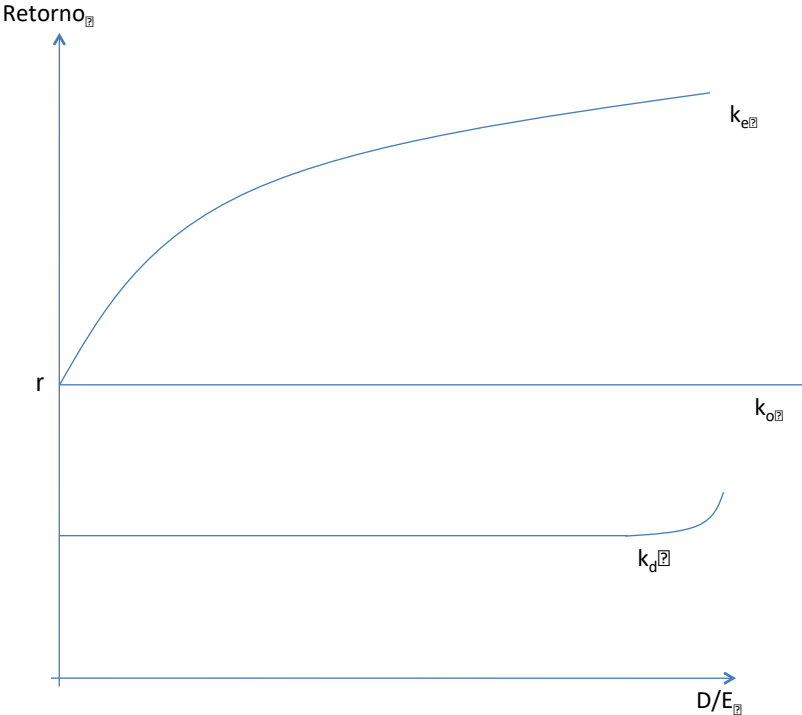
$$k = \left(\frac{D}{A} \right) (1 - \tau_D) k_D + \left(\frac{E}{A} \right) k_E$$

- Weighted average cost of capital
 - Average cost of different debt components
 - Adjust by tax
 - Cost of equity
 - Could use CAPM

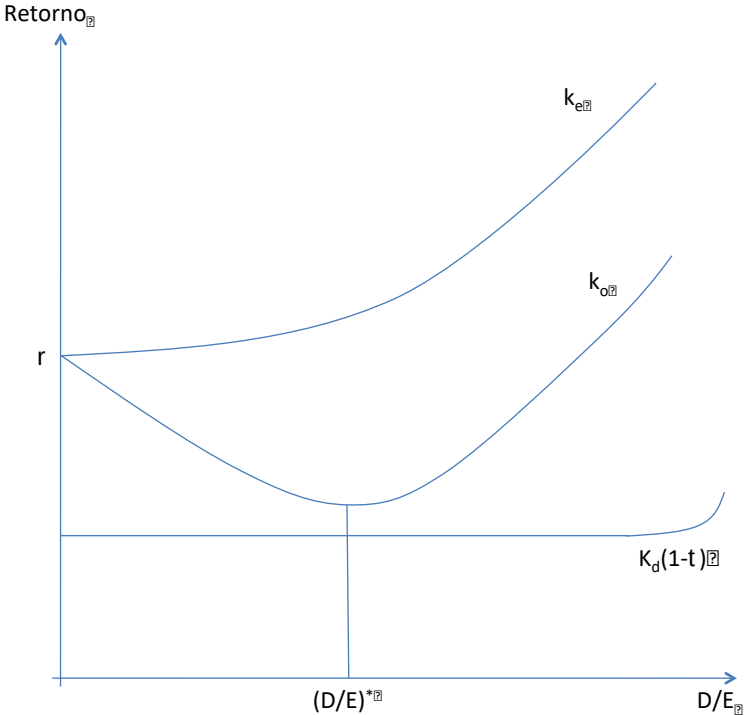
$$R_i = R_f + \beta_i (R_M - R_f)$$

- Get market returns, risk-free rate, and find a reliable estimate of β ...

Modigliani-Miller



Trade-Off



And with international capital markets

- One option is to use ICAPM

$$R_i^{global} = R_f^g + \beta_i^g \left(R_m^g - R_f^g \right)$$

- We push the international diversification discussion to CAPM, and think of a global investor
- Thus, global portfolios and global risk-free rate
- Graphically...

Example - from the book

Nestlé's estimate of its cost of equity will depend upon whether a Swiss investor is thought to hold a domestic portfolio of equity securities or a global portfolio.

Domestic Portfolio for Swiss Investor

$k_{RF} = 3.3\%$ (Swiss bond index yield)

$k_M = 10.2\%$ (Swiss market portfolio in SF)

$\beta_{Nestlé} = 0.885$ (Nestlé versus Swiss market portfolio)

Global Portfolio for Swiss Investor

$k_{RF} = 3.3\%$ (Swiss bond index yield)

$k_M = 13.7\%$ (*Financial Times* Global index in SF)

$\beta_{Nestlé} = 0.585$ (Nestlé versus FTA-Swiss index)

$$k^{Nestlé} = k_{RF} + \beta^{Nestlé} (k_M - k_{RF})$$

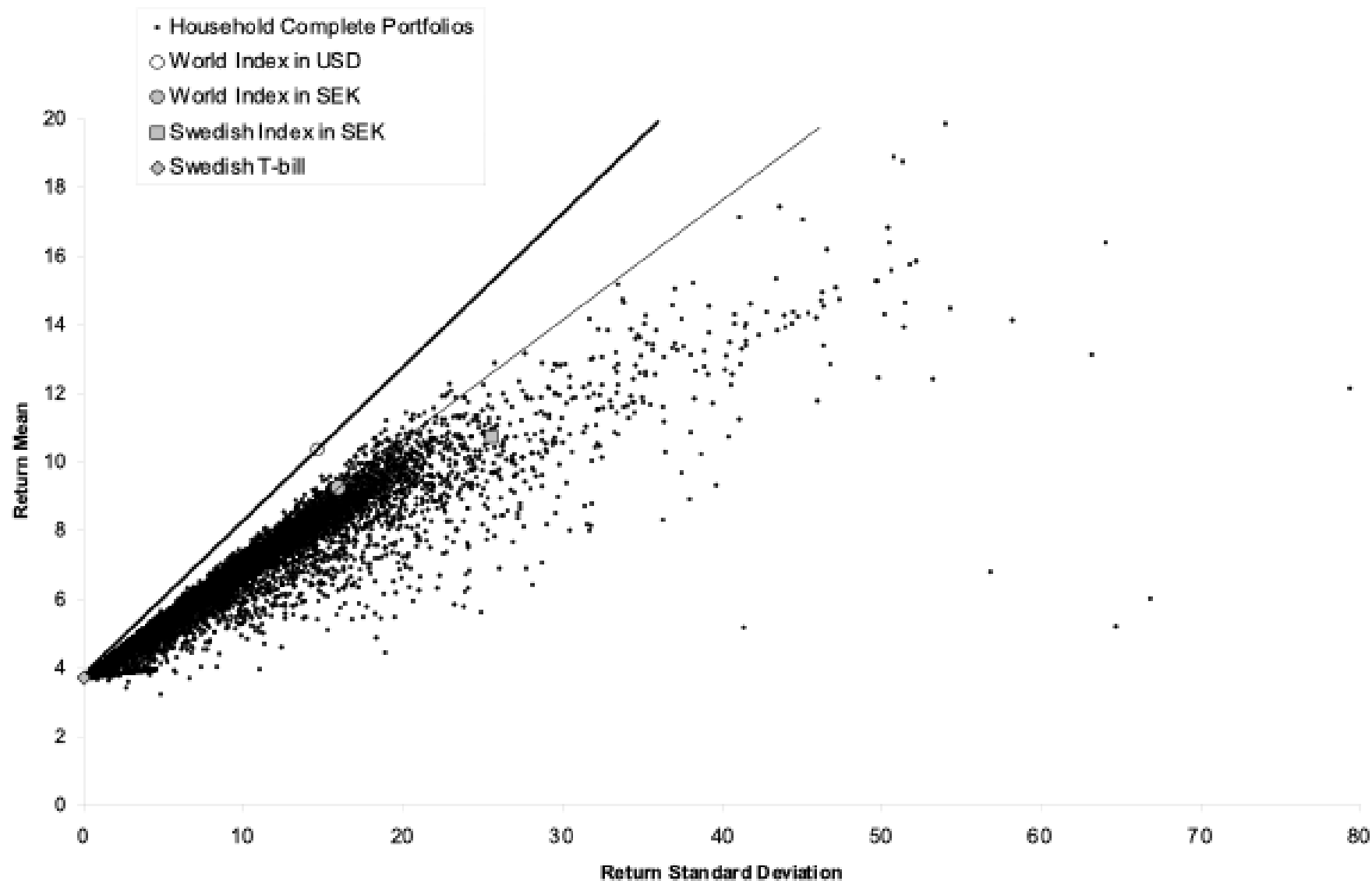
Required return on Nestlé:

$$k_e^{Nestlé} = 9.4065\%$$

Required return on Nestlé:

$$k_e^{Nestlé} = 9.3840\%$$

Source: All values are taken from René Stulz, "The Cost of Capital in Internationally Integrated Markets: The Case of Nestlé," *European Financial Management*, Volume 1, Number 1, March 1995, 11–22.



-
- The weighted average cost of capital is normally used as the risk-adjusted discount rate whenever a firm's new projects are in the same general risk class as its existing projects.
 - On the other hand, a project-specific required rate of return should be used as the discount rate if a new project differs from existing projects in business or financial risk.

EXHIBIT 13.1 Dimensions of the Cost and Availability of Capital Strategy

Local Market Access

Global Market Access

Firm-Specific Characteristics

Firm's securities appeal only to domestic investors

Firm's securities appeal to international portfolio investors

Market Liquidity for Firm's Securities

Illiquid domestic securities market and limited international liquidity

Highly liquid domestic market and broad international participation

Effect of Market Segmentation on Firm's Securities and Cost of Capital

Segmented domestic securities market that prices shares according to domestic standards

Access to global securities market that prices shares according to international standards

The Demand for Foreign Securities: The Role of International Portfolio Investors

- Gradual deregulation of equity markets during the past three decades not only elicited increased competition from domestic players but also opened up markets to foreign competitors.
- *Market liquidity* (observed by noting the degree to which a firm can issue a new security without depressing the existing market price) can affect a firm's cost of capital.
- In the domestic case, a firm's marginal cost of capital will eventually increase as suppliers of capital become saturated with the firm's securities.
- In the multinational case, a firm is able to tap many capital markets above and beyond what would have been available in a domestic capital market only.

The Demand for Foreign Securities: The Role of International Portfolio Investors

- Some imperfections that can affect the efficiency of a national capital market include:
 - Asymmetric information
 - Lack of transparency
 - High transaction costs
 - Political risks
 - Corporate governance issues
 - Regulatory barriers

The Demand for Foreign Securities: The Role of International Portfolio Investors

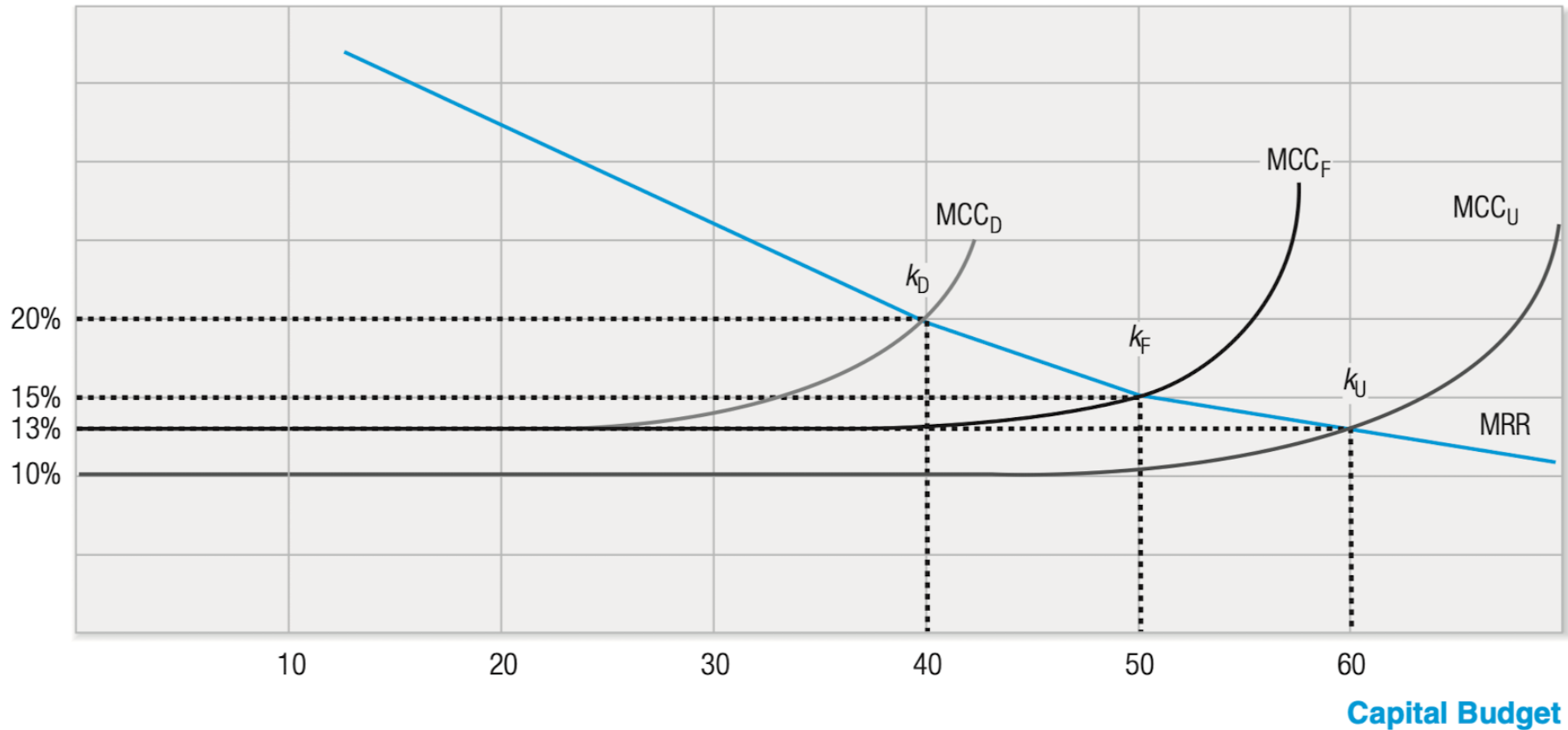
- The degree to which capital markets are illiquid has an important influence on a firm's marginal cost of capital (and thus on its weighted average cost of capital).
- Denote the marginal return on capital at different budget levels as MRR.

- If the firm is limited to raising funds in its domestic market, the line MCC_D shows the marginal domestic cost of capital.
- If the firm has additional sources of capital outside the domestic (illiquid) capital market, the marginal cost of capital shifts right to MCC_F .
- If the MNE is located in a capital market that is both illiquid and segmented, the line MCC_U represents the decreased marginal cost of capital if it gains access to other equity markets.

EXHIBIT 13.5 Market Liquidity, Segmentation, and the Marginal Cost of Capital

Cost of Capital

(marginal cost of capital for users,
rate of return for providers, *percent*)



The Cost of Capital for MNEs Compared to Domestic Firms

- Determining whether a MNEs cost of capital is higher or lower than a domestic counterpart is a function of the marginal cost of capital, the relative after-tax cost of debt, the optimal debt ratio and the relative cost of equity.
- While the MNE is supposed to have a lower marginal cost of capital (MCC) than a domestic firm, empirical studies show the opposite (as a result of the additional risks and complexities associated with foreign operations).

-
- This relationship lies in the link between the cost of capital, its availability, and the opportunity set of projects.
 - As the opportunity set of projects increases, the firm will eventually need to increase its capital budget to the point where its marginal cost of capital is increasing.

The Cost of Capital for MNE and Domestic Counterpart Compared

- The optimal capital budget would still be at the point where the rising marginal cost of capital equals the declining rate of return on the opportunity set of projects.
- This would be at a higher weighted average cost of capital than would have occurred for a lower level of the optimal capital budget.

EXHIBIT 13.6 Cost of Capital for MNE and Domestic Counterpart Compared

Cost of Capital

(marginal cost of capital for users,
rate of return for providers, *percent*)

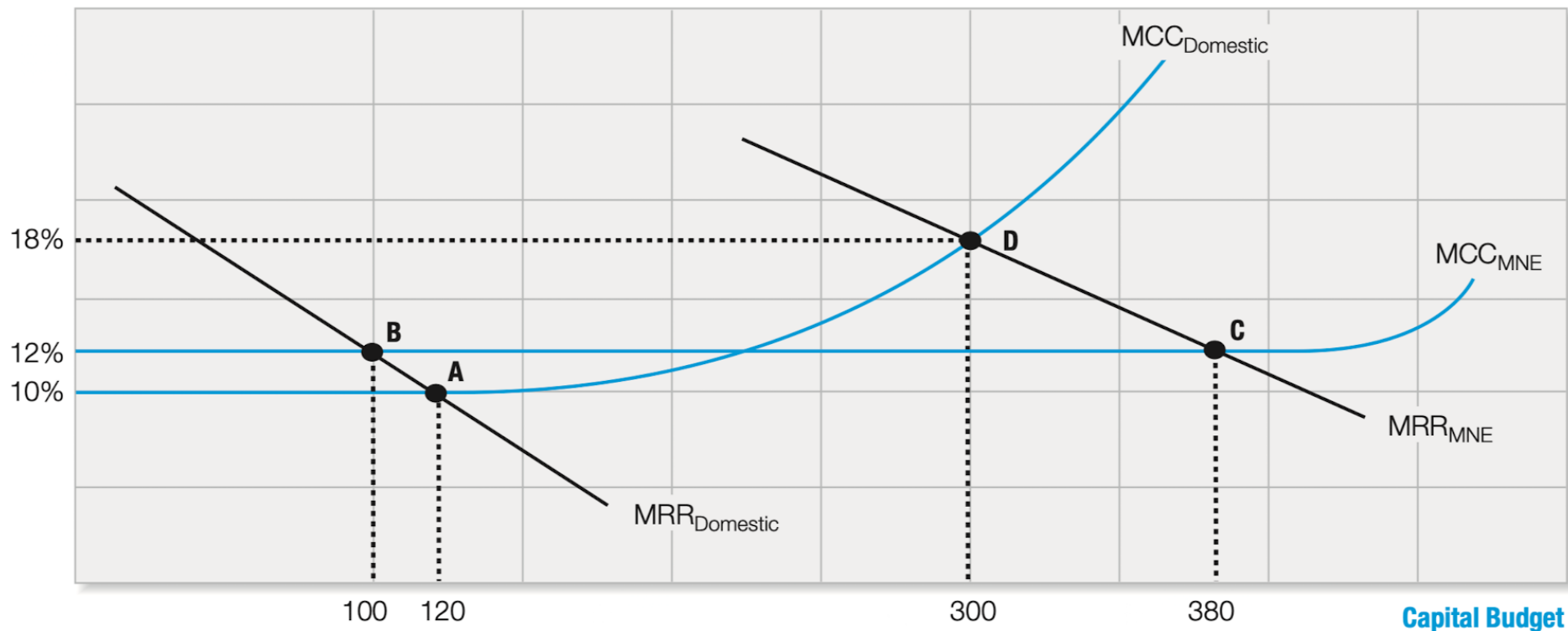


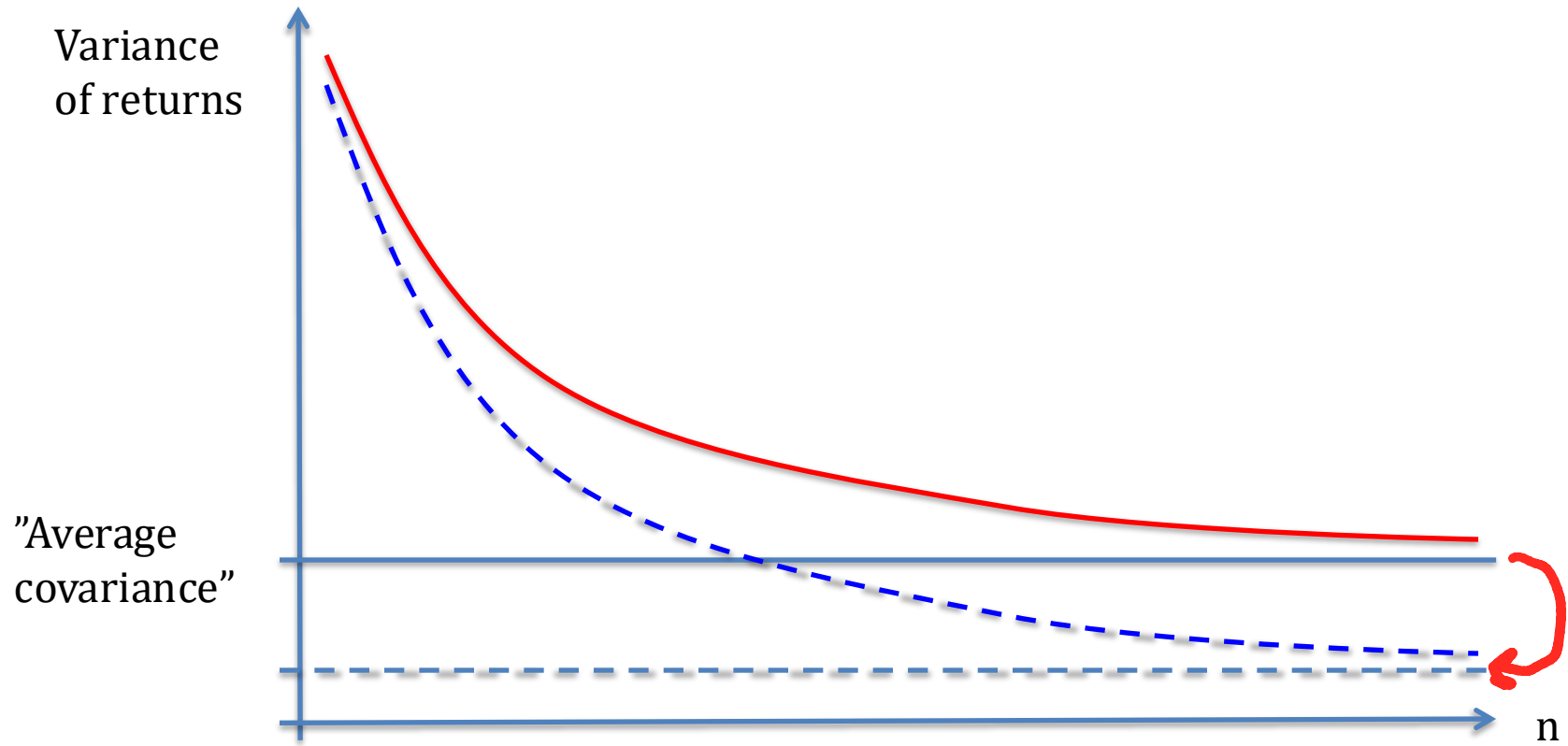
EXHIBIT 13.7 Do MNEs Have a Higher Cost of Capital Than Their Domestic Counterparts?

$$k_{WACC} = \underbrace{k_e \left[\frac{\text{Equity}}{\text{Value}} \right]}_{\text{Equity Cost Component}} + \underbrace{k_d (1 - t) \left[\frac{\text{Debt}}{\text{Value}} \right]}_{\text{Debt Cost Component}}$$

Studies indicate that MNEs have a **higher cost of equity** but *more equity* in their capital structure than domestic counterparts.

Studies indicate that MNEs have a **lower cost of debt** but *less debt* in their capital structure than domestic counterparts.

$$k_{WACC}^{\text{MNE}} > k_{WACC}^{\text{Domestic}}$$



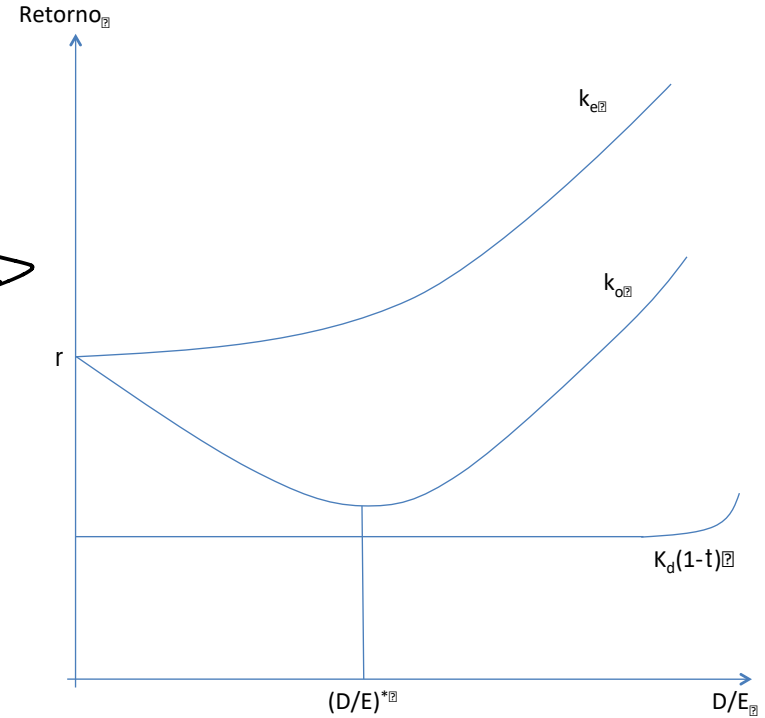
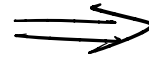
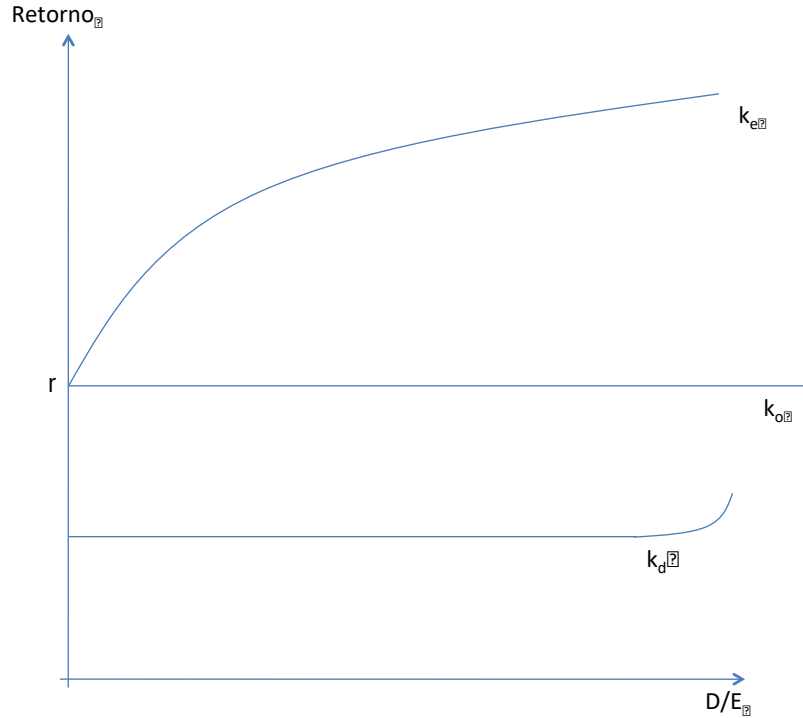
- Weighted average cost of capital
 - Average cost of different debt components
 - Adjust by tax
 - Cost of equity
 - Could use CAPM

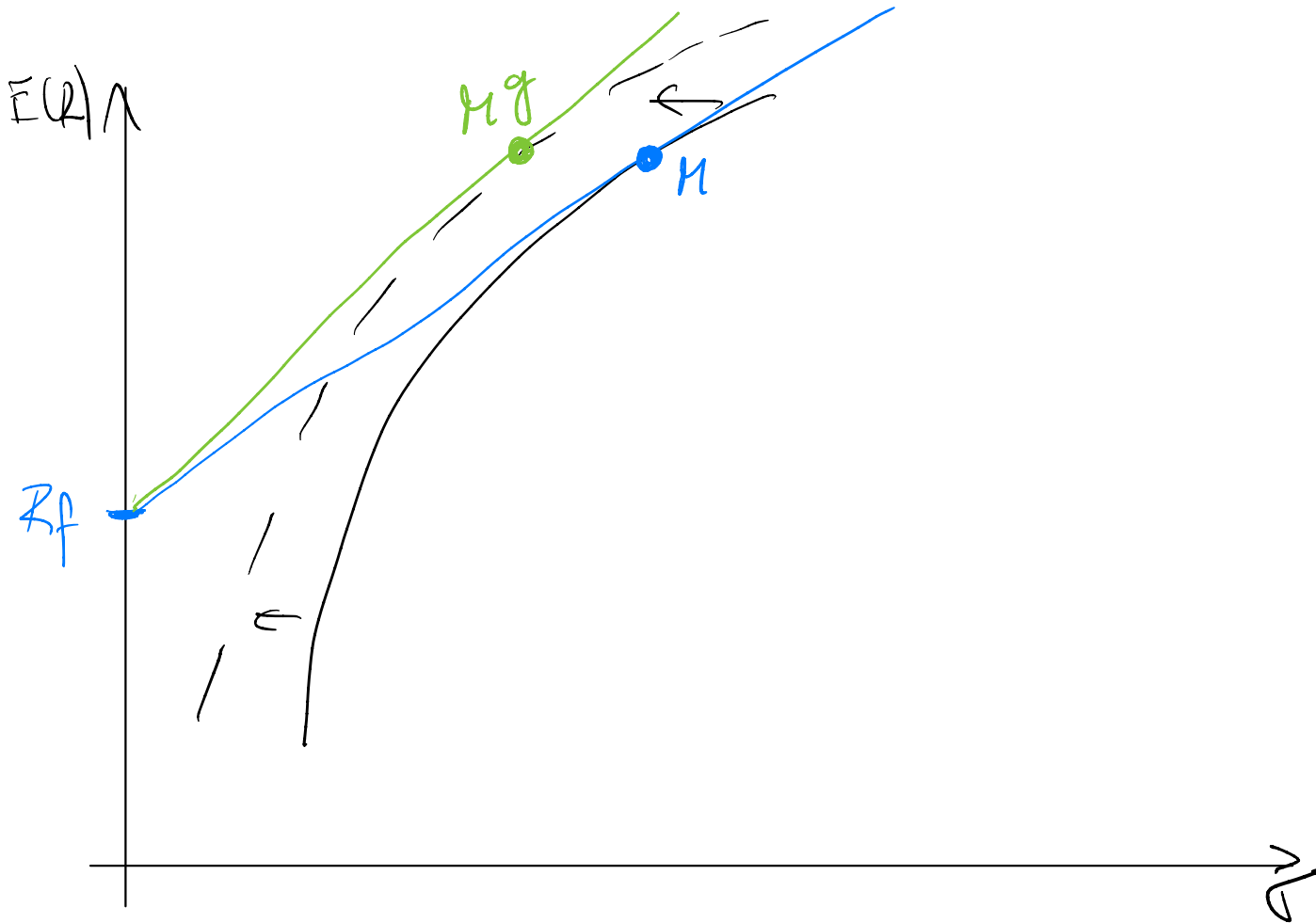
$$E(R_i) = R_f + \beta_i (R_M - R_f)$$

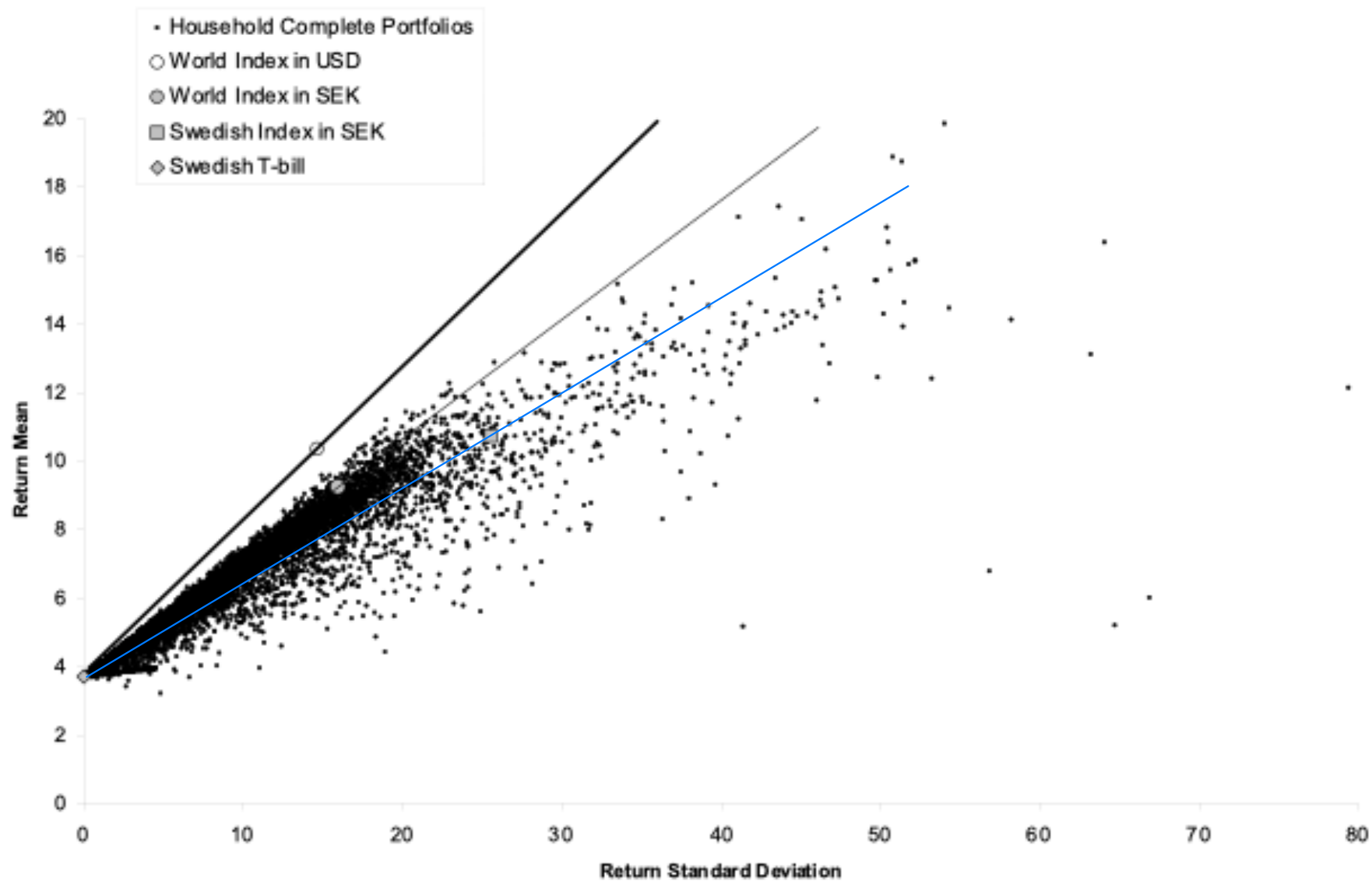

- Get market returns, risk-free rate, and find a reliable estimate of β ...

Modigliani-Miller

Usually...
Trade-Off

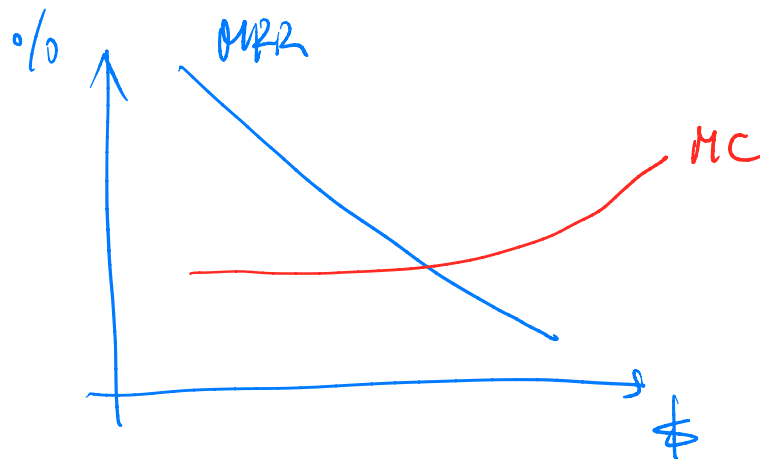






The Demand for Foreign Securities: The Role of International Portfolio Investors

- The degree to which capital markets are illiquid has an important influence on a firm's marginal cost of capital (and thus on its weighted average cost of capital).
- Denote the marginal return on capital at different budget levels as MRR.





ESCUELA DE ADMINISTRACIÓN
FACULTAD DE ECONOMÍA Y ADMINISTRACIÓN

DISCOUNT RATES, FDI AND POLITICAL RISK

José A. Tessada

October 2024

TAKING STOCK

Summing up ...

- Factors that will influence the cost of capital of a company multinational
 - Bigger size
 - Greater (better) access to markets financial international
 - International diversification
 - Exposure to risk exchange rate
 - Exposure to country risk

DISCOUNT RATES

What rates do we need?

- For valuation we need expected returns for: assets, debt and equity
- Depending on the valuation method we use we should calculate the WACC
- The cost of equity (k_E) is the most demanding part
- What we need now is a framework to analyze this cost in a context international
- The conceptual framework for determining (calculating) k_E will be the same: CAPM
- But, we must define if we take a local or an international perspective

Local or International?

- Let's remember the CAPM formula

$$R_i = R_f + \beta_i(R_M - R_f)$$

- For the case of k_E we need $(R_M - R_f)$ and β_E
- If we go local, then we measure systematic risk in the local market
 - We are assuming segmented capital markets
- If we go international , we measure regarding market international
 - We are assuming integrated capital market, ie investors diversify internationally

Local or International? Hypothetical case

- Offer for a business with operation in the US and UK
- Investors based in UK
- What would you do?

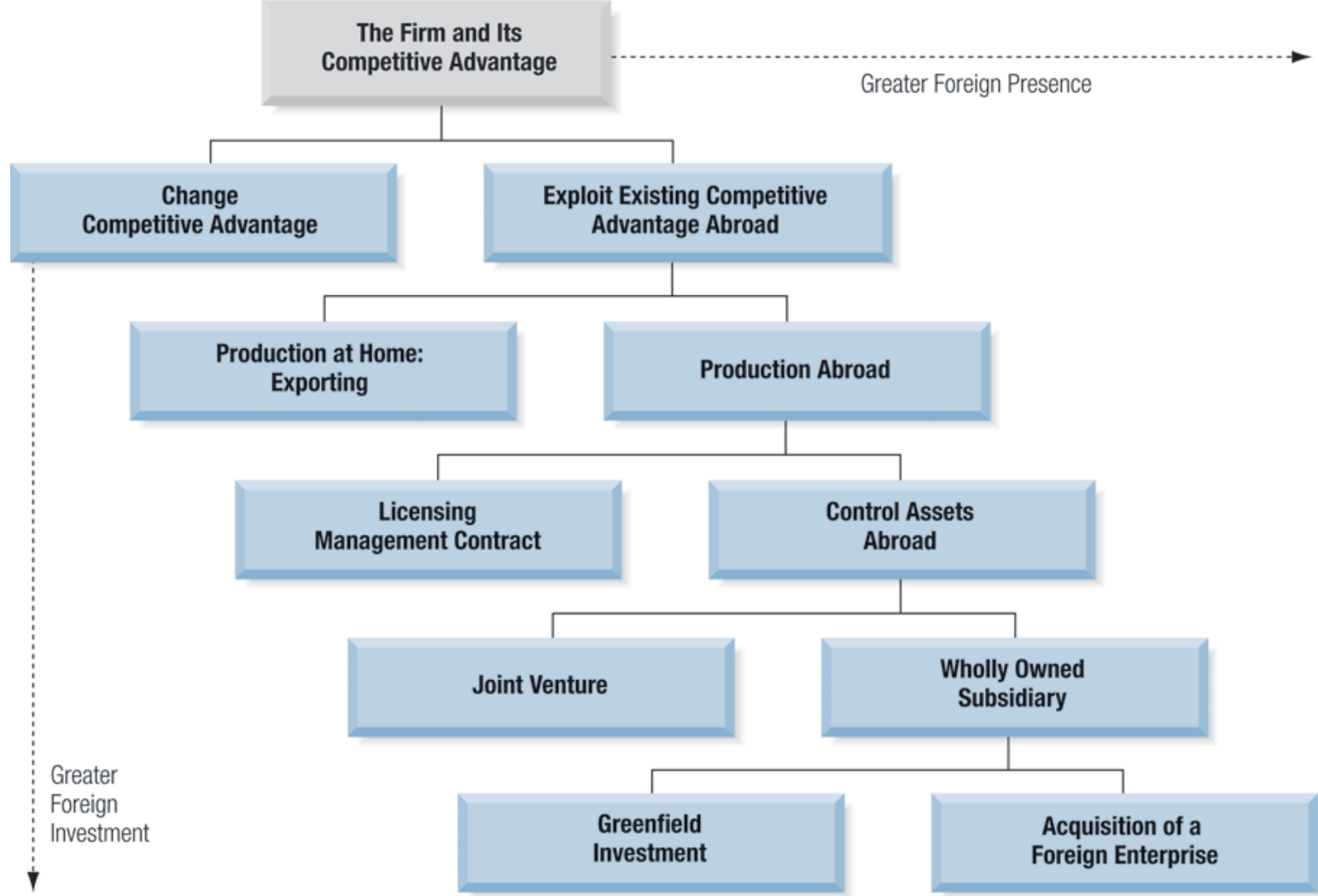
Local or International? Others items important

- Global versus worldwide portfolio
 - Difference subtle but relevant
- Are they very different answers?
 - It depends, one dimension is the degree of integration between markets
 - Other way to ask it, is it very different the cost of capital in different countries?
 - International evidence suggests that integration has increased but we are not still in world completely integrated
- Do multinationals have a financing cost? Higher?
 - We reviewed part of this in the previous topic

A DETOUR: WHY DO COMPANIES DO FOREIGN DIRECT INVESTMENT (FDI)?

Why FDI?

- A company becomes multinational when it carries out investment foreigner direct
 - Not the same as being a company with international operation
- Two ways to do FDI
 - Greenfield investment
 - Cross-border M&A
- What do we know about FDI in the world?
 - Increased until the early 2010s
 - New destinations and new countries of origin
 - Linked to increased trade international?
 - Globalization and the anti-globalization



Why FDI?

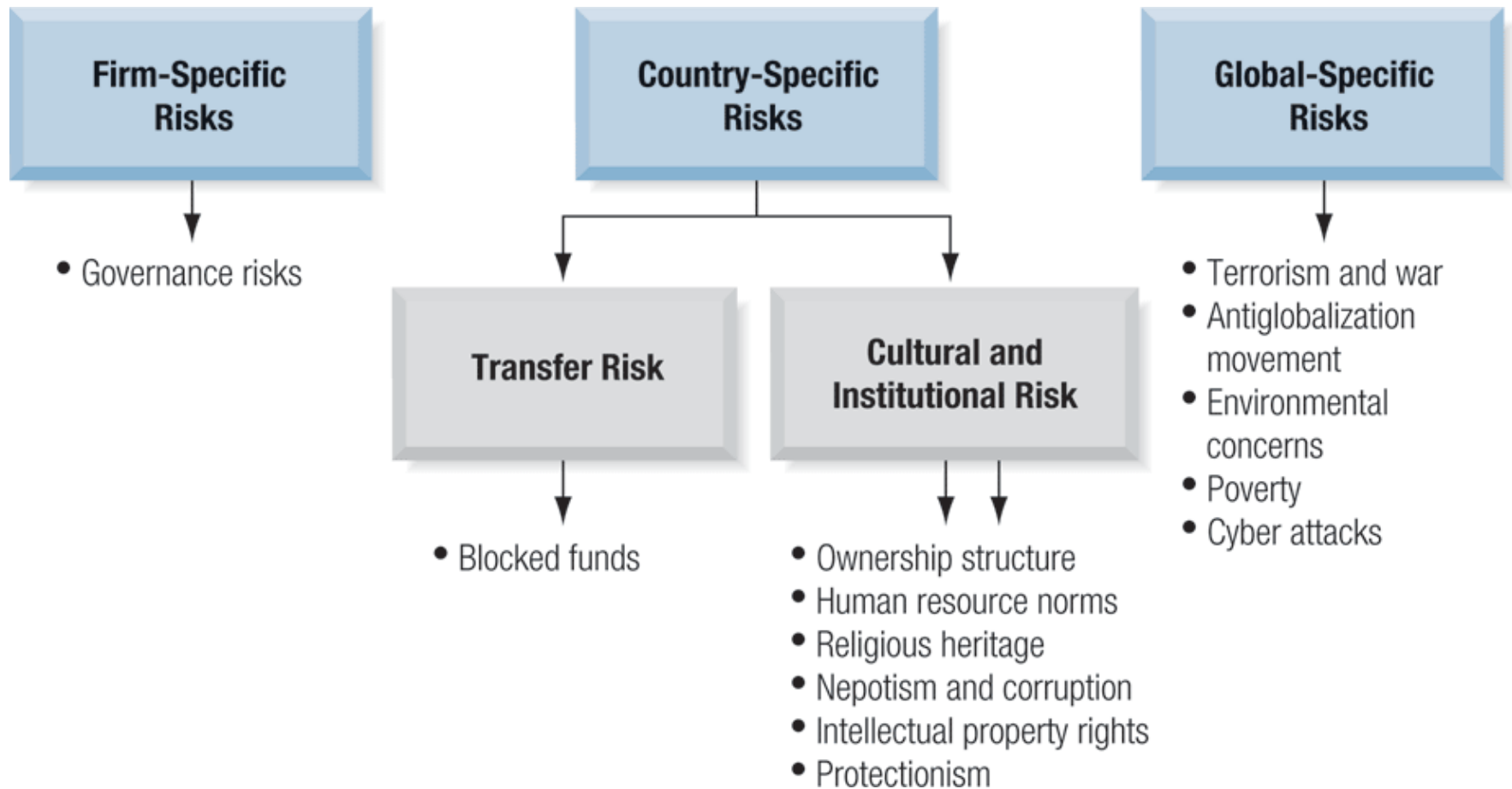
- Trade barriers
- Imperfect labor markets
- Intangible assets
- Vertical integration
- Product life cycle
- Diversification for the shareholders

- More common when firms want to consolidate, protect and deepen their international position
 - Reduce investment in parts that are not essential to its business strategy
 - Invest in assets strategic to improve its position
- Examples :
 - Walmart in Chile
 - Lithium markets
- Synergies are a crucial element for the success of these operations
- Information?

POLITICAL RISK

Risks in a MNE

- Several differences with local company
 - International diversification
 - Exchange rate risk
 - Country risk
- The country risk has two elements
 - Financial risks
 - Economic variables
 - Monetary policy
 - Fiscal policy
 - Political risk



- A different approach
 - Macro risks: the host country faces variations affecting all operations
 - Micro risks: effects are sectoral
- Alternatively
 - Control risk
 - Operational risk

Incorporating country risk

- Important elements: we need incorporate risk country in our valuations
- Alternative 1: adjust the flows
 - We incorporate scenarios
 - Forget “expected” values
 - We compute VPN conditional in some scenarios
 - International default
 - Blocking remittances
 - Certainty equivalence
 - We adjust the expected value taking an adjustment factor less than 1

Incorporating country risk

- Alternative 2:

- We start from a base risk level
- We take the CAPM in international version

$$R_i = R_f + \beta_i(R_M - R_f)$$

- Must consider the expected depreciation
- We add country risk
 - Sovereign spread – rate differentials in government or treasury
 - Adjustment by scenario – probability of obtaining returns
- There is not one unique formula nor a magic trick



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October 2024

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Escenario base: no pasa nada 50%

Escenario 1 : guerra, a partir
de año 6
flujos como la
mitad 50%

$$r_f + \beta(ERP) - R_f$$

por por desc.

VPA
Ajuste de la
base

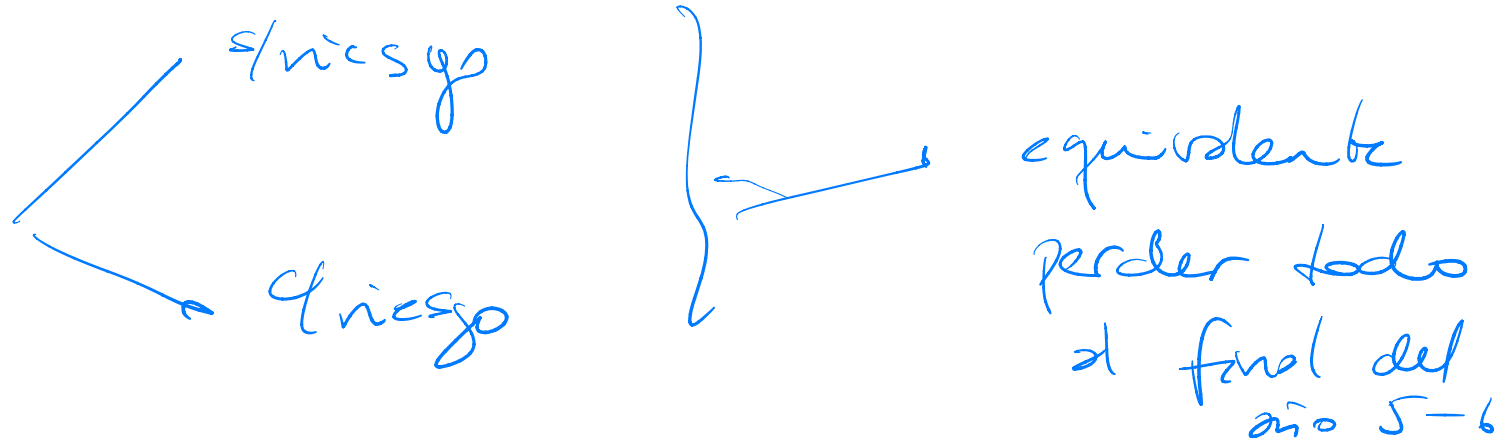


VPA
Promedio

23% $\longrightarrow$ VIN \$277,5 millones Lol Pir

Red Oak $\longrightarrow$ bsr 6,5%

$\longrightarrow$ Lol Pir en 6 años: VIN \$310,6 millones



$$VPA = \frac{100}{1+r_1} + \frac{100}{(1+r_1)^2} + \frac{100}{(1+r_1)^3} + \dots + \frac{100}{(1+r_1)^6} + \dots = 274,5$$

$$VPS = \frac{100}{1+r_2} + \frac{100}{(1+r_2)^2} + \dots + \frac{100}{(1+r_2)^6} + \dots$$

$$\begin{array}{ccc} x & xy & xy \\ \sum_{i=1}^5 & & \uparrow \\ \frac{100}{(1+r_2)^i} & & = 310,6 \end{array}$$

LoL Per CL

3 años de uso 50%
hoy 50%

9 u al 6,5%

$$VPN = \frac{VPN_{hoy}}{2} + \frac{VPN_{permanente}}{2}$$

LoL Per CL

VPN_{ESB}

→ base ajustada



ESCUELA DE ADMINISTRACIÓN
FACULTAD DE ECONOMÍA Y ADMINISTRACIÓN

RAISING EQUITY AND DEBT GLOBALLY

José A. Tessada

November 2024

Global Cost and Availability of Capital

- A firm that must source its long-term debt and equity in a highly illiquid domestic capital market will probably
 - have a relatively high cost of capital...
 - and will face limited availability of such capital...
 - which will, in turn, damage the overall competitiveness of the firm.
- Global integration of capital markets has given many firms access to new and cheaper sources of funds beyond those available in their home markets
 - If a firm is located in a country with illiquid, small capital markets, it can achieve this lower global cost and greater availability of capital by a properly designed and implemented strategy

Sourcing Equity Globally

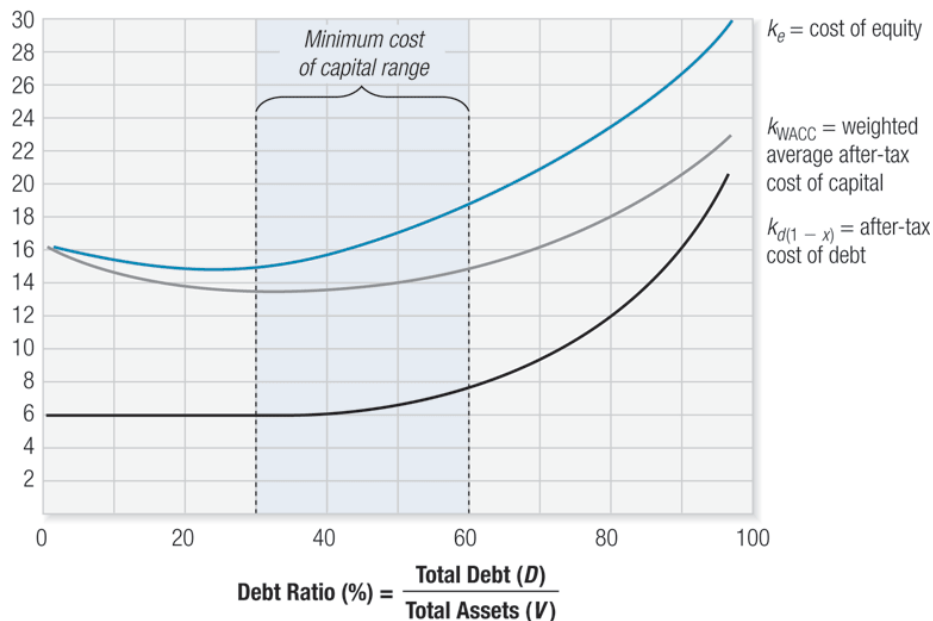
- To implement the goal of gaining access to global capital markets a firm must begin by designing a strategy that will ultimately attract international investors.
 - identifying and choosing alternative paths to access global markets.
 - restructuring of the firm, improving the quality and level of its disclosure, and making its accounting and reporting standards more transparent to potential foreign investors.
- Often, this decision-making process is aided by an early appointment of an investment bank as an official advisor to the firm.
 - Investment bankers are in touch with potential foreign investors and know what they currently require and can also help navigate the numerous institutional and regulatory barriers in place.

Designing a Strategy to Source Equity Globally

- Most firms raise their initial capital in their own domestic market.
- However, most firms that have only raised capital in their domestic market are not well known enough to attract foreign investors.
- Incremental steps to bridge this gap include conducting an international bond offering and/or equity issuing on more highly liquid foreign stock exchanges.

Optimal Financial Structure

Cost of Capital (%)



- When taxes and bankruptcy costs are considered, a firm has an optimal financial structure determined by that particular mix of debt and equity that minimizes the firm's cost of capital for a given level of business risk.
- As the business risk of new projects differs from the risk of existing projects, the optimal mix of debt and equity would change to recognize tradeoffs between business and financial risks.
- The domestic theory of optimal financial structure must be modified considerably to encompass the multinational firm.
- As the debt ratio increases, the overall cost of capital (k_{WACC}) decreases because of the heavier weight of low-cost (due to tax-deductibility) debt ($[k_d(1-t)]$) compared to high-cost equity (k_e).

Optimal Financial Structure and the MNE

- The domestic theory of optimal financial structures needs to be modified by four more variables in order to accommodate the case of the MNE.
- These variables include:
 - **Availability of capital.** A multinational firm's marginal cost of capital is constant for considerable ranges of its capital budget. This statement is not true for most small domestic firms (as they do not have equal access to capital markets).
 - **Diversification of cash flows.** Multinational firms could be in a better position than domestic firms to support higher debt ratios because their cash flows are diversified internationally (much in the same way as portfolio investors who diversify their security holdings globally).
 - **Foreign exchange risk.**
 - **Expectations of international portfolio investors.** The key to gaining a global cost and availability of capital is attracting and retaining international portfolio investors. E.g., by adopting global norms that are close to the U.S.

Raising Equity Globally

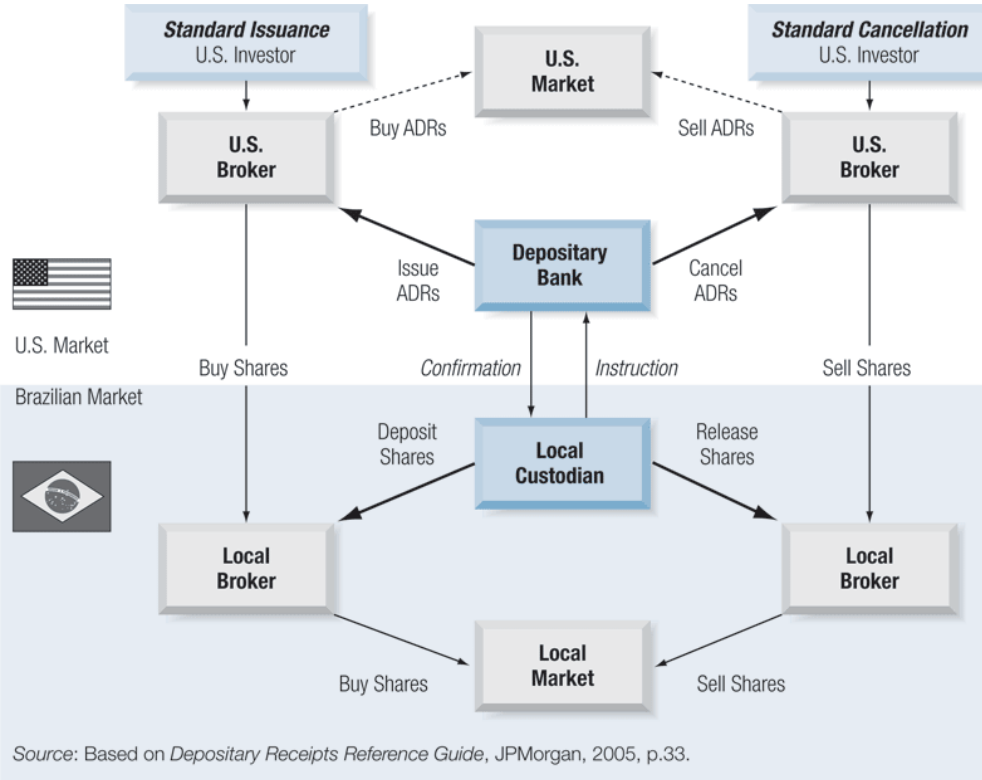
- **Initial Public Offering (IPO)** – the first public issue of a firm’s equity shares.
 - The IPO typically represents 15% to 25% of ownership
 - Later issues by the firm are considered “seasoned offerings”
 - With public issuance of shares comes greater public disclosure
- A firm can now issue equity underwritten and distributed in multiple foreign equity markets, sometimes simultaneously with distribution in the domestic market.
- A **directed public share issue** is defined as one that is targeted at investors in a single country and underwritten in whole or in part by investment institutions from that country.
- The issue might or might not be denominated in the currency of the target market.
- The shares might or might not be cross-listed on a stock exchange in the target market.

Raising Equity Globally

- **Depository receipts (depository shares)** are negotiable certificates issued by a bank to represent the underlying shares of stock, which are held in trust at a foreign custodian bank.
- **American depository receipts (ADRs)** are certificates traded in the United States and denominated in US dollars.
- ADRs are sold, registered, and transferred in the US in the same manner as any share of stock.



Raising Equity Globally



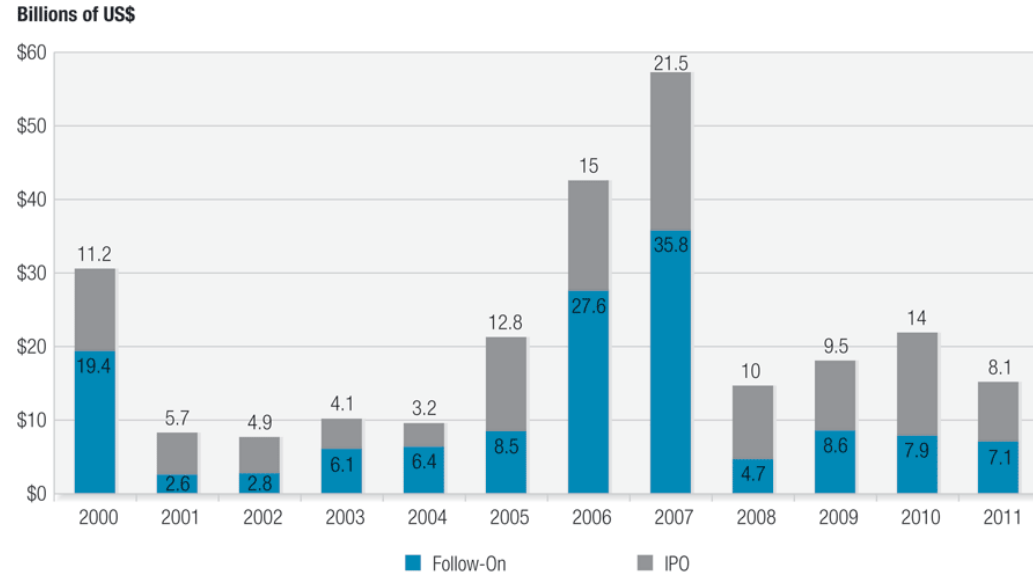
1. The U.S. investor instructs his broker to make a purchase of shares in the publicly traded Brazilian company.
2. The U.S. broker contacts a local broker in Brazil (either through the broker's international offices or directly), placing the order.
3. The Brazilian broker purchases the desired ordinary shares and delivers them to a custodian bank in Brazil.
4. The U.S. broker converts the U.S. dollars received from the investor into Brazilian *reais* to pay the Brazilian broker for the shares purchased.
5. On the same day that the shares are delivered to the Brazilian custodian bank, the custodian notifies the U.S. depository bank of their deposit.
6. Upon notification, the U.S. depository bank issues and delivers Depository Receipts for the Brazilian company shares to the U.S. broker.
7. The U.S. broker then delivers the DRs to the U.S. investor.

Raising Equity Globally

- **ADRs** can be exchanged for the underlying foreign shares, or vice versa, so arbitrage keeps foreign and U.S. prices of any given share the same after adjusting for transfer costs.
- **ADRs** also convey certain technical advantages to U.S. shareholders.
- **Sponsored ADRs** are created at the request of the foreign firm wishing to be listed in the U.S.
- **Unsponsored ADRs** are a result of investor demand and initiated by a U.S. securities firm – these must be approved by the firm whose shares are being listed.

Depository Receipts: Who, What, and Where

- Who – typically rapidly developing companies from emerging markets
- What – IPOs and follow-on offerings
- Where – Mostly New York and London
- Distribution of money raised in the depository receipts market since 2000



Source: "Depository Receipts, Year in Review 2011," JPMorgan, p. 9. Data derived by JPMorgan from other depository banks, Bloomberg, and stock exchanges, January 2012.

Potential (positive) effects of cross-listing

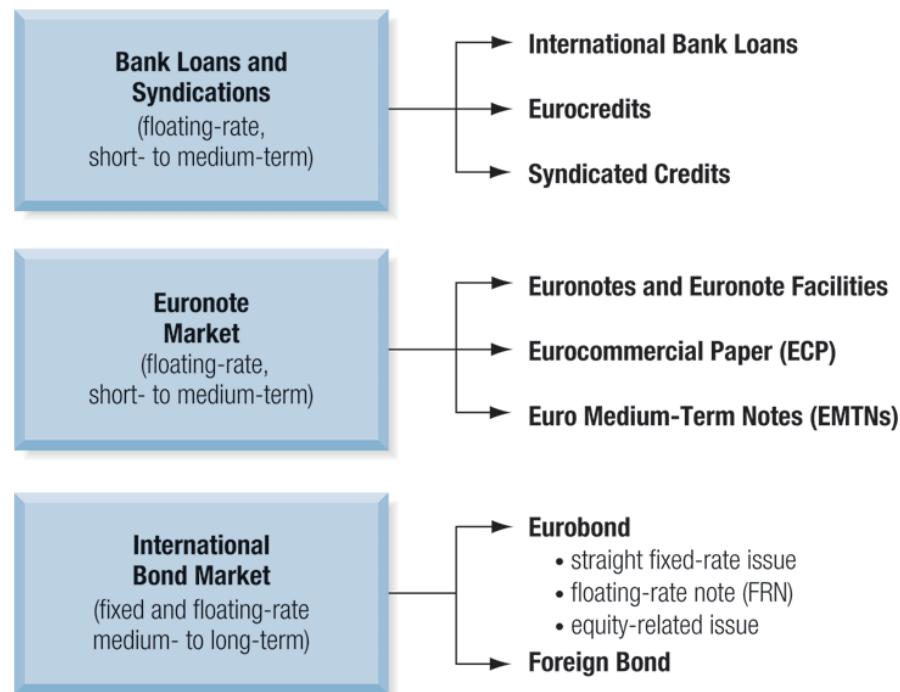
1. The company can expand its potential investor base, which will lead to a higher stock price and a lower cost of capital
2. Creates a secondary market for the company's shares, which facilitates raising new capital in foreign markets
3. It might enhance the liquidity of the company's stock
4. It might enhance the visibility of the company's name and its products in foreign marketplaces
5. Cross-listed shares may be used as the “acquisition currency” for taking over foreign companies
6. It may improve the company's corporate governance and transparency

Potential (negative) effects of cross-listing

1. It can be costly to meet the disclosure and listing requirements imposed by the foreign exchange and regulatory authorities
2. Controlling insiders may find it difficult to continue to derive private benefits once the company is cross-listed on foreign exchanges (flip side of point 6 previous slide)
3. Once a company's stock is traded in overseas markets, there can be volatility spillover from those markets
4. Once a company's stock is made available to foreigners, they might acquire a controlling interest and challenge the domestic control of the company

Raising Debt Globally

- The international debt market offers the borrower a wide variety of different maturities, repayment structures, and currencies of denomination.
- The markets and their many different instruments vary by source of funding, pricing structure, maturity, and subordination.
- The three major sources of debt funding on the international markets:



- **Bank loans and syndications:**
 - International bank loans have traditionally been sourced in the Eurocurrency markets; there is a narrow interest rate spread between deposit and loan rates of less than 1%.
 - Eurocredits are bank loans to MNEs, sovereign governments, international institutions, and banks denominated in Eurocurrencies and extended by banks in countries other than the country in whose currency the loan is denominated.
 - The syndication of loans has enabled banks to spread the risk of very large loans among a number of banks (this is significant for MNEs as they usually need credit in an amount larger than a single bank's loan limit).

- **The Euronote market:**

- Euronotes are short to medium in term and are either underwritten and non-underwritten.
- Euro-commercial paper is a short-term debt obligation of a corporation or bank (usually denominated in US dollars).
- Euro medium-term notes is a new entrant to the world's debt markets, which bridges the gap between Euro-commercial paper and a longer-term and less flexible international bond.

- **The International Bond Market:**
 - A Eurobond is underwritten by an international syndicate of banks and other securities firms and is sold exclusively in countries other than the country in whose currency the issue is denominated
 - A foreign bond is underwritten by a syndicate composed of members from a single country, sold principally within that country, and denominated in the currency of that country
 - The Eurobond markets differ from the Eurodollar markets in that there is an absence of regulatory interference, less stringent disclosure rules and favorable tax treatments for these bonds

SUBSIDIARY VS PARENT AND OTHER RELATED DECISIONS

Foreign Equity Ownership Restrictions

- Several countries impose restrictions on foreign ownership
 - Sometimes, they differ by sector
- Such restrictions are not uncommon
- They limit both the financing decision of the parent company and the subsidiary
 - Raising equity abroad
 - Financing of the subsidiary from within the MNE and mode of operation in foreign countries
- The most common restriction is a limit to % of shares owned by foreign investors

Some examples

Country	Restrictions on Foreigners
Australia	10% in banks, 20% in broadcasting, and 50% in new mining ventures.
Canada	20% in broadcasting, and 25% in bank/insurance companies.
China	Foreigners are restricted to B shares; locals are eligible for A shares.
France	Limited to 20%.
India	Limited to 49%.
Indonesia	Limited to 49%.
Mexico	Limited to 49%.
Japan	Maximum of 25–50% for several major firms; acquisition of over 10% of a single firm subject to approval of the Ministry of Finance.
Korea	Limited to 20%.
Malaysia	20% in banks and 30% in natural resources.
Norway	0% in pulp, paper, and mining, 10% in banks, 20% in industrial and oil shares, and 50% in shipping companies.
Spain	0% in defense industries and mass media. Limited to 50% for other firms.
Sweden	20% of voting shares and 40% of total equity capital.
Switzerland	Foreigners can be restricted to bearer shares.
U.K.	Government retains the veto power over any foreign takeover of British firms.

Source: Various publications of Price Waterhouse.

The effect of some of these restrictions

- Nestle in 1988
 - Registered stocks and bearer stocks
 - Only locals could own registered stocks
- Consider the A-B shares in China
 - What are the effects on financing?
 - What is the effect on prices? Could there be arbitrage?

Financial structure of subsidiaries

- How do we determine the capital structure of a subsidiary?
 - Added layer: international dimension
 - Another layer: financing from the parent or other related firms
- Three ways of looking at it
 - Follow a norm from the parent
 - Follow similar firms in the country where it operates
 - “Opportunistic” view: take advantage of tax structure, financing costs, risks, and potential market imperfections

Financial structure of subsidiaries

- Crucial factor: is the parent company responsible for the financial obligations of the subsidiary?
 - If so, it is the whole MNEs', or at least the parents', structure that matters
- If the connection is weak or non-existent, then the focus is on the subsidiary's structure
 - No legal connection
 - Difficulty to enforce across borders
- Could the parent let a subsidiary default without facing any costs?



ESCUELA DE ADMINISTRACIÓN
FACULTAD DE ECONOMÍA Y ADMINISTRACIÓN

TRADE FINANCE

José A. Tessada

November 2024

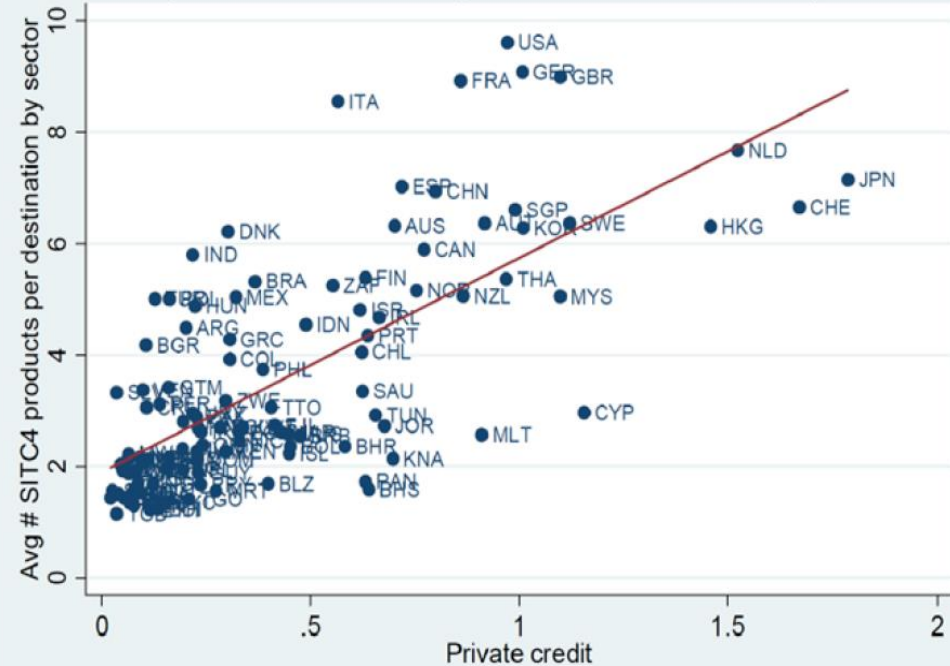
- Trade financing shares a number of common characteristics with the traditional value chain activities conducted by all firms
- All companies must search out suppliers for the many goods and services required as inputs to their own goods production or service provision processes
- Issues to consider in this process include the capability of suppliers to produce the product to adequate specifications, deliver said products in a timely fashion, and to work in conjunction on product enhancements

Some more motivation

Bilateral Exports and Financial Development



Export Product Variety and Financial Development

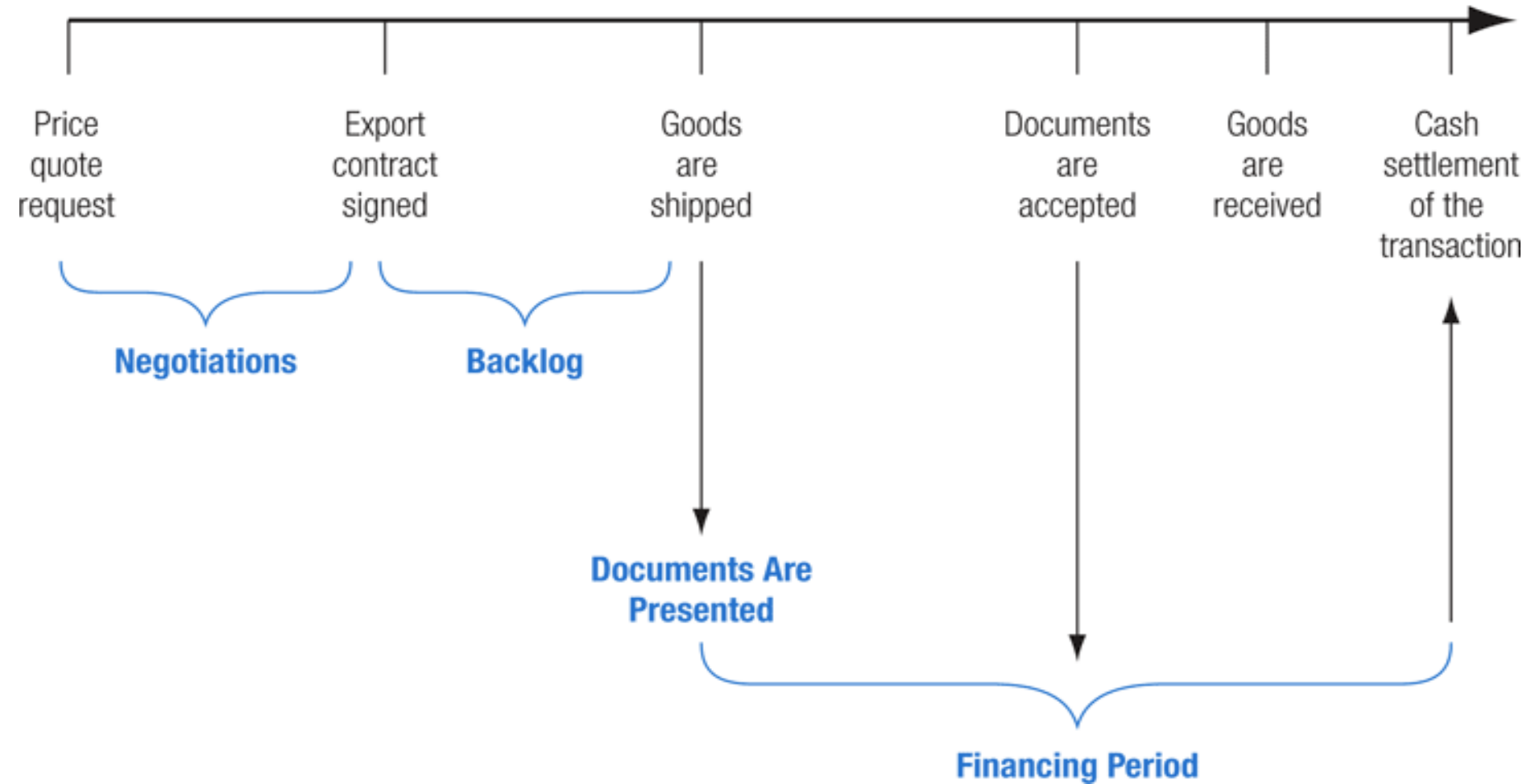


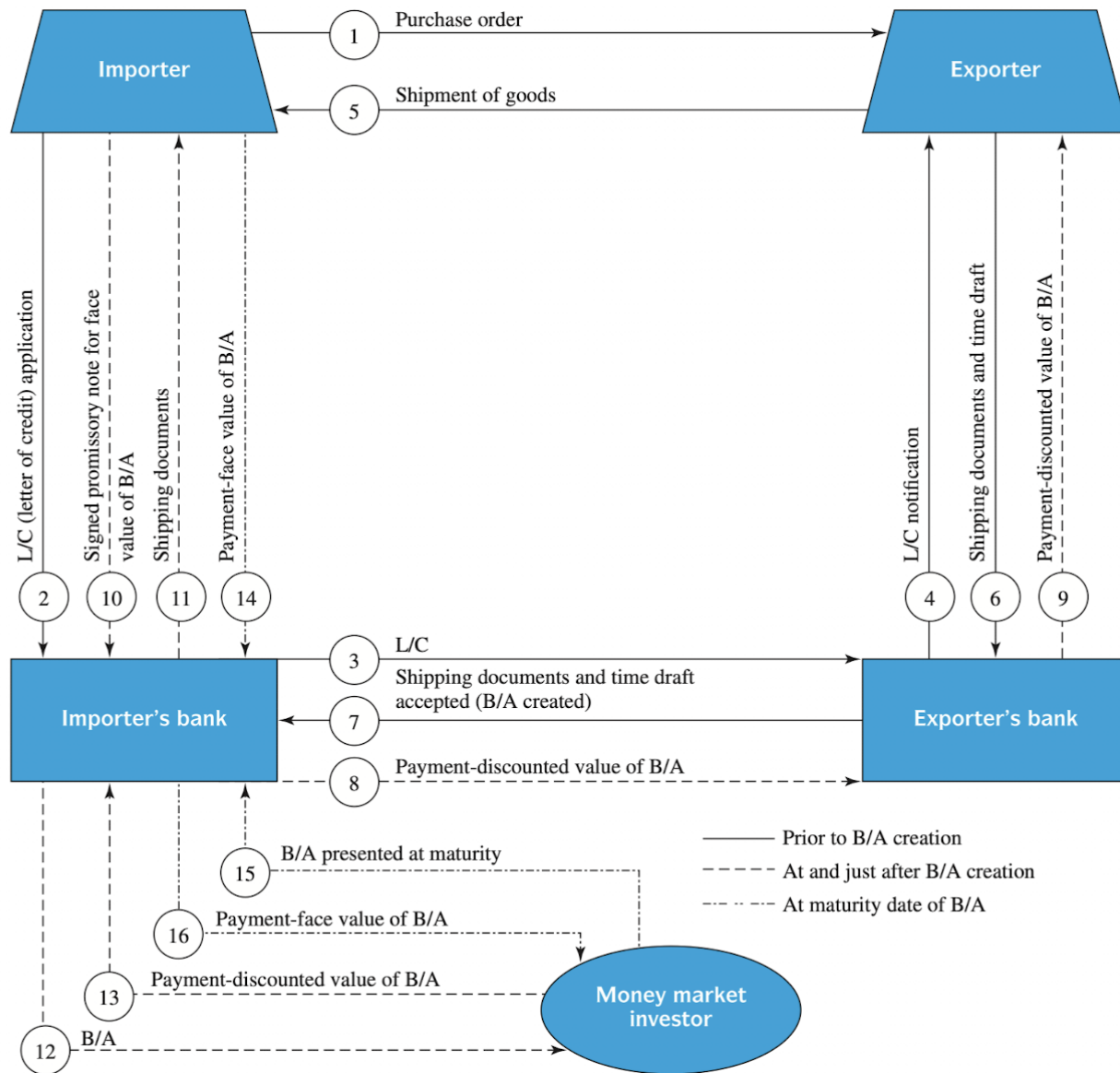
- “Rule of law” in a country affects the development of the sectors that require the intensive use of contracts in the development and production of goods: countries with good rule of law will trade more in these sectors (Nunn 2007)
- The rule of law of the country's neighbors is as much or more important than its own rule of law in international trade of these sectors (Miranda and Wagner 2014)
- “For exporters, any sale is a gift until payment is received... For importers, any payment is a donation until the goods are received” (The US Commerce Department’s International Trade Administration)

Some ideas

- What sectors are these? Can you name an example?
- How can the rule of law affect the production and export of these sectors?
- What is special about trade international in this aspect?
 - Time
 - Distance
 - Cross-border payments

Time and Events





A typical transaction

- Letter of credit: guarantee from the importer's bank that it will act on behalf of the importer and pay the exporter
 - The bank replaces the importer
 - The exporter's bank also plays a role
- Time draft: order to pay a certain amount on a certain date
- Bill of lading: document issued by the carrier stating that it has received the goods
- Time draft + bill of lading: the importer's bank creates a banker's acceptance (B/A), which are negotiable money market instruments
- Notice a (not so) simple alternative: pay in advance
 - Does it work? Could it work?

Prepayment

- To avoid nonpayment, exporters prefer prepayment
- Typically requires the importer to pay the exporter before goods ship
- It may also require payment before the goods are produced
- Prepayment is the least attractive payment option for importers
- Usually uncompetitive for exporters

- Forfaiting
 - Medium-term trade financing
 - Promissory notes signed by the importer in favor of the exporter
 - A bank (forfait) buys the note at a discount
 - Several years, usually associated to capital goods, so amounts are substantial
 - Geographical focus
- Open account
 - Exporter ships and delivers goods before payment, typically 30 to 90 days after delivery
 - Opportunity for the importer to resell goods before having to pay for them
 - Generate working capital needs for exporters
 - Still, they are prevalent because of the highly competitive global market for exports

- Countertrade
 - An umbrella term for some methods that may or may not involve money
 - Includes barter as one of them
 - A two-way flow of goods (or services)
 - Notice that it might work without access to the financial sector
 - One (of many) classifications
 - Barter
 - Clearing arrangement
 - Switch trade
 - Buy-back
 - Counterpurchase
 - Offset transaction